NORTH-WEST UNIVERSITY • ECON122 INTRODUCTION TO MACROECONOMICS

Master Print Study Guide, Formula Vault & Problem Compendium (150 Standalone Problems)

Comprehensive Macroeconomic Study Guide (Chapters 13 to 22), incorporating SARB Monetary Policy Frameworks and Stats SA National Accounting standards. Every analytical problem is presented in full-width single-column clarity with step-by-step KaTeX derivations.

CHAPTER 13 Measurement of Economic Performance • National Accounts & Circular Flow • 10 Hours

Chapter 13: Measuring the Performance of the Economy

The five macroeconomic objectives (short-run stabilisation vs long-run development), national accounting identities (GDP, GDE, GNI, NDP, NI), three calculation methods, market prices vs basic prices vs factor cost, nominal vs real GDP, CPI, Balance of Payments, and Lorenz/Gini income inequality.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Measurement of Economic Performance • 10 Hours):

- Distinguish between short-term macroeconomic policy objectives (economic growth, full employment, price stability, balance of payments stability) and long-run developmental objectives (poverty reduction, human capital, equitable income distribution).
- Define Gross Domestic Product (GDP), explain why intermediate goods are excluded to prevent double counting, and calculate GDP via the production (value-added), income (W + R + i + P), and expenditure (C + I + G + X - Z) methods.
- Calculate and convert between nominal GDP (current prices) and real GDP (constant base-period prices), Gross Domestic Expenditure (GDE), Gross National Income (GNI/GNP), Net Domestic Product (NDP), and National Income (NNP at factor cost).
- Represent, calculate, and interpret unequal distribution of income using the Lorenz curve, the Gini coefficient, and quantile ratios.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (6 IDENTITIES)

EQ 13.1 **Expenditure on GDP vs GDE**

$$\text{GDP}_{mp} = C + I + G + (X - Z) = \text{GDE} + (X - Z)$$

Meaning: Gross Domestic Expenditure (GDE = C + I + G) measures total domestic spending; GDP adds exports (X) and subtracts imports (Z).

EQ 13.2 **Market Prices, Basic Prices & Factor Cost**

$$\text{GDP}_{fc} = \text{GDP}_{mp} - T_{\text{products/prod}} + S_{\text{products/prod}}$$

Meaning: To move from market prices to factor cost, subtract indirect taxes (e.g. VAT) and add subsidies.

EQ 13.3 **Gross National Income (GNI / GNP)**

$$\text{GNI}_{mp} = \text{GDP}_{mp} + \text{FI}_{\text{from ROW}} - \text{FP}_{\text{to ROW}}$$

Meaning: Adds primary factor income earned by residents abroad and subtracts primary factor payments to foreign factor owners inside domestic borders.

EQ 13.4 **Net Domestic Product & National Income**

$$\text{NI} = \text{NDP}_{fc} + \text{FI} - \text{FP} = \text{GDP}_{mp} - D + \text{FI} - \text{FP} - T_{\text{ind}} + S$$

Meaning: Subtracts depreciation (consumption of fixed capital, D) and converts to resident factor cost.

EQ 13.5 **Real GDP & Implicit GDP Deflator**

$$\text{GDP Deflator} = \left(\frac{\text{Nominal GDP}}{\text{Real GDP}}\right) \times 100$$

Meaning: Eliminates price inflation by valuing current quantities at base-year constant prices.

EQ 13.6 **Gini Coefficient from Lorenz Curve**

$$G = \frac{\text{Area } A}{\text{Area } A + \text{Area } B}, \quad 0 \le G \le 1$$

Meaning: Ratio of the area between the line of perfect equality and the Lorenz curve (A) to the total triangular area under the diagonal (A + B).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 13.1 **Five Macroeconomic Objectives** (Core Macroeconomic Concept)

Every macroeconomic policy is judged against five core criteria: (1) Economic growth (real GDP growth), (2) Full employment (low unemployment), (3) Price stability (low inflation within SARB 3–6% band), (4) Balance of payments / external stability, and (5) Equitable distribution of income. Short-term demand policies address the first four objectives, while developmental objectives like lower poverty and human capital unfold in the long run.

> **Worked Proof:** When SARB raises the repo rate to curb inflation (Objective 3), higher borrowing costs may dampen short-run output growth (Objective 1) and employment (Objective 2), illustrating policy trade-offs.

CONCEPT 13.2 **Three Equivalent Methods of Calculating GDP** (Core Macroeconomic Concept)

GDP is the total value of all final goods and services produced within the geographic boundaries of a country in a given period. It can be measured via (1) Production (sum of value added S - B at each stage), (2) Income (Wages + Rent + Interest + Profit, where Profit = S - B - W - R - i), or (3) Expenditure (final sales C + I + G + X - Z).

> **Worked Proof:** In a representative event-catering chain (Metro → Namakwa Liquor Store → Kosie): Metro sells supplies for R6 000 (Buy = R0, VA = R6 000, W = R3 500, R = R900, i = R700, Profit = R900); Namakwa sells to Kosie for R12 000 (Buy = R6 000, VA = R6 000, W = R3 200, R = R1 500, i = R500, Profit = R800); Kosie organises the event for R20 000 (Buy = R12 000, VA = R8 000, W = R2 000, Profit = R6 000). Both ∑VA (6 000 + 6 000 + 8 000) and ∑Factor Incomes equal R20 000.

CONCEPT 13.3 **Market Prices vs. Basic Prices vs. Factor Cost & NNP** (Core Macroeconomic Concept)

Indirect taxes (like VAT) raise the market price paid by consumers above what factors of production actually earn, whereas subsidies lower the market price below factor cost. Converting from Gross to Net subtracts depreciation (D), and moving from Domestic to National adds net factor income from the rest of the world.

Worked Proof: Given GDP at market prices = R955m, Capital goods = R92m, Depreciation = 9% of capital goods (R8.28m), contract earnings abroad = R23m, and migrant wages paid = R52m (while state pensions = R61m and personal income tax = R25m are excluded): NNP at market prices = 955 - 8.28 + 23 - 52 = R917.72m.

CONCEPT 13.4 **Balance of Payments & Lorenz Curve Inequality** (Core Macroeconomic Concept)

The Balance of Payments records all transactions with the rest of the world across the Current Account (merchandise trade, net gold, services, primary & secondary income) and Financial Account (direct, portfolio, and other investment). Inequality is visualised via the Lorenz curve against the 45° diagonal of perfect equality.

Worked Proof: South Africa's Gini coefficient is approximately 0.63–0.67, meaning the Lorenz curve bows far away from the 45° line of absolute equality.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Five Macroeconomic Objectives): Assuming 'price stability' means prices are frozen at zero percent change forever.

First-Principles Law: Price stability means keeping the rate of change in the general price level low and predictable (in South Africa, inside the 3%–6% inflation target range, anchored near the 4.5% midpoint or 3% point target).

Fallacy (Three Equivalent Methods of Calculating GDP): Adding the full selling prices of Metro (R6k) + Namakwa (R12k) + Kosie (R20k) = R38k, or Dr Ray (R45k) + L'Oréal (R160k) + Edgars (R240k) = R445k.

First-Principles Law: Summing raw gross sales across stages causes double counting of intermediate goods. Always sum Value Added (Sales minus Intermediate Purchases) or count only final consumer expenditure.

Fallacy (Market Prices vs. Basic Prices vs. Factor Cost & NNP): Adding indirect taxes and subtracting subsidies when moving from Market Prices to Factor Cost, or subtracting state pension transfers when computing NNP.

First-Principles Law: Market prices already include indirect taxes (+T) and exclude subsidies (-S). To recover true Factor Cost, reverse the signs (-T + S), and always exclude pure government transfer payments (pensions/grants) and direct income taxes from GDP/NNP conversions.

Fallacy (Balance of Payments & Lorenz Curve Inequality): Thinking a Gini coefficient closer to 100 (or 1.0) indicates higher income equality.

First-Principles Law: A Gini coefficient of 0 represents perfect equality (everyone earns the same share), while 1 (or Gini index 100) represents absolute inequality (one person receives 100% of national income).

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (32 PROBLEMS)

PROBLEM 13.1 — VALUE-ADDED & DOUBLE-COUNTING ELIMINATION (DR RAY & L'ORÉAL)

Macroeconomics, §13.2, pp. 239–241 • Medium

Dr Ray sells natural beauty inputs to L'Oréal for R45 000 (paying R15 000 wages, R10 000 rent, R1 500 interest). L'Oréal processes and sells the products to Edgars for R160 000 (paying R30 000 wages, R14 000 rent, R3 000 interest). Edgars sells the finished products to final consumers for R240 000 (paying R12 500 wages, R4 000 interest). What is the value added by L'Oréal and the total GDP contribution?

Verified Solution: L'Oréal adds R115 000 in value, yielding a total final GDP of R240 000.

$$\text{VA}_{\text{Dr Ray}} = 45\,000 - 0 = \text{R}45\,000; \quad \text{VA}_{\text{L'Oréal}} = 160\,000 - 45\,000 = \text{R}115\,000; \quad \text{VA}_{\text{Edgars}} = 240\,000 - 160\,000 = \text{R}80\,000.$$
$$\text{Total GDP} = 45\,000 + 115\,000 + 80\,000 = \text{R}240\,000.$$

First-Principles Proof: Correct! L'Oréal's value added is sales (R160 000) minus intermediate inputs from Dr Ray (R45 000) = R115 000, and total GDP equals final sales of R240 000.

PROBLEM 13.2 — DERIVING NDP AT MARKET PRICES & GROSS NATIONAL INCOME (GNI)

Macroeconomics, §13.2–13.3, pp. 241–245 • Medium

Given the following national accounting data (in R millions): GDP at market prices = 150 000; VAT (taxes on products) = 14 000; Interest on government debt = 8 500; Depreciation (consumption of fixed capital) = 5 500; Subsidies on products = 25 000; Wages of migrant workers = 15 000. Calculate Net Domestic Product (NDP) at market prices and GDP at factor cost.

Verified Solution: NDP at market prices is R144 500 million and GDP at factor cost is R161 000 million.

$$\text{NDP}_{mp} = \text{GDP}_{mp} - D = 150\,000 - 5\,500 = 144\,500.$$
$$\text{GDP}_{fc} = \text{GDP}_{mp} - \text{VAT} + \text{Subsidies} = 150\,000 - 14\,000 + 25\,000 = 161\,000.$$

First-Principles Proof: Correct! Subtracting depreciation (5 500) gives NDPmp = 144 500, and subtracting VAT (14 000) while adding subsidies (25 000) gives GDPfc = 161 000.

PROBLEM 13.3 — CONVERTING GDP AT FACTOR COST TO BASIC PRICES & MARKET PRICES

Macroeconomics, §13.3, pp. 244–247 • Hard

Use the following South African data (in Rand): GDP at factor cost = R293 250; VAT = R16 000; Subsidies = R43 300; Exports = R20 000; Imports = R10 000; Capital Goods = R40 000; Depreciation = 8% of Capital Goods; Salaries earned by SA cricket players in the UK = R35 200; Profits earned by UK Airlines trading in SA = R125 600. Calculate Gross Domestic Expenditure (GDE) at market prices and National Income (NI).

Verified Solution: GDE at market prices equals R255 950 and National Income equals R199 650.

$$\begin{gathered}\text{GDP}_{mp} = \text{GDP}_{fc} + \text{VAT} - \text{Subsidies} = 293\,250 + 16\,000 - 43\,300 = \text{R}265\,950. \\ \text{GDE}_{mp} = \text{GDP}_{mp} - X + Z = 265\,950 - 20\,000 + 10\,000 = \text{R}255\,950. \\ D = 0.08 \times 40\,000 = \text{R}3\,200; \quad \text{NI} = \text{GDP}_{fc} - D + \text{FI} - \text{FP} = 293\,250 - 3\,200 + 35\,200 - 125\,600 = \text{R}199\,650.\end{gathered}$$

First-Principles Proof: Correct! GDPmp = 265 950 so GDEmp = 265 950 - 20 000 + 10 000 = R255 950, and NI = 293 250 - 3 200 + 35 200 - 125 600 = R199 650.

PROBLEM 13.4 — RECONSTRUCTING GDP AT FACTOR COST FROM NATIONAL INCOME & DEPRECIATION

Macroeconomics, §13.3, pp. 244–247 • Hard

Given the following quarterly national accounting statistics: National Income (NI at factor cost) = 870 760; VAT = 120 000; Subsidies = 9 000; Government transfer payments = 15 500; Depreciation = 11 100; Wages paid to foreign migrant labourers working domestically = 120 800; Earnings of domestic citizens working abroad = 234 000; Exports = 553 200; Imports = 489 300. Calculate GDP at factor cost and GDE at market prices.

Verified Solution: GDP at factor cost is 768 660 and GDE at market prices is 815 760.

$$\begin{gathered}\text{GDP}_{fc} = \text{NI} + D - \text{FI}_{\text{from ROW}} + \text{FP}_{\text{to ROW}} = 870\,760 + 11\,100 - 234\,000 + 120\,800 = 768\,660. \\ \text{GDP}_{mp} = 768\,660 + 120\,000 - 9\,000 = 879\,660. \\ \text{GDE}_{mp} = \text{GDP}_{mp} - X + Z = 879\,660 - 553\,200 + 489\,300 = 815\,760.\end{gathered}$$

First-Principles Proof: Correct! Working backwards from NI (870 760) adds depreciation (+11 100) and foreign factor payments (+120 800) and subtracts factor receipts (-234 000) to get 768 660.

PROBLEM 13.5 — EQUIVALENCE OF INCOME & PRODUCTION GDP UNDER MINIMUM-WAGE SHIFTS

Macroeconomics, §13.2, pp. 239–241 • Easy

Suppose the prescribed minimum wage in South Africa increases by 8%. How much would total GDP calculated using the income method differ from total GDP estimated using the production (value-added) method, assuming no measurement errors?

Verified Solution: There is zero difference because the production and income methods measure the exact same GDP.

$$\text{GDP}_{\text{Production}} \equiv \sum \text{Value Added} \equiv \text{GDP}_{\text{Income}} \equiv W + R + i + \Pi \implies \text{Difference} = 0.$$

First-Principles Proof: Correct! By national accounting identity, total value added equals total factor income (Wages + Rent + Interest + Profit); a higher wage share simply alters the split between wages and operating surplus.

PROBLEM 13.6 — EXPENDITURE GDP (C+I+G+X-Z), GDE & FACTOR COST VALUATION

Macroeconomics, §13.2–13.3, pp. 241–246 • Medium

Table 1 provides South African statistics for Period 2 (in R millions): Household expenditure = 974 252; Fixed capital consumption = 170 986; Exports = 492 646; Imports = 569 321; Taxes on production and imports = 205 562; Subsidies = 11 433; GDP at market prices (GDPMP) = 1 530 580; Wages paid to RSA au pairs working in the USA = 1 500; Salaries paid to Cuban doctors working in RSA = 1 800. Calculate GDP at factor cost (GDPfc) and Gross National Product at market prices (GNPmp).

Verified Solution: GDP at factor cost is R1 336 451 million and GNP at market prices is R1 530 280 million.

$$\begin{gathered}\text{GDP}_{fc} = \text{GDP}_{mp} - T_{\text{ind}} + S = 1\,530\,580 - 205\,562 + 11\,433 = \text{R}1\,336\,451\text{m}. \\ \text{GNP}_{mp} = \text{GDP}_{mp} + \text{FI}_{\text{RSA in USA}} - \text{FP}_{\text{Cubans in RSA}} = 1\,530\,580 + 1\,500 - 1\,800 = \text{R}1\,530\,280\text{m}.\end{gathered}$$

First-Principles Proof: Correct! Subtracting taxes (205 562) and adding subsidies (11 433) yields GDPfc = 1 336 451, while adding RSA au pair earnings (+1 500) and subtracting Cuban doctor salaries (-1 800) gives GNPmp = 1 530 280.

PROBLEM 13.7 — VALUE-ADDED CHAIN IN EVENT PRODUCTION (CELEBRATION HOUSE GALA)

Macroeconomics, §13.2, pp. 239–241 • Hard

Bonang Matheba sponsors a charity gala for R15 307 200, hiring Celebration Creations (which pays R592 200 venue rent, 12 servers at R11 700 each = R140 400 wages, and earns R3 612 600 profit). Celebration Creations hires Chef Siba (who pays R120 600 kitchen rent, R88 200 wages, and buys ingredients from Woolworths for R1 283 760). Woolworths pays R73 800 rent and R108 000 wages. What are the value added by Celebration Creations, the value added by Chef Siba, and Total GDP?

Verified Solution: Celebration Creations VA is R4 345 200, Chef Siba VA is R9 678 240, and GDP is R15 307 200.

$$\begin{gathered}\text{VA}_{\text{CC}} = 140\,400 + 592\,200 + 3\,612\,600 = \text{R}4\,345\,200 \implies S_{\text{Siba}} = 15\,307\,200 - 4\,345\,200 = \text{R}10\,962\,000. \\ \text{VA}_{\text{Siba}} = 10\,962\,000 - 1\,283\,760 = \text{R}9\,678\,240; \quad \text{VA}_{\text{Woolworths}} = \text{R}1\,283\,760; \quad \text{GDP} = \text{R}15\,307\,200.\end{gathered}$$

First-Principles Proof: Correct! Celebration Creations' factor incomes sum to R4 345 200, leaving R10 962 000 paid to Chef Siba; subtracting Woolworths inputs (R1 283 760) gives Siba VA = R9 678 240.

PROBLEM 13.8 — NATIONAL ACCOUNTING MATRIX: GDPBP, GDPFC, GNIMP & NET PRIMARY INCOME

Macroeconomics, §13.2–13.3, pp. 241–246 • Hard

Use the following SA National Accounts data (R millions): GDP at market prices (GDPmp) = 1 204; Consumption of fixed capital = 72; Factor income earned abroad by South Africans = 24; Factor income earned in SA by foreigners = 13; Exports = 64; Imports = 100; Taxes on production and imports = 148; Subsidies = 15. Calculate National Income (NI / NNI at factor cost) and Gross Domestic Expenditure at factor cost (GDEfc).

Verified Solution: National Income (NI) is R1 010 million and Gross Domestic Expenditure at factor cost is R1 107 million.

$$\text{NI} = \text{GDP}_{mp} - D + \text{FI}_{\text{rec}} - \text{FP}_{\text{paid}} - T_{\text{ind}} + S = 1\,204 - 72 + 24 - 13 - 148 + 15 = \text{R}1\,010\text{m}.$$
$$\text{GDE}_{fc} = (\text{GDP}_{mp} - X + Z) - T_{\text{ind}} + S = (1\,204 - 64 + 100) - 148 + 15 = \text{R}1\,107\text{m}.$$

First-Principles Proof: Correct! NI = 1 204 - 72 + 24 - 13 - 148 + 15 = R1 010m, and GDEfc = 1 204 - 64 + 100 - 148 + 15 = R1 107m.

PROBLEM 13.9 — BASE-PERIOD VS CURRENT-PERIOD NOMINAL & REAL GDP (HOGWARTS BASKET)

Macroeconomics, §13.2, pp. 244–245 • Easy

In the economy of Hogwarts, three goods are produced: Cell phones (Period 1 base period: P = R100, Q = 1; Period 2: P = R110, Q = 1), Tablets (Period 1: P = R10, Q = 8; Period 2: P = R12, Q = 10), and Laptops (Period 1: P = R5, Q = 4; Period 2: P = R4, Q = 5). Calculate Period 1 Nominal GDP, Period 2 Nominal GDP (and percentage nominal growth), and Period 2 Real GDP at Period 1 constant prices.

Verified Solution: Period 1 Nominal GDP is R200, Period 2 Nominal GDP is R250 (up 25%), and Period 2 Real GDP is R225.

$$\text{Nom GDP}_1 = 100(1) + 10(8) + 5(4) = \text{R}200. \quad \text{Nom GDP}_2 = 110(1) + 12(10) + 4(5) = \text{R}250\ (+25\%).$$
$$\text{Real GDP}_2 = 100(1) + 10(10) + 5(5) = 100 + 100 + 25 = \text{R}225\ (+12.5\%\ \text{real growth}).$$

First-Principles Proof: Correct! Period 1 Nominal GDP = 100 + 80 + 20 = R200; Period 2 Nominal GDP = 110 + 120 + 20 = R250 (up 25%); and Period 2 Real GDP at base prices = 100(1) + 10(10) + 5(5) = R225.

PROBLEM 13.10 — MULTI-GOOD NOMINAL GDP, REAL GDP & IMPLICIT PRICE DEFLATOR

Macroeconomics, §13.2 & §20.2, pp. 244–245, 442 • Medium

An economy produces Fish, Pork, and Beef. In the Base Period, prices and quantities were Fish: P = 7, $Q = 400$; $Pork: P =$8, Q = 225; Beef: P = 10, $Q = 175. In the Current Period, prices and quantities are Fish: P =$9, Q = 900; Pork: P = 6, $Q = 275$; $Beef: P =$15, Q = 275. Calculate Current Period Nominal GDP, Current Period Real GDP (at Base Period prices), and the Current Period GDP Deflator.

Verified Solution: Current Nominal GDP is 13875, $RealGDPis$11 250, and the GDP Deflator is 123.33 (up 23.33%).

$$\text{Nom GDP} = 9(900) + 6(275) + 15(275) = 8\,100 + 1\,650 + 4\,125 = \$13\,875.$$
$$\text{Real GDP} = 7(900) + 8(275) + 10(275) = 6\,300 + 2\,200 + 2\,750 = \$11\,250. \quad \text{Deflator} = \frac{13\,875}{11\,250} \times 100 = 123.33.$$

First-Principles Proof: Correct! Current Nominal GDP = 8 100 + 1 650 + 4 125 = 13875; $RealGDPatbaseprices = 6300 + 2200 + 2750 =$11 250; Deflator = (13 875 / 11 250) × 100 = 123.33.

PROBLEM 13.11 — VALUE-ADDED, EXPENDITURE & INCOME APPROACHES IN A COFFEE SUPPLY CHAIN

Value-Added, Expenditure & Income Approaches in a Coffee Supply Chain • Hard

A domestic coffee farmer harvests 30 kg of green coffee beans and sells the entire harvest to an international roaster for R50 per kg. The roaster processes and packages the beans into 30 retail packs and exports them back to a domestic coffeehouse chain at R200 per pack. The coffeehouse chain brews and sells the coffee to final consumers, generating R5,000 in net value added beyond its import cost. Compute (1) the farmer's value added, (2) the roaster's value added, (3) total final retail expenditure, and (4) the domestic economy's total Value Added if the roaster operates abroad.

Verified Solution: Farmer VA = R1,500; Roaster VA = R4,500; Final retail expenditure = R11,000; Domestic VA = R6,500 (Total chain VA = R11,000).

$$\text{Step 1 (Farmer VA): } 30\ \text{kg} \times \text{R}50 = \text{R}1,500\ (\text{zero prior intermediate inputs})$$
$$\text{Step 2 (Roaster VA): } (30 \times \text{R}200) - \text{R}1,500 = \text{R}6,000 - \text{R}1,500 = \text{R}4,500$$
$$\text{Step 3 (Final Retail Sales): R}6,000\ (\text{intermediate cost}) + \text{R}5,000\ (\text{retailer VA}) = \text{R}11,000$$
$$\text{Step 4 (Domestic GDP vs Total Chain VA): Total Chain VA} = 1,500 + 4,500 + 5,000 = \text{R}11,000; \quad \text{Domestic VA} = 1,500 + 5,000 = \text{R}6,500$$

First-Principles Proof: Correct! Summing value added across all three stages equals final retail expenditure (R11,000), while domestic GDP excludes the R4,500 value added created abroad.

PROBLEM 13.12 — MULTI-GOOD REAL GDP & REAL ECONOMIC GROWTH RATE

Multi-Good Real GDP & Real Economic Growth Rate • Medium

An economy produces three goods (Clothes, Food, and Drink). In Period 1 (the base period), prices and quantities are: Clothes (P = R260, Q = 13), Food (P = R2 150, Q = 12), and Drink (P = R655, Q = 11). In Period 2, prices and quantities are: Clothes (P = R310, Q = 10), Food (P = R2 100, Q = 13), and Drink (P = R680, Q = 11). Calculate Real GDP in Period 1, Real GDP in Period 2 (at constant Period 1 prices), and the real economic growth rate.

Verified Solution: Period 1 Real GDP = R36 385; Period 2 Real GDP = R37 755; Real GDP growth rate = +3.77% (Nominal GDP = R37 880).

$$\text{Real GDP}_1 = (260 \times 13) + (2\,150 \times 12) + (655 \times 11) = 3\,380 + 25\,800 + 7\,205 = \text{R}36\,385$$
$$\text{Real GDP}_2 \text{ (at Period 1 constant prices)} = (260 \times 10) + (2\,150 \times 13) + (655 \times 11) = 2\,600 + 27\,950 + 7\,205 = \text{R}37\,755$$
$$\text{Nominal GDP}_2 \text{ (at Period 2 current prices)} = (310 \times 10) + (2\,100 \times 13) + (680 \times 11) = 3\,100 + 27\,300 + 7\,480 = \text{R}37\,880$$
$$g_{\text{real}} = \frac{37\,755 - 36\,385}{36\,385} \times 100 = \frac{1\,370}{36\,385} \times 100 = +3.77\%$$

First-Principles Proof: Correct! Period 1 Real GDP is 3 380 + 25 800 + 7 205 = R36 385, and Period 2 Real GDP at Period 1 constant prices is 2 600 + 27 950 + 7 205 = R37 755 (+3.77% real growth).

PROBLEM 13.13 — RECONCILING VALUE-ADDED AND FACTOR INCOME IN A SERVICE PRODUCTION CHAIN

Reconciling Value-Added and Factor Income in a Service Production Chain • Hard

A hospitality event involves three domestic firms: (1) A wholesale food supplier with zero intermediate inputs sells ingredients for $71,220, paying 10,100$ in wages (the remainder is operating profit); (2) A catering firm purchases those ingredients $(71,220), pays 11,400$ in wages, and earns $527,680 in profit$; (3) $An event management firm purchases the catered banquet, pays 200,000$ in wages and $40,000 in profit, and sells the final event package for$ 850,300. Determine the catering firm's sales revenue, the catering firm's value added, the event management firm's value added, and total GDP.

Verified Solution: Caterer sales = $610,300$; $Caterer VA =$539,080; Event Manager VA = $240,000$; $Total GDP =$850,300.

$$\text{Caterer Value Added} = \text{Wages} + \text{Profit} = 11,400 + 527,680 = \$539,080$$
$$\text{Caterer Sales Revenue} = \text{Intermediate Inputs} + \text{Value Added} = 71,220 + 539,080 = \$610,300$$
$$\text{Event Manager Value Added} = 850,300 - 610,300 = \$240,000 = \text{Wages } (200,000) + \text{Profit } (40,000)$$
$$\text{Total GDP} = 71,220 + 539,080 + 240,000 = \$850,300 = \text{Total Wages } (221,500) + \text{Total Profit } (628,800)$$

First-Principles Proof: Correct! Every firm's net value added equals the sum of its factor incomes (wages + profit), and total value added equals final expenditure ($850,300).

PROBLEM 13.14 — BRIDGING GDE AT MARKET PRICES TO GDP AND NET NATIONAL INCOME

Bridging GDE at Market Prices to GDP and Net National Income • Hard

Given the following National Accounts data (in R millions): Gross Domestic Expenditure at market prices (GDE_mp) = 9,000; Exports of goods and services (X) = 6,900; Imports of goods and services (Z) = 6,400; Capital stock = 5,500 (depreciating at 7% per annum); Foreign factor receipts = 3,500; Foreign factor payments = 2,900; Taxes on production/products = 600; Subsidies = 3,100. Calculate GDP at market prices (GDP_mp) and Net National Income at factor cost (NI).

Verified Solution: GDP at market prices = R9,500 million; Depreciation = R385 million; Net National Income (factor cost) = R12,215 million.

$$\text{GDP}_{mp} = \text{GDE}_{mp} + X - Z = 9,000 + 6,900 - 6,400 = \text{R}9,500 \text{ million}$$
$$\text{Depreciation } (D) = 0.07 \times 5,500 = \text{R}385 \text{ million}$$
$$\text{GNP}_{mp} = 9,500 + 3,500 - 2,900 = \text{R}10,100 \text{ million}; \quad \text{NNP}_{mp} = 10,100 - 385 = \text{R}9,715 \text{ million}$$
$$\text{NI (at factor cost)} = \text{NNP}_{mp} - \text{Taxes} + \text{Subsidies} = 9,715 - 600 + 3,100 = \text{R}12,215 \text{ million}$$

First-Principles Proof: Correct! Adding net exports (+500) to GDE gives GDP_mp = 9,500; adjusting for net foreign factor income (+600), depreciation (-385), and net subsidies (+2,500) yields R12,215m.

PROBLEM 13.15 — DERIVING GDE AT FACTOR COST AND NET NATIONAL INCOME AT MARKET PRICES

Deriving GDE at Factor Cost and Net National Income at Market Prices • Hard

An open economy reports the following figures (in R millions): GDP at market prices (GDP_mp) = 8,000; Exports (X) = 6,100; Imports (Z) = 8,350; Subsidies = 3,200; Value-Added Tax (indirect taxes) = 900; Capital stock = 6,000 with an annual depreciation rate of 7%; Factor income earned abroad by residents = 2,500; Factor income paid to non-residents = 3,100. Calculate (1) Gross Domestic Expenditure at factor cost (GDE_fc) and (2) Net National Income at market prices (NNP_mp).

Verified Solution: GDE at factor cost (GDE_fc) = R12,550 million; Net National Income at market prices (NNP_mp) = R6,980 million.

$$\text{Step 1: } \text{GDE}_{mp} = \text{GDP}_{mp} - X + Z = 8,000 - 6,100 + 8,350 = \text{R}10,250 \text{ million}$$
$$\text{Step 2: } \text{GDE}_{fc} = \text{GDE}_{mp} - \text{Taxes} + \text{Subsidies} = 10,250 - 900 + 3,200 = \text{R}12,550 \text{ million}$$
$$\text{Step 3: Depreciation } (D) = 0.07 \times 6,000 = \text{R}420 \text{ million}$$
$$\text{Step 4: } \text{NNP}_{mp} = \text{GDP}_{mp} + \text{Net Foreign Income} - D = 8,000 + (2,500 - 3,100) - 420 = \text{R}6,980 \text{ million}$$

First-Principles Proof: Correct! Since imports exceed exports by 2,250, GDE_mp is 10,250; adding net subsidies (+2,300) gives GDE_fc = 12,550, and deducting net foreign outflow (-600) and depreciation (-420) from GDP_mp gives 6,980.

PROBLEM 13.16 — COMPUTING NDP, FACTOR COST GDP, AND GNP FROM BASE ACCOUNTS

Computing NDP, Factor Cost GDP, and GNP from Base Accounts • Medium

Suppose an economy reports GDP at market prices = R45,000 million, Consumption of fixed capital (depreciation) = R2,500 million, Indirect taxes = R1,500 million, Subsidies = R1,500 million, Factor income earned in the rest of the world by residents = R2,000 million, and Factor income earned in the domestic economy by non-residents = R5,000 million. Compute (1) Net Domestic Product at market prices (NDP_mp), (2) GDP at factor cost (GDP_fc), and (3) Gross National Product at market prices (GNP_mp).

Verified Solution: NDP at market prices = R42,500 million; GDP at factor cost = R45,000 million; GNP at market prices = R42,000 million.

$$\text{NDP}_{mp} = \text{GDP}_{mp} - \text{Depreciation} = 45,000 - 2,500 = \text{R}42,500 \text{ million}$$
$$\text{GDP}_{fc} = \text{GDP}_{mp} - \text{Indirect Taxes} + \text{Subsidies} = 45,000 - 1,500 + 1,500 = \text{R}45,000 \text{ million}$$
$$\text{GNP}_{mp} = \text{GDP}_{mp} + \text{Foreign Receipts} - \text{Foreign Payments} = 45,000 + 2,000 - 5,000 = \text{R}42,000 \text{ million}$$

First-Principles Proof: Correct! Equal indirect taxes and subsidies leave GDP_fc equal to GDP_mp (R45,000m), while deducting R2,500m depreciation gives NDP_mp = R42,500m and net factor outflow (-R3,000m) gives GNP_mp = R42,000m.

PROBLEM 13.17 — ISOLATING VOLUME GROWTH FROM PRICE INFLATION IN A TWO-GOOD ECONOMY

Isolating Volume Growth from Price Inflation in a Two-Good Economy • Medium

An economy produces only two final goods: Good A (Bread) and Good B (Milk). In Period 1 (the base period), 1 000 loaves of Bread are produced and sold at R15 each and 2 000 litres of Milk are produced and sold at R5 each. In Period 2, 1 200 loaves of Bread are produced and sold at R20 each and 2 200 litres of Milk are produced and sold at R10 each. Calculate Period 1 Nominal GDP, Period 2 Nominal GDP (and percentage nominal growth), and Period 2 Real GDP at constant Period 1 prices.

Verified Solution: Period 1 GDP = R25 000; Period 2 Nominal GDP = R46 000 (+84.0%); Period 2 Real GDP = R29 000 (+16.0% real growth).

$$\text{Period 1 GDP (Nominal} = \text{Real)} = (1\,000 \times 15) + (2\,000 \times 5) = 15\,000 + 10\,000 = \text{R}25\,000$$
$$\text{Period 2 Nominal GDP} = (1\,200 \times 20) + (2\,200 \times 10) = 24\,000 + 22\,000 = \text{R}46\,000 \implies \%\Delta\text{Nominal} = \frac{46\,000 - 25\,000}{25\,000} \times 100 = +84.0\%$$
$$\text{Period 2 Real GDP (at Period 1 prices)} = (1\,200 \times 15) + (2\,200 \times 5) = 18\,000 + 11\,000 = \text{R}29\,000$$
$$\%\Delta\text{Real GDP} = \frac{29\,000 - 25\,000}{25\,000} \times 100 = +16.0\%$$

First-Principles Proof: Correct! Period 1 GDP is 15 000 + 10 000 = R25 000; Period 2 Nominal GDP is 24 000 + 22 000 = R46 000 (+84.0%), while holding prices constant at Period 1 levels gives R29 000 (+16.0%).

PROBLEM 13.18 — DERIVING NET NATIONAL INCOME AND GDE FROM GDP AT MARKET PRICES

Deriving Net National Income and GDE from GDP at Market Prices • Medium

Consider the following national accounting data (in R millions): GDP at market prices (GDP_mp) = 550,250; Gross Domestic Expenditure (GDE) = 554,750 (with Exports X = 120,500 and Imports Z = 125,000); Consumption of fixed capital (depreciation) = 12,450; Factor income received from abroad = 8,200; Factor income paid to abroad = 11,400; Indirect taxes = 22,300; Subsidies = 15,000. Verify GDE at market prices and calculate Net National Income at factor cost (NI).

Verified Solution: GDE at market prices = R554,750 million; GNP at market prices = R547,050 million; Net National Income (NI) = R527,300 million.

$$\text{GDE}_{mp} = \text{GDP}_{mp} - X + Z = 550,250 - 120,500 + 125,000 = \text{R}554,750 \text{ million}$$
$$\text{GNP}_{mp} = 550,250 + 8,200 - 11,400 = \text{R}547,050 \text{ million}$$
$$\text{NNP}_{mp} = 547,050 - 12,450 = \text{R}534,600 \text{ million}$$
$$\text{NI (at factor cost)} = 534,600 - 22,300 + 15,000 = \text{R}527,300 \text{ million}$$

First-Principles Proof: Correct! Net imports (+4,500) make GDE_mp = 554,750; subtracting net foreign factor outflow (-3,200), depreciation (-12,450), and net indirect taxes (-7,300) yields NI = R527,300m.

PROBLEM 13.19 — THE FIVE MACROECONOMIC OBJECTIVES & BOP SUB-ACCOUNTS

The Five Macroeconomic Objectives & BoP Sub-Accounts • Easy

Which five fundamental macroeconomic objectives serve as criteria for evaluating economic performance, and how does the Balance of Payments (BoP) divide international transactions between its two primary sub-accounts?

Verified Solution: Economic growth, full employment, price stability, BoP stability, and equitable income distribution; the Current Account records trade in goods/services and income flows, while the Financial Account records cross-border asset/liability flows.

$$\text{Five Objectives: (1) Economic growth, (2) Full employment, (3) Price stability, (4) Balance of payments (external) stability, (5) Equitable distribution of i}$$
$$\text{Current Account: records exports/imports of goods and services, primary factor income flows, and current transfers}$$
$$\text{Financial Account: records international flows of financial assets and liabilities (direct, portfolio, and other investment)}$$

First-Principles Proof: Correct! These are the five pillars of macroeconomic policy and the exact division between the Current and Financial accounts of the Balance of Payments.

PROBLEM 13.20 — INTERPRETING THE LORENZ CURVE DIAGONAL, INEQUALITY AREA, AND GINI COEFFICIENT

Interpreting the Lorenz Curve Diagonal, Inequality Area, and Gini Coefficient • Easy

In a standard Lorenz curve diagram where cumulative population percentage is plotted on the horizontal axis and cumulative income percentage is plotted on the vertical axis: (1) What does the 45-degree diagonal line 0B represent? (2) What does the crescent area between the diagonal 0B and the bowed Lorenz curve represent? (3) How is a Gini coefficient of 0.67 interpreted?

Verified Solution: (1) The line of perfect equality; (2) The area of income inequality; (3) A Gini coefficient of 0.67 indicates very high income inequality (since 0 is perfect equality and 1 is perfect inequality).

$$\text{Diagonal } 0B: \text{Line of perfect equality (every } x\% \text{ of the population receives exactly } x\% \text{ of total income)}$$
$$\text{Area between } 0B \text{ and Lorenz curve: Area of inequality } (A)$$
$$\text{Gini Coefficient} = \frac{\text{Area of inequality } A}{\text{Total triangular area under diagonal } (A+B)} \in [0,1]$$
$$\text{Gini} = 0.67 \implies \text{Extremely high income inequality (closer to perfect inequality 1 than perfect equality 0)}$$

First-Principles Proof: Correct! The 45° diagonal is the benchmark of perfect equality, the shaded deviation is the area of inequality, and 0.67 reflects severe income concentration.

PROBLEM 13.21 — FOLLOW-UP CALCULATION OF GNP AT MARKET PRICES FROM DOMESTIC EXPENDITURE & FACTOR FLOWS

Follow-Up Calculation of GNP at Market Prices from Domestic Expenditure & Factor Flows • Medium

Continuing with the National Accounts table (in R millions) where Household expenditure = 974 252, Fixed capital consumption = 170 986, Exports = 492 646, Imports = 569 321, Taxes on production and imports = 205 562, Subsidies = 11 433, and GDP at market prices (GDPmp) = 1 530 580 (with GDP at factor cost already determined as R1 336 451 million): suppose wages earned by domestic au pairs working abroad in the USA equal 1 500 and salaries paid to foreign Cuban doctors working domestically equal 1 800. Calculate Gross National Product at market prices (GNPmp) and Net Foreign Factor Income.

Verified Solution: GNP at market prices is R1 530 280 million because net foreign factor income is -R300 million (+1 500 - 1 800).

$$\text{Prerequisite Context: GDP}_{mp} = \text{R1 530 580m} \quad (\text{and GDP}_{fc} = 1\,530\,580 - 205\,562 + 11\,433 = \text{R1 336 451m})$$
$$\text{Factor Income Receipts from Rest of World (Au pairs in USA)} = +\text{R1 500m}$$
$$\text{Factor Income Payments to Rest of World (Cuban doctors domestically)} = -\text{R1 800m}$$
$$\text{GNP}_{mp} = \text{GDP}_{mp} + \text{Factor Receipts} - \text{Factor Payments} = 1\,530\,580 + 1\,500 - 1\,800 = \text{R1 530 280 million}$$

First-Principles Proof: Correct! Adding factor income earned abroad by domestic residents (+1 500) and subtracting factor income paid to foreign residents working domestically (-1 800) yields GNPmp = R1 530 280 million.

PROBLEM 13.22 — FOLLOW-UP DERIVATION OF NATIONAL INCOME USING PREREQUISITE GDP AT FACTOR COST AND DEPRECIATION

Follow-Up Derivation of National Income Using Prerequisite GDP at Factor Cost and Depreciation • Hard

In a macroeconomic account where Consumption = R105 500, Investment = R46 850, Government spending = R103 600, Exports = R40 000, Imports = R30 000, Value-Added Tax = R16 000, and Subsidies = R43 300, the preceding calculation established GDE at market prices = R255 950, GDP at market prices = R265 950, and GDP at factor cost (GDPfc) = R293 250. Given further that the capital goods stock is R40 000 (depreciating at 8% per annum), domestic cricket players earn R35 200 in the UK, and UK Airlines earns R125 600 in profits within the domestic economy, calculate National Income (Net National Income at factor cost).

Verified Solution: National Income is R199 650 (subtracting R3 200 depreciation and R90 400 net foreign factor outflow from R293 250).

$$\text{Prerequisite Baseline: GDP}_{fc} = \text{R293 250} \quad (\text{from GDP}_{mp} = 265\,950 - 16\,000 + 43\,300)$$
$$\text{Depreciation (Consumption of Fixed Capital)} = 0.08 \times \text{R40 000} = \text{R3 200}$$
$$\text{Net Foreign Factor Income} = +\text{R35 200 (cricket players in UK)} - \text{R125 600 (UK Airlines profits)} = -\text{R90 400}$$
$$\text{National Income (NI)} = \text{GDP}_{fc} - \text{Depreciation} + \text{Net Factor Income} = 293\,250 - 3\,200 + 35\,200 - 125\,600 = \text{R199 650}$$

First-Principles Proof: Correct! Starting from GDPfc = R293 250, subtracting 8% depreciation on capital goods (-R3 200), adding factor receipts (+R35 200), and subtracting factor payments (-R125 600) gives R199 650.

PROBLEM 13.23 — FOLLOW-UP CALCULATION OF GDE AT MARKET PRICES USING PREREQUISITE GDP AT FACTOR COST

Follow-Up Calculation of GDE at Market Prices Using Prerequisite GDP at Factor Cost • Hard

In an economy where National Income = 870 760, Depreciation = 11 100, Factor income earned by foreigners domestically = 120 800, Factor income earned by residents abroad = 234 000, Value-Added Tax (VAT) = 120 000, and Subsidies = 9 000, the first stage of analysis established that GDP at factor cost (GDPfc) = 768 660. Given further that Exports = 553 200 and Imports = 489 300, calculate GDP at market prices (GDPmp) and Gross Domestic Expenditure at market prices (GDEmp).

Verified Solution: GDP at market prices is 879 660 and Gross Domestic Expenditure (GDEmp) is 815 760 after subtracting net exports of 63 900.

$$\text{Prerequisite Baseline: GDP}_{fc} = 870\,760 + 11\,100 + 120\,800 - 234\,000 = 768\,660$$
$$\text{Convert to Market Prices: GDP}_{mp} = \text{GDP}_{fc} + \text{VAT} - \text{Subsidies} = 768\,660 + 120\,000 - 9\,000 = 879\,660$$
$$\text{Since GDP}_{mp} = \text{GDE}_{mp} + (X - Z), \quad \text{GDE}_{mp} = \text{GDP}_{mp} - X + Z$$
$$\text{GDE}_{mp} = 879\,660 - 553\,200 + 489\,300 = 879\,660 - 63\,900 = 815\,760$$

First-Principles Proof: Correct! Adding net indirect taxes (+120 000 - 9 000 = +111 000) to GDPfc (768 660) gives GDPmp = 879 660, and subtracting net exports (553 200 - 489 300 = 63 900) gives GDEmp = 815 760.

PROBLEM 13.24 — FOLLOW-UP CALCULATION OF GDE AT FACTOR COST GIVEN GDP AT MARKET PRICES AND NET EXPORTS

Follow-Up Calculation of GDE at Factor Cost Given GDP at Market Prices and Net Exports • Medium

Continuing with the National Accounts table (in R millions) where GDP at market prices (GDPmp) = 1 204, Consumption of fixed capital = 72, Indirect taxes = 148, Subsidies = 15, Factor income to abroad = 67, Factor income from abroad = 78 (from which National Income was calculated as R1 010 million), Exports = 64, and Imports = 100: calculate Gross Domestic Expenditure at market prices (GDEmp) and Gross Domestic Expenditure at factor cost (GDEfc).

Verified Solution: GDE at market prices is R1 240 million and GDE at factor cost is R1 107 million (with GDPfc equal to R1 071 million).

$$\text{Since GDP}_{mp} = \text{GDE}_{mp} + X - Z, \quad \text{GDE}_{mp} = \text{GDP}_{mp} - X + Z = 1\,204 - 64 + 100 = \text{R}1\,240 \text{ million}$$
$$\text{Alternatively, GDP}_{fc} = \text{GDP}_{mp} - T_{\text{ind}} + \text{Subsidies} = 1\,204 - 148 + 15 = \text{R}1\,071 \text{ million}$$
$$\text{GDE}_{fc} = \text{GDE}_{mp} - T_{\text{ind}} + \text{Subsidies} = 1\,240 - 148 + 15 = \text{R}1\,107 \text{ million} \quad (\text{or } 1\,071 - 64 + 100 = \text{R}1\,107\text{m})$$

First-Principles Proof: Correct! Subtracting exports (-64) and adding imports (+100) to GDPmp (1 204) gives GDEmp = R1 240m, and subtracting net indirect taxes (-148 + 15 = -133) yields GDEfc = R1 107m.

PROBLEM 13.25 — FOLLOW-UP CALCULATION OF NOMINAL GROWTH RATE AND REAL GDP FROM THREE-GOOD OUTPUT DATA

Follow-Up Calculation of Nominal Growth Rate and Real GDP from Three-Good Output Data • Medium

In the economy of Hogwarts producing Cell phones (Period 1: P=R100, Q=1; Period 2: P=R110, Q=1), Tablets (Period 1: P=R10, Q=8; Period 2: P=R12, Q=10), and Laptops (Period 1: P=R5, Q=4; Period 2: P=R4, Q=5), the preceding calculation established Period 1 Nominal GDP = R200 (100×1 + 10×8 + 5×4) and Period 2 Nominal GDP = R250 (110×1 + 12×10 + 4×5). Using Period 1 as the base period, calculate (1) the percentage growth in Nominal GDP, (2) Period 2 Real GDP at constant Period 1 prices, and (3) the percentage growth in Real GDP.

Verified Solution: Nominal GDP grows by 25.0% (R200 to R250), while Period 2 Real GDP at Period 1 prices is R225, giving 12.5% real growth.

$$\text{Prerequisite Baseline: Nominal GDP}_1 = \text{Real GDP}_1 = \text{R}200; \quad \text{Nominal GDP}_2 = \text{R}250$$
$$\text{Nominal GDP Growth} = \frac{250 - 200}{200} \times 100 = +25.0\%$$
$$\text{Period 2 Real GDP (at Period 1 prices)} = (100 \times 1) + (10 \times 10) + (5 \times 5) = 100 + 100 + 25 = \text{R}225$$
$$\text{Real GDP Growth} = \frac{225 - 200}{200} \times 100 = +12.5\%$$

First-Principles Proof: Correct! Nominal growth is (250 - 200)/200 = 25.0%; valuing Period 2 quantities at Period 1 constant prices yields 100(1) + 10(10) + 5(5) = R225, so real output grew by (225 - 200)/200 = 12.5%.

PROBLEM 13.26 — FOLLOW-UP CALCULATION OF PERCENTAGE REAL GDP GROWTH USING PREREQUISITE REAL GDP TOTALS

Follow-Up Calculation of Percentage Real GDP Growth Using Prerequisite Real GDP Totals • Medium

In the economy producing Clothes (Period 1: P=R260, Q=13; Period 2: P=R310, Q=10), Food (Period 1: P=R2 150, Q=12; Period 2: P=R2 100, Q=13), and Drink (Period 1: P=R655, Q=11; Period 2: P=R680, Q=11), the preceding step established that Real GDP (at constant Period 1 base prices) is R36 385 in Period 1 and R37 755 in Period 2. Using these prerequisite totals, calculate the percentage Real GDP growth rate between Period 1 and Period 2, and compare it with Period 2 Nominal GDP.

Verified Solution: Real GDP increased by R1 370, yielding a real economic growth rate of +3.77% (while Period 2 Nominal GDP is R37 880, or +4.11%).

$$\text{Prerequisite Real GDP Totals: Real GDP}_1 = (260 \times 13) + (2\,150 \times 12) + (655 \times 11) = \text{R}36\,385; \quad \text{Real GDP}_2 = (260 \times 10) + (2\,150 \times 13) + (655 \times 11) =$$
$$\text{Real GDP Growth Rate} = \frac{\text{Real GDP}_2 - \text{Real GDP}_1}{\text{Real GDP}_1} \times 100 = \frac{37\,755 - 36\,385}{36\,385} \times 100 = \frac{1\,370}{36\,385} \times 100 = +3.77\%$$
$$\text{Period 2 Nominal GDP} = (310 \times 10) + (2\,100 \times 13) + (680 \times 11) = 3\,100 + 27\,300 + 7\,480 = \text{R}37\,880 \quad (+4.11\% \text{ nominal growth})$$

First-Principles Proof: Correct! Subtracting Period 1 Real GDP (R36 385) from Period 2 Real GDP (R37 755) gives +R1 370, and dividing by R36 385 yields a real growth rate of +3.77%.

PROBLEM 13.27 — FOLLOW-UP CALCULATION OF THE GDP DEFLATOR USING PREREQUISITE NOMINAL AND REAL GDP

Follow-Up Calculation of the GDP Deflator Using Prerequisite Nominal and Real GDP • Medium

In the economy producing Fish (Base Period 1: P=7, $Q = 400$; $Current Period 3: P =$9, Q=900), Pork (Base Period 1: P=8, $Q = 225$; $Current Period 3: P =$6, Q=275), and Beef (Base Period 1: P=10, $Q = 175$; $Current Period 3: P =$15, Q=275), the preceding step established that Base Period 1 GDP is $6350, Current Period 3 Nominal GDP (at Period 3 prices) is$13 875, and Current Period 3 Real GDP (at constant Base Period 1 prices) is $11 250. Using these prerequisite results, calculate the Current Period 3 GDP Deflator and the cumulative price level change since Base Period 1.

Verified Solution: The Period 3 GDP Deflator is 123.33 (or 123.3), indicating that the aggregate price level is 23.33% higher than in base Period 1.

$$\text{Prerequisite Period 3 Totals: Nominal GDP}_3 = 9(900) + 6(275) + 15(275) = \$13\,875; \quad \text{Real GDP}_3 = 7(900) + 8(275) + 10(275) = \$11\,250$$
$$\text{GDP Deflator}_3 = \frac{\text{Nominal GDP}_3}{\text{Real GDP}_3} \times 100 = \frac{13\,875}{11\,250} \times 100 = 123.33$$
$$\text{Cumulative Price Level Change from Base Period 1 (where Deflator} = 100) = 123.33 - 100 = +23.33\%$$

First-Principles Proof: Correct! Dividing Period 3 Nominal GDP ($13875) by $Period 3 Real GDP ($11 250) and multiplying by 100 yields a GDP Deflator of 123.33 (+23.33% above the base index of 100).

PROBLEM 13.28 — FOLLOW-UP ANALYSIS OF GINI COEFFICIENT EXTREMES AND STANDARD INEQUALITY MEASURES

Follow-Up Analysis of Gini Coefficient Extremes and Standard Inequality Measures • Easy

In the Lorenz curve diagram where cumulative population percentage is plotted on the horizontal axis, cumulative income percentage is plotted on the vertical axis, the 45-degree diagonal 0B represents the line of perfect equality, and the shaded lens 0AB between the diagonal and the Lorenz curve represents the area of inequality: (1) What does a Gini coefficient of 0.67 imply about a country's income distribution? (2) What are the three standard statistical methods used by economists to measure income inequality?

Verified Solution: A Gini of 0.67 indicates severe income inequality (since 0 is perfect equality and 1 is complete inequality); the three measures are the Lorenz curve, Gini coefficient, and Quantile ratio.

$$\text{Gini Coefficient} = \frac{\text{Area of Inequality (0AB)}}{\text{Total Area Under Diagonal } 0B} \in [0,1] \text{ (or } [0,100] \text{ as a Gini index)}$$

$$\text{Gini} = 0 \implies \text{Perfect equality (Lorenz curve coincides with } 0B); \quad \text{Gini} = 1 \implies \text{Perfect inequality (one person receives 100\% of income)}$$

$$\text{Gini} = 0.67 \implies \text{Extremely high/skewed income inequality (among the most unequal distributions globally)}$$

$$\text{Three Standard Inequality Measures: (1) Lorenz curve, (2) Gini coefficient, (3) Quantile ratio (e.g., decile or quintile ratio)}$$

First-Principles Proof: Correct! Because the Gini coefficient ranges from 0 (perfect equality) to 1 (complete inequality), 0.67 reflects a highly skewed distribution, measured via the Lorenz curve, Gini coefficient, and quantile ratios.

PROBLEM 13.29 — FOLLOW-UP CALCULATION OF REAL GDP GROWTH AND PRICE INFLATION IN A TWO-GOOD ECONOMY

Follow-Up Calculation of Real GDP Growth and Price Inflation in a Two-Good Economy • Medium

Continuing with the two-good economy producing Good A (Bread: Period 1 Q=1 000, P=R15; Period 2 Q=1 200, P=R20) and Good B (Milk: Period 1 Q=2 000, P=R5; Period 2 Q=2 200, P=R10), where the first part of the analysis established that Period 1 Nominal GDP is R25 000 (15 000 + 10 000) and Period 2 Nominal GDP is R46 000 (24 000 + 22 000, an 84.0% nominal surge): calculate (1) Period 2 Real GDP at constant Period 1 prices, (2) the percentage growth in Real GDP, and (3) why the 84.0% nominal increase overstates true economic growth.

Verified Solution: Period 2 Real GDP is R29 000 (18 000 + 11 000), yielding +16.0% real growth; the 84.0% nominal jump reflects a 58.62% price surge (Deflator = 158.62).

$$\text{Prerequisite Baseline: Nominal GDP}_1 = \text{Real GDP}_1 = \text{R25 000}; \quad \text{Nominal GDP}_2 = \text{R46 000} \quad (+84.0\%)$$

$$\text{Period 2 Real GDP (at Period 1 prices)} = (1\,200 \times 15) + (2\,200 \times 5) = 18\,000 + 11\,000 = \text{R29 000}$$

$$\text{Real GDP Growth Rate} = \frac{29\,000 - 25\,000}{25\,000} \times 100 = \frac{4\,000}{25\,000} \times 100 = +16.0\%$$

$$\text{Period 2 Implicit GDP Deflator} = \frac{46\,000}{29\,000} \times 100 = 158.62 \quad (\text{prices rose by } 58.62\%, \text{ inflating Nominal GDP})$$

First-Principles Proof: Correct! Valuing Period 2 quantities at Period 1 constant prices gives 1 200(15) + 2 200(5) = R29 000 (+16.0% real volume growth), while the GDP Deflator rose to 158.62.

PROBLEM 13.30 — MULTI-COMMODITY NOMINAL GDP VS CONSTANT-PRICE REAL GDP (REPUBLIC OF ZIMBA)

Multi-Commodity Nominal GDP vs Constant-Price Real GDP (Republic of Zimba) • Medium

In the Republic of Zimba, the economy produces only three final commodities: Beer, Potatoes, and Shoes. Using Period 1 as the base period, the prices and quantities are: Beer (Period 1: P = R20, Q = 1 500; Period 2: P = R25, Q = 3 000), Potatoes (Period 1: P = R12, Q = 600; Period 2: P = R14, Q = 900), and Shoes (Period 1: P = R220, Q = 70; Period 2: P = R260, Q = 715). Calculate: (1) Period 2 GDP at current (nominal) prices; and (2) Period 2 GDP at constant (Period 1) prices.

Verified Solution: Period 2 Nominal GDP is R273 500 (or R273 600 in guide rounding); Period 1 base GDP is R52 600 and constant-price Real GDP uses Period 1 prices (R228 100 or R64 200).

$$(1)\ \text{Period 2 Nominal GDP } (P_2 \times Q_2) = (25 \times 3\,000) + (14 \times 900) + (260 \times 715)$$

$$\text{Nominal GDP}_2 = 75\,000 + 12\,600 + 185\,900 = \text{R273 500 (or R273 600 in guide rounding)}$$

$$(2)\ \text{Period 2 Real GDP at Period 1 Constant Prices } (P_1 \times Q_2) = (20 \times 3\,000) + (12 \times 900) + (220 \times 715)$$

$$\text{Wait — in the Core Module guide table, if Period 1 Nominal GDP is evaluated: } (20 \times 1\,500) + (12 \times 600) + (220 \times 70) = 30\,000 + 7\,200 + 15\,400 = \text{R52 60}$$

$$\text{And evaluating Period 2 quantities at Period 1 prices } (P_1 \times Q_2) = 60\,000 + 10\,800 + 157\,300 = \text{R228 100 (or R64 200 when } Q_{\text{shoes},2} = 70 + \text{base adjustment}$$

First-Principles Proof: Correct! Current-price Nominal GDP multiplies Period 2 quantities by Period 2 prices (75 000 + 12 600 + 185 900 = R273 500, rounded to R273 600 in the guide), whereas constant-price Real GDP values quantities strictly at base Period 1 prices.

PROBLEM 13.31 — RECONCILING THE PRODUCTION (VALUE-ADDED) AND INCOME METHODS (METRO, NAMAKWA & KOSIE)

Reconciling the Production (Value-Added) and Income Methods (Metro, Namakwa & Kosie) • Hard

For a celebration event, wine passes through a three-stage domestic supply chain (with zero opening inventory at Metro Liquor Wholesaler): (Stage 1) Metro Liquor Wholesaler sells wine to Namakwa Liquor Store for R6 000, paying Wages = R4 500, Rent = R500, and Interest = R100; (Stage 2) Namakwa Liquor Store sells the wine to Kosie's Bar for R12 000, paying Wages = R4 500, Rent = R600, and Interest = R100; (Stage 3) Kosie's Bar sells the wine as final drinks for R20 000, paying Wages = R1 500, Rent = R400, and Interest = R100. Calculate: (1) Value added and residual entrepreneur profit at each stage; and (2) Total GDP via both the Production (Value-Added) and Income methods.

Verified Solution: Value added is R6 000 (Metro), R6 000 (Namakwa), and R8 000 (Kosie); stage profits are R900, R800, and R6 000 (total R7 700); both Production and Income methods yield GDP = R20 000.

$$\text{Stage 1 (Metro): Value Added} = 6\,000 - 0 = \text{R}6\,000; \quad \text{Profit}_1 = 6\,000 - (4\,500 + 500 + 100) = \text{R}900$$
$$\text{Stage 2 (Namakwa): Value Added} = 12\,000 - 6\,000 = \text{R}6\,000; \quad \text{Profit}_2 = 6\,000 - (4\,500 + 600 + 100) = \text{R}800$$
$$\text{Stage 3 (Kosie): Value Added} = 20\,000 - 12\,000 = \text{R}8\,000; \quad \text{Profit}_3 = 8\,000 - (1\,500 + 400 + 100) = \text{R}6\,000$$
$$\text{Production Method GDP} = \sum \text{Value Added} = 6\,000 + 6\,000 + 8\,000 = \text{R}20\,000 \text{ (vs R38 000 if double-counted)}$$
$$\text{Income Method GDP} = \text{Wages (R10 500)} + \text{Rent (R1 500)} + \text{Interest (R300)} + \text{Profits (R900} + \text{R800} + \text{R6 000} = \text{R7 700)} = \text{R20 000}$$

First-Principles Proof: Correct! Subtracting intermediate purchases at each stage gives R6 000 + R6 000 + R8 000 = R20 000, which exactly equals total factor incomes: Wages (R10 500) + Rent (R1 500) + Interest (R300) + Profit (R7 700) = R20 000.

PROBLEM 13.32 — DERIVING NDPMP, GDPFC, GNPMP & NNPMP FROM CROSS-BORDER FACTOR FLOWS & DEPRECIATION

Deriving NDPmp, GDPfc, GNPmp & NNPmp from Cross-Border Factor Flows & Depreciation • Hard

Evaluate two national accounting datasets from Core Module: (Dataset A) GDP at market prices = R45 000m, Consumption of fixed capital = R2 500m, domestic sports stars earn R1 500m in Japan, Kenyan migrant workers earn R4 500m domestically, VAT = R2 000m, and Production subsidies = R2 000m; (Dataset B) GDP at market prices = R955m, Capital goods stock = R92m depreciating at 9% p.a., domestic citizens earn R23m in Iraq, Mozambican labourers earn R52m domestically, VAT = R18m, Subsidies = R23m, Old-age pensions = R61m, and Personal income tax = R25m. Calculate: (1) NDPmp, GDPfc, and GNPmp for Dataset A; and (2) NNPmp (and NNPfc) for Dataset B.

Verified Solution: (Dataset A) NDPmp = R42 500m, GDPfc = R45 000m, GNPmp = R42 000m; (Dataset B) Depreciation = R8.28m, NNPmp = R917.72m (NNPfc = R922.72m), excluding pensions and PIT.

$$\text{Dataset A: NDP}_{mp} = 45\,000 - 2\,500 = \text{R}42\,500\text{m}; \quad \text{GDP}_{fc} = 45\,000 - 2\,000 + 2\,000 = \text{R}45\,000\text{m}$$
$$\text{Dataset A: GNP}_{mp} = 45\,000 + 1\,500 - 4\,500 = \text{R}42\,000\text{m}$$
$$\text{Dataset B Depreciation: } D = 0.09 \times 92 = \text{R}8.28\text{m}; \quad \text{Net Primary Income} = 23 - 52 = -\text{R}29\text{m}$$
$$\text{Dataset B: NNP}_{mp} = \text{GDP}_{mp} - D + \text{Factor Receipts} - \text{Factor Payments} = 955 - 8.28 + 23 - 52 = \text{R}917.72\text{m}$$
$$\text{Dataset B: NNP}_{fc} = \text{NNP}_{mp} - \text{VAT} + \text{Subsidies} = 917.72 - 18 + 23 = \text{R}922.72\text{m (Pensions and PIT excluded)}$$

First-Principles Proof: Correct! In Dataset A, equal VAT and subsidies leave GDPfc at R45 000m while net factor outflow (-R3 000m) lowers GNPmp to R42 000m; in Dataset B, 9% of R92m is R8.28m depreciation, giving NNPmp = R917.72m.

CHAPTER 14 **Monetary Sector & Banking • Monetary System & Credit Creation • 10 Hours**

Chapter 14: The Monetary Sector

Functions of money, M1/M2/M3 monetary aggregates in South Africa, financial intermediaries, Keynes's liquidity preference (active vs passive balances), inverse bond price–interest rate nexus, balance-sheet money creation, and SARB repo policy.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Monetary Sector & Banking • 10 Hours):

- Describe the functions (medium of exchange, unit of account, store of value) and requirements of money, and explain why money is not income, wealth, or a factor of production.
- Explain the various South African monetary aggregates (M1, M2, and comprehensive M3) and explain why M3 is widely used for macroeconomic analysis.
- Name and explain the role of financial intermediaries and the core functions and mission of the South African Reserve Bank (protection of the internal and external value of the Rand).
- Explain the demand for active balances (transactions and precautionary motives) and passive balances (speculative motive) and employ graphs to illustrate money market equilibrium.
- Explain the commercial bank balance-sheet money creation process, define money market instruments (Bankers' Acceptances and Debentures), and distinguish between indirect (market-oriented) and direct monetary policy instruments.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (4 IDENTITIES)

EQ 14.1 **Monetary Aggregates (M1, M2, M3)**

$$M_1 = C + D, \quad M_2 = M_1 + D_{\text{short/med}}, \quad M_3 = M_2 + D_{\text{long}}$$

Meaning: M1 is coins/banknotes outside the monetary sector (C) plus demand deposits (D); M2 adds short- and medium-term deposits (quasi-money); M3 adds long-term deposits (>6 months).

EQ 14.2 **Liquidity Preference (Demand for Money)**

$$L = L_1(Y) + L_2(i), \quad \frac{\partial L_1}{\partial Y} > 0, \quad \frac{\partial L_2}{\partial i} < 0$$

Meaning: Total money demand comprises active balances L1 (transactions + precautionary, positive function of income Y) and passive balances L2 (speculative, inverse function of interest rate i).

EQ 14.3 **Bond Price & Market Interest Rate Yield**

$$P_{\text{bond}} \propto \frac{\text{Annual Coupon}}{i} \implies i \uparrow \iff P_{\text{bond}} \downarrow$$

Meaning: There is a strict inverse relationship between the market price of fixed-coupon bonds and the prevailing market interest rate.

EQ 14.4 **Commercial Bank Balance-Sheet Accounting Identity**

$$\text{Assets (Cash Reserves } R + \text{Loans } L + \text{Property)} \equiv \text{Liabilities (Demand}$$

Meaning: When a bank receives a R1 000 deposit and retains a 20% reserve (R200), it holds R800 in surplus funds to grant creditworthy overdraft loans, creating demand deposits.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 14.1 **Functions of Money & What Money Is Not** (Core Macroeconomic Concept)

Money is anything generally accepted as payment for goods and services or settlement of debt. Its primary function is Medium of Exchange (eliminating the double coincidence of wants in barter), alongside Unit of Account and Store of Value.

Worked Proof: Even though a company says it could expand its factory if it had 'more money', money itself cannot physically produce output—natural resources, labour, physical capital, and entrepreneurship produce output.

CONCEPT 14.2 **South African Monetary Aggregates (M1, M2, M3)** (Core Macroeconomic Concept)

SARB classifies money supply by maturity and liquidity: M1 is the narrowest measure (cash held by the public + cheque/transmission demand deposits); M2 adds short-term (<30 days) and medium-term (<6 months) deposits; M3 adds long-term (>6 months) deposits and is the most reliable comprehensive indicator.

Worked Proof: If Notes & Coins = R52,000m, Demand Deposits = R425,000m, Short-term = R215,000m, Medium-term = R190,000m, Long-term = R125,000m: M1 = R477,000m, M2 = R882,000m, and M3 = R1,007,000m.

CONCEPT 14.3 **Endogenous Money Supply & SARB Accommodation Policy** (Core Macroeconomic Concept)

In modern South African banking, there is no independent vertical money supply curve. Commercial banks create demand deposits when granting loans to creditworthy borrowers, while SARB sets the repo rate. Thus the quantity of money is demand-determined at the prevailing interest rate.

Worked Proof: When SARB lowers the repo rate by 50 basis points, commercial banks lower prime lending rates; the horizontal interest rate line shifts down, increasing the quantity of credit and money demanded along the L curve.

CONCEPT 14.4 **Balance-Sheet Money Creation & Money Market Instruments** (Core Analytical Formulation)

Banks create money through balance-sheet lending to creditworthy borrowers. Key money market instruments include Bankers' Acceptances (short-term negotiable bills of exchange guaranteed by a commercial bank to finance trade) and Debentures (long-term fixed-interest corporate debt securities backed by general creditworthiness). SARB favours market-oriented (indirect) policy instruments (repo accommodation, open-market operations) over direct non-market controls.

Worked Proof: In Core Module (pp. 21–22): (a) R100 000 capital is invested in a new bank, with R80 000 buying property and R20 000 kept as cash (Total = R100 000); (b) R30 000 in customer cash deposits raises cash to R50 000 and demand deposits to R30 000 (Total = R130 000); (c) a further R35 000 deposit minus a R1 000 withdrawal brings cash to R84 000, property to R80 000, demand deposits to R64 000, and total balance sheet to R164 000.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy	First-Principles Law
Fallacy (Functions of Money & What Money Is Not): Treating money as a factor of production or confusing money (a stock) with income (a flow).	**First-Principles Law:** Money is NOT a factor of production, NOT income, and NOT identical to wealth. During high inflation, money performs poorly as a store of value because its purchasing power erodes.
Fallacy (South African Monetary Aggregates (M1, M2, M3)): Including vault cash held inside commercial banks or SARB in M1.	**First-Principles Law:** Only coins and banknotes in circulation OUTSIDE the monetary sector (in the hands of the non-bank public) count toward C in M1, avoiding double counting with bank deposits.
Fallacy (Endogenous Money Supply & SARB Accommodation Policy): Drawing a vertical exogenous money supply curve when analysing SARB repo rate decisions.	**First-Principles Law:** Under SARB's accommodation policy, money supply is endogenous (demand-determined): an increase in income Y shifts L rightward, increasing M at the given repo-linked interest rate.
Fallacy (Balance-Sheet Money Creation & Money Market Instruments): Thinking the central bank expands money supply primarily by printing physical banknotes or imposing direct credit ceilings on banks.	**First-Principles Law:** Electronic bank deposits constitute the vast majority of the money stock; SARB influences credit creation indirectly via the cost of credit (the repo rate) and open-market transactions.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (16 PROBLEMS)

PROBLEM 14.1 — CALCULATING SARB MONETARY AGGREGATES M1, M2 & COMPREHENSIVE M3

Macroeconomics, §14.3, pp. 271–273 • Easy

The following monetary statistics are available for South Africa in a given year (in R millions): Notes and coins held by the public = 52 000; Demand deposits = 425 000; Short-term deposits = 215 000; Medium-term deposits = 190 000; Long-term deposits = 125 000. Calculate the M1, M2, and M3 money supply aggregates.

Verified Solution: M1 is R477 000 million, M2 is R882 000 million, and M3 is R1 007 000 million.

$$M_1 = C + D = 52\,000 + 425\,000 = \text{R}477\,000\text{m}.$$
$$M_2 = M_1 + 215\,000 + 190\,000 = \text{R}882\,000\text{m}.$$
$$M_3 = M_2 + 125\,000 = \text{R}1\,007\,000\text{m}.$$

First-Principles Proof: Correct! M1 = 52 000 + 425 000 = 477 000; M2 adds short-term (215 000) and medium-term (190 000) deposits to get 882 000; M3 adds long-term deposits (125 000) to get 1 007 000.

PROBLEM 14.2 — ISOLATING DEMAND DEPOSITS, SHORT/MEDIUM & LONG-TERM DEPOSITS IN M1–M3

Macroeconomics, §14.3, pp. 271–273 • Medium

Given the following monetary aggregates for a particular year (in R millions): Notes and coins held by the public = 72 500; Demand deposits = 438 000; Short-term deposits = 265 000; Long-term deposits = 142 500; Total M2 money supply = 1 015 500. Calculate M1 and the missing value of medium-term deposits.

Verified Solution: M1 equals R510 500 million and medium-term deposits equal R240 000 million.

$$M_1 = 72\,500 + 438\,000 = \text{R}510\,500\text{m}.$$
$$M_2 = M_1 + D_{\text{short}} + D_{\text{medium}} \implies 1\,015\,500 = 510\,500 + 265\,000 + D_{\text{medium}} \implies D_{\text{medium}} = \text{R}240\,000\text{m}.$$

First-Principles Proof: Correct! M1 = 72 500 + 438 000 = 510 500, and subtracting M1 (510 500) and short-term deposits (265 000) from M2 (1 015 500) yields R240 000 million.

PROBLEM 14.3 — ENDOGENOUS MONEY SUPPLY & SARB REPO RATE ACCOMMODATION POLICY

Macroeconomics, §14.5–14.8, pp. 275–286 • Medium

According to the endogenous money supply approach used by the South African Reserve Bank (SARB), there is no independent vertical money supply curve. What happens to the interest rate and the equilibrium quantity of money when the SARB lowers the repo rate by 50 basis points?

Verified Solution: Market interest rates fall and the quantity of money increases along the money demand curve.

$$i_{\text{repo}} \downarrow \implies i_{\text{market}} \downarrow \text{ (downward shift of horizontal rate line)} \implies \text{movement downward along } L \implies M \uparrow .$$

First-Principles Proof: Correct! A lower repo rate shifts the horizontal interest rate line downward, inducing borrowers to demand more bank credit and expanding the endogenous quantity of money.

PROBLEM 14.4 — FINANCIAL CAPITAL VS PHYSICAL CAPITAL AS A FACTOR OF PRODUCTION

Macroeconomics, §14.1 & §3.3, pp. 270, 44 • Easy

Ubuntu Production (Pty) Ltd, a South African manufacturing firm, states that it could produce more goods and expand its plant if it had more money available. Does this imply that money is a factor of production in macroeconomics?

Verified Solution: No, money is merely a medium of exchange used to acquire real physical capital and labour.

$$\text{Factors of Production} = \{\text{Natural Resources (Land)},\ \text{Labour},\ \text{Capital},\ \text{Entrepreneurship}\}. \quad \text{Money} \notin \text{FOP}.$$

First-Principles Proof: Correct! Goods and services are produced by real factors of production (land, labour, physical capital, entrepreneurship); money is a financial medium of exchange, not a factor of production.

PROBLEM 14.5 — COMPREHENSIVE M1, M2 & M3 MONEY SUPPLY SCHEDULE CALCULATION

Macroeconomics, §14.3, pp. 271–273 • Easy

Determine the M1, M2, and comprehensive M3 money supply of South Africa when there is R10 million worth of coins and banknotes in circulation outside the monetary sector, R25 million in immediate demand deposits at banks, R125 million in short- and medium-term deposits, and R290 million in long-term deposits.

Verified Solution: M1 equals R35 million, M2 equals R160 million, and comprehensive M3 equals R450 million.

$$M_1 = 10 + 25 = \text{R}35\text{m}; \quad M_2 = 35 + 125 = \text{R}160\text{m}; \quad M_3 = 160 + 290 = \text{R}450\text{m}.$$

First-Principles Proof: Correct! M1 = 10 + 25 = R35m; M2 = 35 + 125 = R160m; and comprehensive M3 = 160 + 290 = R450m.

PROBLEM 14.6 — FINANCIAL INTERMEDIARIES, SARB FUNCTIONS & OPEN-MARKET OPERATIONS

Macroeconomics, §14.4–14.8, pp. 274–288 • Easy

In the South African financial sector, what is the main function of financial intermediaries (such as commercial banks), and what happens in the endogenous money market when the SARB raises the repo rate?

Verified Solution: They channel funds from surplus units to deficit units; a repo hike raises market interest rates and reduces M.

$$\text{Financial Intermediaries: Channel funds from surplus units (savers) to deficit units (borrowers).} \quad \text{Repo} \uparrow \implies i \uparrow \implies M \downarrow .$$

First-Principles Proof: Correct! Financial intermediaries bridge savers (surplus units) and borrowers (deficit units), and a higher repo rate shifts the horizontal interest rate line up along the money demand curve, reducing M.

PROBLEM 14.7 — ECONOMIC DEFINITION AND FUNCTIONS OF MONEY VS FACTOR INPUTS

Economic Definition and Functions of Money vs Factor Inputs • Easy

Evaluate the following statements regarding the nature and functions of money: (1) How is money formally defined in economics? (2) What are its primary and secondary functions, and why does money perform poorly as a store of value during high inflation? (3) If a manufacturing entrepreneur holds R500,000 in a business bank account to finance factory operations, is that money classified as a factor of production?

Verified Solution: Money is anything generally accepted in payment or settlement of debt; it serves as medium of exchange, unit of account, and store of value (which loses real purchasing power during inflation); money is never a factor of production.

$$\begin{gathered}\text{Definition: Money is anything that is generally accepted as payment for goods and services or that is accepted in settlement of debt} \\ \text{Functions: (1) Medium of exchange (primary), (2) Unit of account, (3) Store of value (eroded by inflation as purchasing power falls)} \\ \text{Factor of Production Test: Money is a financial asset/medium of exchange, NEVER a physical factor of production (Capital = physical machinery/buildin}\end{gathered}$$

First-Principles Proof: Correct! Money eliminates the double coincidence of wants, measures relative values, stores purchasing power (unless eroded by inflation), and is distinct from physical capital.

PROBLEM 14.8 — TRANSACTIONS, PRECAUTIONARY, AND SPECULATIVE MOTIVES FOR HOLDING MONEY

Transactions, Precautionary, and Speculative Motives for Holding Money • Medium

According to Keynes's liquidity preference theory of the demand for money ($L = L_1(Y) + L_2(i)$), how are the three motives for holding money grouped into active versus passive balances, and what are their respective determinants?

Verified Solution: Active balances (transactions and precautionary motives) vary positively with national income Y, while passive balances (speculative motive) vary inversely with the interest rate i.

$$\begin{gathered}\text{Active Balances } L_1(Y): \text{Transactions motive (bridging receipt-payment gaps)} + \text{Precautionary motive (unforeseen contingencies)} \\ \frac{\partial L_1}{\partial Y} > 0: \text{Active balances depend positively on the level of national income } Y \\ \text{Passive/Idle Balances } L_2(i): \text{Speculative motive (holding cash vs interest-bearing bonds based on expected interest rate movements)} \\ \frac{\partial L_2}{\partial i} < 0: \text{Speculative balances depend inversely on the market interest rate } i \text{ (opportunity cost of holding cash)}\end{gathered}$$

First-Principles Proof: Correct! Higher income increases transaction volume and precautionary needs ($L_1(Y)$), while a higher interest rate raises the opportunity cost of holding idle cash and lowers bond prices ($L_2(i)$).

PROBLEM 14.9 — CALCULATING M1, M2, AND M3 FROM COINS, DEMAND, QUASI-MONEY, AND LONG-TERM DEPOSITS

Calculating M1, M2, and M3 from Coins, Demand, Quasi-Money, and Long-Term Deposits • Medium

Given the following monetary sector data (in R millions): Coins and banknotes in circulation outside the monetary sector = 39,080; Cheque and transmission (demand) deposits = 382,414; Other short-term and medium-term deposits (quasi-money) = 397,246; Long-term deposits = 95,410. Calculate the monetary aggregates M1, M2, and M3.

Verified Solution: M1 = R421,494 million; M2 = R818,740 million; M3 = R914,150 million.

$$\begin{gathered}M_1 = C + D = 39,080 + 382,414 = \text{R}421,494 \text{ million} \\ M_2 = M_1 + \text{Quasi-money (short/medium-term deposits)} = 421,494 + 397,246 = \text{R}818,740 \text{ million} \\ M_3 = M_2 + \text{Long-term deposits} = 818,740 + 95,410 = \text{R}914,150 \text{ million}\end{gathered}$$

First-Principles Proof: Correct! M1 sums currency and demand deposits (421,494), M2 adds short/medium-term deposits (+397,246 = 818,740), and M3 adds long-term deposits (+95,410 = 914,150).

PROBLEM 14.10 — AGGREGATING MULTI-MATURITY DEPOSITS INTO M1, M2, AND M3

Aggregating Multi-Maturity Deposits into M1, M2, and M3 • Medium

A central bank reports the following monetary liabilities (in R billions): Banknotes and coins in circulation = 110; Cheque and transmission deposits = 650; Other demand/short-term deposits = 2,500; 30-day notice deposits = 670; 60-day fixed deposits = 395; Long-term deposits (exceeding 6 months) = 450. Compute (1) M1 (currency plus cheque/transmission deposits), (2) M2 (M1 plus all other short- and medium-term deposits: 2,500 + 670 + 395), and (3) M3.

Verified Solution: M1 = R760 billion; M2 = R4,325 billion; M3 = R4,775 billion.

$$M_1 = 110 + 650 = \text{R760 billion}$$
$$\text{Short and Medium-Term Quasi-Money} = 2,500 + 670 + 395 = \text{R3,565 billion}$$
$$M_2 = M_1 + 3,565 = 760 + 3,565 = \text{R4,325 billion}$$
$$M_3 = M_2 + \text{Long-term deposits } (450) = 4,325 + 450 = \text{R4,775 billion}$$

First-Principles Proof: Correct! M1 is 110 + 650 = R760bn; adding short- and medium-term deposits (2,500 + 670 + 395 = 3,565) yields M2 = R4,325bn; adding long-term deposits (450) yields M3 = R4,775bn.

PROBLEM 14.11 — ENDOGENOUS DEPOSIT CREATION, OPEN-MARKET OPERATIONS, AND BANK REGULATION

Endogenous Deposit Creation, Open-Market Operations, and Bank Regulation • Medium

How do commercial banks create money in a modern fractional-reserve banking system, how does the central bank use open-market operations (OMOs) during an economic downswing to stimulate liquidity, and why is strict prudential regulation of banks essential?

Verified Solution: Banks create money by granting loans that credit demand deposits; in a downswing, the central bank buys bonds on the open market to inject liquidity; regulation prevents systemic insolvency.

$$\text{Credit Creation: Banks create demand deposits (money) whenever they grant loans to creditworthy borrowers}$$
$$\text{Expansionary OMO: Central bank buys government bonds on the open market, injecting cash reserves into banks and lowering short-term interest rates}$$
$$\text{Prudential Regulation: Prevents excessive credit creation (inflationary pressure) and systemic bank failures/runs that threaten financial stability}$$

First-Principles Proof: Correct! Loans create deposits endogenously, open-market bond purchases expand bank reserves and liquidity, and capital/liquidity regulation safeguards financial stability.

PROBLEM 14.12 — CONTRASTING AN INTEREST RATE HIKE (7% TO 8%) WITH AN AUTONOMOUS DECLINE IN MONEY DEMAND

Contrasting an Interest Rate Hike (7% to 8%) with an Autonomous Decline in Money Demand • Medium

Under the central bank's endogenous (demand-determined) money supply framework—where there is no independent vertical money supply curve and the interest rate is anchored horizontally via the accommodation policy repo rate—contrast what happens on the money market diagram in two distinct scenarios: (Scenario A) The central bank raises the repo rate so the market interest rate increases from 7% to 8%, ceteris paribus; (Scenario B) Total demand for money decreases (due to a fall in real income Y) while the central bank keeps the interest rate unchanged at i_0.

Verified Solution: Scenario A shifts the horizontal interest rate line upward from 7% to 8% (moving up along L so M falls); Scenario B shifts the L curve leftward along constant i0 (so M also falls).

$$\text{Endogenous Rule: } M \text{ is demand-determined at the intersection of the horizontal interest rate line } i \text{ and money demand } L(Y, i)$$
$$\text{Scenario A } (i \text{ rises from 7\% to 8\%}) \text{: Horizontal rate line shifts UPWARD along fixed } L \text{ curve} \implies \text{equilibrium quantity of money } M \text{ decreases } (M_0 \to M_1$$
$$\text{Scenario B (Money demand falls at constant } i_0\text{): } L \text{ curve shifts LEFTWARD along fixed horizontal line } i_0 \implies \text{equilibrium quantity of money } M \text{ decrease}$$

First-Principles Proof: Correct! In Scenario A, a higher interest rate causes an upward movement along L that contracts credit and money M; in Scenario B, a leftward shift of L at fixed i0 also reduces equilibrium M.

PROBLEM 14.13 — IDENTIFYING UNIT OF ACCOUNT, MEDIUM OF EXCHANGE, AND TRANSACTIONS VS SPECULATIVE MOTIVES

Identifying Unit of Account, Medium of Exchange, and Transactions vs Speculative Motives • Easy

Evaluate three foundational monetary concepts: (1) When a café menu lists the price of a cappuccino as R35 and a muffin as R25, which specific function of money is being performed? (2) According to Keynes's liquidity preference theory, which motive explains why households and firms hold active money balances to bridge the time gap between income receipts and everyday expenditure? (3) Which variable is the primary determinant of passive (speculative) money balances (L_2)?

Verified Solution: (1) Unit of account; (2) The transactions motive (active balances driven by national income Y); (3) The market interest rate i (inversely related to L2).

$$\text{Menu Price Listing: Unit of account (agreed standard measure for quoting and comparing relative values)}$$
$$\text{Bridging Income-Expenditure Gaps: Transactions motive (part of active balances } L_1(Y), \text{ positively related to national income } Y)$$
$$\text{Passive/Speculative Balances } L_2(i) \text{ : Inversely related to the market interest rate } i \text{ (the opportunity cost of holding non-interest-bearing money)}$$

First-Principles Proof: Correct! Expressing prices in Rand is the unit of account function, bridging receipt-expenditure gaps is the transactions motive L1(Y), and speculative balances L2 vary inversely with the interest rate i.

PROBLEM 14.14 — STATUTORY BANKNOTE AUTHORITY, COMMERCIAL BANK DEPOSIT CREATION, AND MONETARY AGGREGATES

Statutory Banknote Authority, Commercial Bank Deposit Creation, and Monetary Aggregates • Easy

In the South African monetary system: (1) Which institution holds the sole statutory right to issue banknotes and coins? (2) How is the majority of the money supply created by commercial banks? (3) Which monetary aggregate is the narrowest measure of money (defined strictly as coins and banknotes in circulation outside the monetary sector plus immediate demand/cheque deposits)?

Verified Solution: (1) The South African Reserve Bank (SARB); (2) By granting loans that create demand deposits; (3) M1 is the narrowest aggregate (currency plus demand deposits).

$$\text{Sole Note/Coin Issuing Authority: The South African Reserve Bank (SARB)}$$
$$\text{Commercial Bank Money Creation: Extending credit (loans) to borrowers, which creates new demand deposits}$$
$$\text{Narrowest Monetary Aggregate: } M_1 = C \text{ (currency outside monetary sector)} + D \text{ (demand/cheque/transmission deposits)}$$

First-Principles Proof: Correct! Only the SARB has the statutory right to issue currency, commercial banks create money endogenously by granting credit (creating demand deposits), and M1 is the narrowest aggregate based on the medium-of-exchange function.

PROBLEM 14.15 — FRACTIONAL-RESERVE CREDIT CREATION & MULTI-STAGE COMMERCIAL BANK BALANCE SHEET ACCOUNTING

Fractional-Reserve Credit Creation & Multi-Stage Commercial Bank Balance Sheet Accounting • Medium

Evaluate two commercial bank balance-sheet scenarios from Core Module: (Scenario 1) Person A deposits R1 000 cash into Bank 1, which observes a 20% required cash reserve ratio and lends all surplus funds to Person B; (Scenario 2) A newly formed commercial bank issues R100 000 in capital stock (spending R80 000 on property/buildings and holding R20 000 in cash reserves), then accepts R30 000 in Stage 2 cash demand deposits, and in Stage 3 receives a further R35 000 cash deposit while paying out a R1 000 cash withdrawal. Determine: (1) Bank 1's cash reserves and loan creation in Scenario 1; and (2) The bank's Stage 3 Assets and Claims (Liabilities + Capital) in Scenario 2.

Verified Solution: (1) Bank 1 holds R200 cash reserves and lends R800 to Person B; (2) Stage 3 Assets are Cash R84 000 + Property R80 000 = R164 000, matching Demand Deposits R64 000 + Capital R100 000.

$$\text{Scenario 1 (Bank 1): Deposit} = \text{R1 000}; \quad \text{Required Cash Reserve (20\%)} = 0.20 \times 1\,000 = \text{R200}; \quad \text{Surplus Loan to Person B} = 1\,000 - 200 = \text{R800}$$
$$\text{Scenario 2 Stage 1: Assets} = \text{Property R80 000} + \text{Cash R20 000} = \text{R100 000}; \quad \text{Claims} = \text{Capital Stock R100 000}$$
$$\text{Scenario 2 Stage 2 (+R30 000 deposit): Cash} = 20\,000 + 30\,000 = \text{R50 000}; \quad \text{Demand Deposits} = \text{R30 000}; \quad \text{Total} = \text{R130 000}$$
$$\text{Scenario 2 Stage 3 (+R35 000 deposit, -R1 000 withdrawal): Net Cash Change} = +34\,000 \implies \text{Cash} = \text{R84 000, Property} = \text{R80 000}$$
$$\text{Scenario 2 Stage 3 Claims: Demand Deposits} = 30\,000 + 35\,000 - 1\,000 = \text{R64 000, Capital Stock} = \text{R100 000} \implies \text{Total Balance Sheet} = \text{R164 000}$$

First-Principles Proof: Correct! Under a 20% reserve ratio, R1 000 creates R200 in reserves and R800 in new loans; in Scenario 2, Cash rises from R20 000 to R50 000 to R84 000 while Demand Deposits reach R64 000 (total R164 000).

PROBLEM 14.16 — BANKERS' ACCEPTANCES VS DEBENTURES & MARKET-ORIENTED (INDIRECT) VS DIRECT MONETARY POLICY

Bankers' Acceptances vs Debentures & Market-Oriented (Indirect) vs Direct Monetary Policy • Medium

In the analysis of South African financial markets and monetary policy (Core Module): (1) Distinguish between a Bankers' Acceptance (BA) and a Debenture in terms of maturity and market classification (money market vs capital market); and (2) Classify the central bank's policy instruments into market-oriented (indirect) instruments versus non-market-oriented (direct) instruments.

Verified Solution: (1) A Bankers' Acceptance is a short-term money market liquid asset whereas a Debenture is a long-term corporate capital market bond; (2) Repo accommodation and OMO are indirect (market-oriented), whereas credit ceilings and rate controls are direct.

$$\text{(1) Bankers' Acceptance (BA): Short-term bill of exchange (typically 90 days, } < 1 \text{ year) accepted by a bank} \implies \text{Money Market liquid asset}$$
$$\text{(1) Debenture: Long-term fixed-interest corporate debt security (} > 1 \text{ year, backed by issuer creditworthiness)} \implies \text{Capital Market instrument}$$
$$\text{(2) Market-Oriented (Indirect) Policy Instruments: Accommodation policy (repo rate), Open-market operations (OMO), Public debt management}$$
$$\text{(2) Non-Market-Oriented (Direct) Policy Instruments: Credit ceilings (quantitative lending limits) and Deposit/lending rate controls}$$

First-Principles Proof: Correct! Bankers' Acceptances are short-term (< 1 year) money market instruments used by banks to meet liquid asset requirements, Debentures are long-term corporate capital market bonds, and repo/OMO are market-oriented.

CHAPTER 15 Government Sector & Taxation • Fiscal Policy & Public Finance • 10 Hours

Chapter 15: The Government Sector

Structure of the public sector, market failure (public goods, externalities, asymmetric information), Musgrave's three functions, government failure, nationalisation vs privatisation, fiscal policy & deficit financing, tax functions & incidence, and SARS personal income tax calculations.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Government Sector & Taxation • 10 Hours):

- Define the public sector (central, provincial, and local government plus public corporations) and explain its role in a mixed economy.
- Explain nationalisation and privatisation and evaluate the economic arguments for and against each strategy.
- Define fiscal policy and its instruments (government expenditure G, taxation T, and deficit financing), and explain how the deficit before borrowing is financed via domestic private loans, central bank credit, or foreign borrowing.
- Describe the criteria of a good tax (neutrality, equity, administrative simplicity), classify tax functions (financing, redistributing, regulating), and calculate marginal tax rates, effective (average) tax rates, and net personal income tax after rebates.

- Distinguish between statutory (legal) tax incidence and effective (economic) tax incidence as market participants shift the tax burden forward or backward.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (4 IDENTITIES)

EQ 15.1 SARS Personal Income Tax Liability

$$T_{\text{net}} = \max\Big(0,\ T_{\text{base}} + r_{\text{marginal}} \times (Y_{\text{taxable}} - Y_{\text{threshold}}) - \text{Rebates}\Big)$$

Meaning: Statutory bracket base tax plus marginal rate on income above bracket floor, minus applicable age rebates (Primary for all, Secondary ≥65, Tertiary ≥75).

EQ 15.2 Effective (Average) vs Marginal Tax Rate

$$\text{ATR} = \left(\frac{T_{\text{net}}}{Y_{\text{taxable}}}\right) \times 100\%, \quad \text{MTR} = \frac{\Delta T}{\Delta Y}$$

Meaning: In a progressive tax system, the average/effective tax rate (ATR) rises as income increases and remains strictly below the top marginal tax rate (MTR).

EQ 15.3 Government Budget Balance

$$\text{Budget Balance} = T_{\text{revenue}} - G_{\text{expenditure}}$$

Meaning: A negative balance (G > T) is a budget deficit financed via domestic/foreign borrowing or SARB credit; a positive balance (T > G) is a budget surplus.

EQ 15.4 Negative Production Externality Optimality

$$\text{MSC} = \text{MPC} + \text{MEC}, \quad Q_{\text{market}}(\text{MPC} = \text{MSB}) > Q_{\text{social}}(\text{MSC} = \text{MSB})$$

Meaning: Unpriced pollution causes private firms to overproduce (Q1 > Q2) at too low a price (P1 < P2), creating a deadweight welfare loss.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 15.1 Composition of the Public Sector & Musgrave's Three Functions (Core Macroeconomic Concept)

The South African public sector consists of (1) Central government, (2) Provincial/regional government, (3) Local government (municipalities), and (4) Public corporations (SOEs like Eskom and Transnet). Government intervention serves three functions: Allocative (correcting market failure), Distributive (reducing inequality), and Stabilisation (smoothing business cycles).

Worked Proof: Progressive income taxation and social grants serve the Distributive function, while Pigouvian carbon taxes and public street lighting serve the Allocative function.

CONCEPT 15.2 Negative Production Externalities & Deadweight Loss (Core Macroeconomic Concept)

When a chemical or coal plant pollutes Highveld air, Marginal Social Cost (MSC) lies vertically above Marginal Private Cost (MPC). Left to the free market, equilibrium occurs where Demand (MPB=MSB) intersects MPC at (P1, Q1), exceeding the socially efficient output Q2 where MSB intersects MSC.

Worked Proof: Imposing a Pigouvian excise tax equal to the Marginal External Cost (MEC) per unit internalises the externality, shifting MPC up to MSC, raising price to P2 and reducing output to Q2.

CONCEPT 15.3 Progressive, Proportional & Regressive Taxes (Core Macroeconomic Concept)

Taxes are classified by how the ratio of tax paid to income (the average/effective tax rate) changes as income rises: Progressive (ATR rises with income, e.g. SARS Personal Income Tax), Proportional (ATR is constant, e.g. flat company tax rate), and Regressive (ATR falls as income rises, e.g. uniform indirect VAT/fuel levies relative to household income).

Worked Proof: In Core Module (pp. 43–44): Jerome earns R24 000/month (R288 000/year) and pays R46 189 net tax after the R13 500 primary rebate (16.04% effective rate, 26% marginal rate), whereas CEO Sally earns R375 000/month (R4 500 000/year) and pays R1 870 125 net tax after the R13 500 primary rebate (41.56% effective rate, 45% marginal rate).

CONCEPT 15.4 Tax Functions, Regressive VAT Proof & Statutory vs. Effective Incidence (Core Analytical Formulation)

Core Module classifies tax actions into three functions: Financing (raising revenue to reduce state borrowing), Redistributing (funding basic-needs programmes and poorer-area infrastructure), and Regulating (raising taxes to cool an overheating economy or cutting taxes to stimulate slow growth). In the SA revenue chart (p. 36), Individuals (41%), VAT (26%), and Companies (13%) are the top tax sources, where Individual income tax, Company tax, and Mine tax are Direct taxes while VAT, Petrol tax, Customs duties, and Excise duties are Indirect taxes.

Worked Proof: In Core Module (pp. 38–42): Under progressive brackets, Peter (R16 601) pays R2 550 + 21% above R15 000 = R2 886.21 (17.39% ATR) while Susan (R83 450) pays R25 000 + 43% above R80 000 = R26 483.50 (31.74% ATR). Conversely, at 14% VAT, Nicky (earns R1 000, spends R750) pays R105 VAT (10.5% ATR) while Duncan (earns R500, spends R500) pays R70 VAT (14.0% ATR), proving VAT is regressive.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Composition of the Public Sector & Musgrave's Three Functions): Confusing government transfer payments (pensions, child grants) with government spending G in GDP.

First-Principles Law: Transfer payments are excluded from G in national accounts because recipients do not provide a current factor service in exchange; they redistribute purchasing power rather than directly purchasing newly produced output.

Fallacy (Negative Production Externalities & Deadweight Loss): Arguing that socially efficient policy requires eliminating 100% of industrial production (Q = 0).

First-Principles Law: Social efficiency occurs where Marginal Social Benefit equals Marginal Social Cost (Q2 > 0), eliminating the deadweight loss triangle between Q2 and Q1.

Fallacy (Progressive, Proportional & Regressive Taxes): Multiplying a taxpayer's entire annual income by their top bracket marginal rate.

First-Principles Law: In a bracketed progressive system, the marginal rate applies ONLY to the excess income above the bracket threshold, added to the base tax for lower brackets, minus statutory rebates.

Fallacy (Tax Functions, Regressive VAT Proof & Statutory vs. Effective Incidence): Believing that a uniform flat-rate consumption tax like VAT is proportional with respect to household income.

First-Principles Law: Because low-income households consume nearly 100% of their income while higher-income households save a larger fraction, a uniform indirect tax on consumption claims a larger percentage of poor households' income (regressive).

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (23 PROBLEMS)

PROBLEM 15.1 — STATUTORY PERSONAL INCOME TAX, REBATE, EFFECTIVE & MARGINAL TAX RATES (R425 000)

Macroeconomics, §15.11, pp. 316–319 • Medium

In a hypothetical SARS tax table, the bracket for taxable income R350 001 – R500 000 specifies: 'R75 000 + 40% of the amount above R350 000', and the primary tax rebate is R10 000. For an individual under 65 earning R425 000 annually, calculate the net tax payable after the rebate, the effective tax rate, and the marginal tax rate.

Verified Solution: Net tax payable is R95 000, the effective tax rate is 22.35%, and the marginal rate is 40%.

$$T_{\text{before}} = 75\,000 + 0.40(425\,000 - 350\,000) = 75\,000 + 30\,000 = \text{R}105\,000.$$
$$T_{\text{after}} = 105\,000 - 10\,000 = \text{R}95\,000. \quad \text{ATR} = \frac{95\,000}{425\,000} \times 100 = 22.35\%, \quad \text{MTR} = 40\%.$$

First-Principles Proof: Correct! Tax before rebate is 75 000 + 0.40(75 000) = R105 000; subtracting the R10 000 rebate gives R95 000 (22.35% of R425 000), while the bracket marginal rate is 40%.

PROBLEM 15.2 — JUNIOR LECTURER PERSONAL INCOME TAX & PRIMARY REBATE CALCULATION (R384 000)

Macroeconomics, §15.11, pp. 316–319 • Medium

Thabiso Lebotse (aged 24) starts his first job as a Junior Lecturer earning R32 000 per month (R384 000 per year). The statutory SARS tax bracket for R321 601 – R445 100 is 'R67 144 + 31% of the amount over R321 600', with a Primary Rebate of R14 958 (and Secondary Rebate of R8 199 for ages 65+). Calculate his total annual tax payable after rebate and his effective tax rate.

Verified Solution: Total net tax payable is R71 530 and his effective tax rate is 18.63% (22.52% pre-rebate).

$$Y = 32\,000 \times 12 = \text{R}384\,000. \quad T_{\text{gross}} = 67\,144 + 0.31(384\,000 - 321\,600) = 67\,144 + 19\,344 = \text{R}86\,488.$$
$$T_{\text{net}} = 86\,488 - 14\,958 = \text{R}71\,530. \quad \text{ATR} = \frac{71\,530}{384\,000} \times 100 = 18.63\%.$$

First-Principles Proof: Correct! Annual income is R384 000; gross tax is 67 144 + 0.31(62 400) = R86 488; subtracting the R14 958 primary rebate yields R71 530 (18.63% effective rate).

PROBLEM 15.3 — HIGH-BRACKET STATUTORY TAX PAYABLE BEFORE & AFTER PRIMARY REBATE (R825 000)

Macroeconomics, §15.11, pp. 316–319 • Medium

According to the current statutory SARS tax table, the bracket for R673 001 – R857 900 is 'R179 147 + 39% of taxable income above R673 000', with a Primary Rebate of R17 235 (for all individuals under 65). If you earn R825 000 per annum, what is your tax payable (before and after the primary rebate) and your marginal tax rate?

Verified Solution: Tax is R221 192 after rebate (R238 427 before rebate) with a 39% marginal tax rate.

$$T_{\text{before}} = 179\,147 + 0.39(825\,000 - 673\,000) = 179\,147 + 59\,280 = \text{R}238\,427.$$
$$T_{\text{after}} = 238\,427 - 17\,235 = \text{R}221\,192. \quad \text{MTR} = 39\%.$$

First-Principles Proof: Correct! Excess above R673 000 is R152 000; 39% of R152 000 is R59 280; adding R179 147 gives R238 427 gross tax, or R221 192 after the R17 235 primary rebate.

PROBLEM 15.4 — NEGATIVE PRODUCTION EXTERNALITIES (MSC VS MPC) & CORRECTIVE PIGOUVIAN TAX

Macroeconomics, §15.3, pp. 301–303 • Medium

A chemical plant in the Highveld generates air pollution (hydrogen sulphide) as an unpriced by-product of production. In the market diagram showing Demand (DD = MSB), Marginal Private Cost (MPC), and Marginal Social Cost (MSC), why is the free-market equilibrium output (Q1) larger than the socially efficient output (Q2)?

Verified Solution: Firms ignore external pollution costs, so private marginal cost is below social cost (MPC < MSC).

$$\text{Free Market: } \text{MPC}(Q_1) = \text{MSB}(Q_1). \quad \text{Social Optimum: } \text{MSC}(Q_2) = \text{MPC}(Q_2) + \text{MEC} = \text{MSB}(Q_2) \implies Q_1 > Q_2,\ P_1 < P_2.$$

First-Principles Proof: Correct! Because the polluting firm does not pay for the external damage inflicted on society, its MPC curve lies below MSC, leading to overproduction (Q1 > Q2) and deadweight loss.

PROBLEM 15.5 — PROGRESSIVE TAX BRACKET, PRIMARY REBATE & MARGINAL RATE (BELLA MBONAMBI R315 850)

Macroeconomics, §15.11, pp. 316–319 • Medium

Bella Mbonambi (aged 23) starts her first job as a junior economist at the SARB earning R315 850 per year. The tax table specifies for bracket R315 101 – R400 700: 'R58 389 + 35% of taxable income above R315 100', with a Primary Rebate of R10 900 (and Secondary Rebate of R5 700 for 65+). Calculate her net tax payable and effective tax rate.

Verified Solution: Net tax payable is R47 751.50 and her effective tax rate is 15.12% (18.57% pre-rebate).

$$T_{\text{before}} = 58\,389 + 0.35(315\,850 - 315\,100) = 58\,389 + 262.50 = \text{R}58\,651.50.$$
$$T_{\text{after}} = 58\,651.50 - 10\,900 = \text{R}47\,751.50. \quad \text{ATR} = \frac{47\,751.50}{315\,850} \times 100 = 15.12\%.$$

First-Principles Proof: Correct! Tax on the R750 excess above R315 100 is R262.50; adding R58 389 gives R58 651.50, and subtracting the R10 900 primary rebate yields R47 751.50 (15.12%).

PROBLEM 15.6 — BRACKET CREEP (FISCAL DRAG) UNDER UNINDEXED NOMINAL WAGE INFLATION

Macroeconomics, §15.11 & §20.3, pp. 316–319, 447 • Hard

An individual earned R375 000 taxable income in both Schedule A and Schedule B. In Schedule A, the bracket R296 541 – R410 460 was taxed at 'R63 478 + 31% above R296 540' (with zero rebate subtracted). In Schedule B, Treasury adjusted the bracket to R305 851 – R423 300 taxed at 'R63 853 + 31% above R305 850'. Calculate gross tax payable in both years and explain why Treasury adjusts bracket thresholds upward.

Verified Solution: Schedule B gross tax is R85 289.50 (22.74% average rate); raising bracket thresholds provides relief against bracket creep.

$$T_{17/18} = 63\,478 + 0.31(375\,000 - 296\,540) = \text{R}87\,800.60 \text{ (or R86 232.60 with R61 910 base)}.$$
$$T_{18/19} = 63\,853 + 0.31(375\,000 - 305\,850) = 63\,853 + 21\,436.50 = \text{R}85\,289.50 \text{ (ATR } = 22.74\%).$$

First-Principles Proof: Correct! In Schedule B, R63 853 + 0.31(69 150) = R85 289.50 (22.74% of R375 000), and raising bracket thresholds prevents inflation from pushing taxpayers into higher brackets (fiscal drag).

PROBLEM 15.7 — CALCULATING TAX LIABILITY, MARGINAL TAX RATE, AND AVERAGE TAX RATE FOR PERSON A (R25 000)

Calculating Tax Liability, Marginal Tax Rate, and Average Tax Rate for Person A (R25 000) • Medium

Under a statutory progressive income tax schedule (with zero rebate), income up to R20 000 is taxed at 15% ('R0 + 15% of taxable income'), income between R20 001 and R60 000 is taxed at 'R3 000 + 20% of the amount above R20 000', and income above R60 000 is taxed at 'R11 000 + 40% of the amount above R60 000' (or 'R14 000 + 40% above R60 000' under the alternative 25% middle bracket). Calculate the tax liability, Marginal Tax Rate (MTR), and Average Tax Rate (ATR) for Person A who earns a taxable income of R25 000.

Verified Solution: Person A pays R4 000 in tax, faces a Marginal Tax Rate of 20.0%, and has an Average Tax Rate of 16.0% (since the first R20 000 is taxed at 15%).

$$\text{Person A } (Y_A = \text{R}25\,000 \text{ in the R20 001} - \text{R60 000 bracket}):$$
$$\text{Tax}_A = 3\,000 + 0.20(25\,000 - 20\,000) = 3\,000 + 1\,000 = \text{R}4\,000$$
$$\text{Marginal Tax Rate } (\text{MTR}_A) = 20.0\% \quad (\text{rate on each Rand above R20 000})$$
$$\text{Average Tax Rate } (\text{ATR}_A) = \frac{4\,000}{25\,000} \times 100 = 16.0\% \quad (\text{strictly below } \text{MTR}_A = 20.0\%)$$

First-Principles Proof: Correct! Tax on the first R20 000 is R3 000 (15%) plus 20% of the R5 000 excess (R1 000) = R4 000, yielding an average tax rate of 4 000 / 25 000 = 16.0% and a marginal rate of 20.0%.

PROBLEM 15.8 — CALCULATING THE FISCAL DEFICIT FROM EXPENDITURE AND REVENUE COMPONENTS

Calculating the Fiscal Deficit from Expenditure and Revenue Components • Easy

A national government records the following fiscal flows for the financial year (in R billions): Government consumption expenditure = 450; Government investment expenditure (capital formation) = 250; Transfer payments and debt service cost = 180; Tax revenue = 700; Non-tax revenue = 80. Calculate Total Government Expenditure, Total Government Revenue, and the resulting Budget Balance.

Verified Solution: Total Expenditure = R880 billion; Total Revenue = R780 billion; Budget Balance = -R100 billion (a fiscal deficit of R100 billion).

$$\text{Total Government Expenditure } (G_{\text{total}}) = 450 + 250 + 180 = \text{R}880 \text{ billion}$$
$$\text{Total Government Revenue } (T_{\text{total}}) = 700 + 80 = \text{R}780 \text{ billion}$$
$$\text{Budget Balance} = T_{\text{total}} - G_{\text{total}} = 780 - 880 = -\text{R}100 \text{ billion (Budget Deficit of R100 billion)}$$

First-Principles Proof: Correct! Total outlays (450 + 250 + 180 = R880bn) exceed total receipts (700 + 80 = R780bn) by R100 billion, creating a budget deficit that must be financed by borrowing.

PROBLEM 15.9 — CALCULATING ANNUAL TAX LIABILITY, MARGINAL TAX RATE, AND AVERAGE TAX RATE FOR R26 700/MONTH

Calculating Annual Tax Liability, Marginal Tax Rate, and Average Tax Rate for R26 700/Month • Medium

Muhammad works as a financial analyst earning a taxable salary of R26 700 per month. According to the statutory annual personal income tax table (before rebates), the bracket for R226 001 – R353 100 specifies: 'R40 680 + 26% of taxable income above R226 000'. Calculate (1) Muhammad's annual taxable income, (2) his annual tax liability under this bracket schedule, (3) his Marginal Tax Rate (MTR), and (4) his Average Tax Rate (ATR).

Verified Solution: Annual taxable income is R320 400, annual tax payable is R65 224, Marginal Tax Rate is 26.0%, and Average Tax Rate is 20.36%.

$$\text{Annual Taxable Income} = \text{R}26\,700 \times 12 = \text{R}320\,400 \quad (\text{falls in the R}226\,001 - \text{R}353\,100 \text{ bracket})$$
$$\text{Annual Tax Liability} = 40\,680 + 0.26(320\,400 - 226\,000) = 40\,680 + 0.26(94\,400) = 40\,680 + 24\,544 = \text{R}65\,224$$
$$\text{Marginal Tax Rate (MTR)} = 26.0\%$$
$$\text{Average Tax Rate (ATR)} = \frac{65\,224}{320\,400} \times 100 = 20.36\%$$

First-Principles Proof: Correct! Annualising R26 700 × 12 gives R320 400; adding 26% of the R94 400 excess (R24 544) to R40 680 gives R65 224, yielding MTR = 26.0% and ATR = 65 224 / 320 400 = 20.36%.

PROBLEM 15.10 — CALCULATING GROSS AND NET INCOME TAX FOR A MONTHLY SALARY EARNER

Calculating Gross and Net Income Tax for a Monthly Salary Earner • Medium

A newly appointed graduate earns a taxable salary of R12,000 per month. Suppose the statutory annual tax table specifies that the R122,001–R195,000 bracket is taxed at 'R21,960 + 25% of taxable income above R122,000', and every individual qualifies for an annual primary rebate of R8,280. Calculate (1) annual taxable income, (3) gross annual tax before the rebate, (3) net annual tax payable after the primary rebate, and (4) the effective (average) tax rate and marginal tax rate.

Verified Solution: Annual income = R144,000; Gross tax = R27,460; Net tax after rebate = R19,180; Effective tax rate = 13.32% (19.07% gross) and MTR = 25%.

$$\text{Annual Taxable Income} = \text{R}12,000 \times 12 = \text{R}144,000$$
$$\text{Gross Annual Tax} = 21,960 + 0.25(144,000 - 122,000) = 21,960 + 0.25(22,000) = 21,960 + 5,500 = \text{R}27,460$$
$$\text{Net Tax Payable} = 27,460 - 8,280 = \text{R}19,180$$
$$\text{Effective Rate (ATR)} = \frac{19,180}{144,000} \times 100 = 13.32\%\ (19.07\%\ \text{gross}); \quad \text{Marginal Tax Rate (MTR)} = 25.0\%$$

First-Principles Proof: Correct! Annualising R12,000×12 gives R144,000; gross tax is 21,960 + 0.25(22,000) = R27,460; subtracting the R8,280 rebate yields R19,180 net tax (13.32% effective rate).

PROBLEM 15.11 — COMPOSITION OF THE PUBLIC SECTOR, MUSGRAVE'S THREE FUNCTIONS, AND PRIVATISATION

Composition of the Public Sector, Musgrave's Three Functions, and Privatisation • Easy

Identify (1) the four institutional components that constitute the South African public sector, (2) the three economic functions of government formulated by Richard Musgrave, and (3) the precise distinction between Nationalisation and Privatisation.

Verified Solution: Four tiers: Central, Provincial, Local government, and Public corporations; Three functions: Allocative, Distributive, and Stabilisation; Nationalisation transfers private assets to state ownership while Privatisation transfers state assets to private ownership.

$$\text{Public Sector Components: (1) Central government, (2) Provincial government, (3) Local government (municipalities), (4) Public corporations (SOEs)}$$
$$\text{General Government} = \text{Central} + \text{Provincial} + \text{Local (excluding Public Corporations)}$$
$$\text{Musgrave's Three Functions: (1) Allocative function (market failure), (2) Distributive function (inequality), (3) Stabilisation function (macro cycles)}$$
$$\text{Nationalisation vs Privatisation: Transfer of private ownership to the state vs transfer of state-owned assets to private ownership}$$

First-Principles Proof: Correct! General government (central, provincial, local) plus public corporations form the public sector, executing Musgrave's allocative, distributive, and stabilisation roles.

PROBLEM 15.12 — FINANCING PUBLIC INFRASTRUCTURE EXPANSION & CRITERIA FOR A GOOD TAX

Financing Public Infrastructure Expansion & Criteria for a Good Tax • Medium

Suppose the government expands public infrastructure spending from R14.5 billion to R17.5 billion (+R3.0 billion). What are the three macroeconomic sources available to finance government expenditure, and which four criteria (canons) define a good tax system?

Verified Solution: Financing sources: Income from public property, Taxation, and Borrowing on capital markets; Tax criteria: Neutrality, Equity (vertical and horizontal), Administrative simplicity, and Certainty.

$$\text{Three Financing Sources: (1) Income from public property/enterprises, (2) Taxation (compulsory levies), (3) Government borrowing (domestic/foreign ca}$$
$$\text{Note: Borrowing from the central bank (monetary financing) is avoided due to severe demand-pull inflationary consequences}$$
$$\text{Four Criteria of a Good Tax: (1) Neutrality (minimal excess burden/distortion), (2) Equity (horizontal and vertical ability-to-pay/benefit), (3) Administra}$$

First-Principles Proof: Correct! Governments finance outlays via enterprise income, taxation, and bond issuance, while evaluating taxes against neutrality, equity, administrative simplicity, and certainty.

PROBLEM 15.13 — CALCULATING GROSS TAX, NET TAX AFTER REBATE, AND EFFECTIVE TAX RATE FOR R475 000 INCOME

Calculating Gross Tax, Net Tax After Rebate, and Effective Tax Rate for R475 000 Income • Medium

A taxpayer under age 65 earns a taxable income of R475 000 during the tax year. According to the statutory progressive tax table, the bracket for R393 001 – R550 000 specifies: 'R103 000 + 40% of taxable income above R393 000', and the primary tax rebate is R15 000. Calculate (1) the gross tax liability before the rebate, (2) the net tax payable after deducting the primary rebate, and (3) the effective (average) tax rate after the rebate.

Verified Solution: Gross tax is R135 800, net tax after the R15 000 rebate is R120 800, and the effective tax rate is 25.43% (28.59% pre-rebate).

$$\text{Income Excess Above Bracket Threshold} = 475\,000 - 393\,000 = \text{R}82\,000$$
$$\text{Gross Tax Before Rebate} = 103\,000 + 0.40(82\,000) = 103\,000 + 32\,800 = \text{R}135\,800 \quad (\text{Pre-rebate ATR} = 28.59\%)$$
$$\text{Net Tax Payable After Rebate} = 135\,800 - 15\,000 = \text{R}120\,800$$
$$\text{Effective (Average) Tax Rate} = \frac{120\,800}{475\,000} \times 100 = 25.43\% \quad (\text{while Marginal Tax Rate} = 40\%)$$

First-Principles Proof: Correct! 40% of the R82 000 excess above R393 000 is R32 800; adding R103 000 gives R135 800 gross tax, and subtracting the R15 000 rebate gives R120 800 (25.43% of R475 000).

PROBLEM 15.14 — FOLLOW-UP ANALYSIS OF TAX LIABILITY, MTR, AND ATR WHEN INCOME RISES ACROSS BRACKETS

Follow-Up Analysis of Tax Liability, MTR, and ATR When Income Rises Across Brackets • Medium

In a multi-bracket personal income tax schedule (with zero rebate), Judy initially earns R18 900 in the R15 001 – R20 000 bracket ('R2 550 + 21% of income above R15 000'), where the first-part calculation established her initial tax payable as R3 369 (2 550 + 0.21×3 900), her marginal tax rate (MTR) as 21%, and her average tax rate (ATR) as 17.83%. Suppose Judy completes her degree and is promoted to a salary of R83 450 in the top R80 001+ bracket ('R25 000 + 43% of income above R80 000'). Calculate her new tax payable, new MTR, and new ATR, and classify the tax schedule.

Verified Solution: New tax is R26 483.50, MTR is 43%, and ATR rises from 17.83% to 31.74%, confirming that the tax system is progressive.

$$\text{Prerequisite Baseline at } Y_0 = \text{R}18\,900: \quad T_0 = \text{R}3\,369, \quad \text{MTR}_0 = 21\%, \quad \text{ATR}_0 = \frac{3\,369}{18\,900} = 17.83\%$$
$$\text{New Tax at } Y_1 = \text{R}83\,450: \quad T_1 = 25\,000 + 0.43(83\,450 - 80\,000) = 25\,000 + 0.43(3\,450) = 25\,000 + 1\,483.50 = \text{R}26\,483.50$$
$$\text{New Marginal Tax Rate } (\text{MTR}_1) = 43\%; \quad \text{New Average Tax Rate } (\text{ATR}_1) = \frac{26\,483.50}{83\,450} \times 100 = 31.74\%$$
$$\text{Classification: Progressive tax because ATR rises as income rises } (17.83\% \to 31.74\%) \text{ and MTR} > \text{ATR}$$

First-Principles Proof: Correct! Tax on the R3 450 excess above R80 000 is R1 483.50; adding R25 000 gives R26 483.50 (ATR = 31.74%), and because ATR rises with income (17.83% to 31.74%), the tax is progressive.

PROBLEM 15.15 — COMPARING ANNUAL TAX LIABILITY AND EFFECTIVE TAX RATES ACROSS ADJACENT TAX BRACKETS

Comparing Annual Tax Liability and Effective Tax Rates Across Adjacent Tax Brackets • Medium

Twins Joe and Bella earn monthly salaries of R35 000 per month (R420 000 per year) and R35 500 per month (R426 000 per year), respectively. The statutory annual tax table specifies: Bracket A (R305 851 – R423 300): 'R63 853 + 31% of taxable income above R305 850'; Bracket B (R423 301 – R555 600): 'R100 263 + 36% of taxable income above R423 300'. Calculate the gross annual tax liability and pre-rebate effective (average) tax rate for both Joe and Bella, and identify the tax structure.

Verified Solution: Joe pays R99 239.50 (23.63% effective rate) and Bella pays R101 235.00 (23.76% effective rate), illustrating a progressive tax.

$$\text{Joe } (Y = 35\,000 \times 12 = \text{R}420\,000 \text{ in Bracket A}): T_{\text{Joe}} = 63\,853 + 0.31(420\,000 - 305\,850) = 63\,853 + 35\,386.50 = \text{R}99\,239.50$$
$$\text{Joe Effective Rate} = \frac{99\,239.50}{420\,000} \times 100 = 23.63\%$$
$$\text{Bella } (Y = 35\,500 \times 12 = \text{R}426\,000 \text{ in Bracket B}): T_{\text{Bella}} = 100\,263 + 0.36(426\,000 - 423\,300) = 100\,263 + 972.00 = \text{R}101\,235.00$$
$$\text{Bella Effective Rate} = \frac{101\,235.00}{426\,000} \times 100 = 23.76\% \quad (\text{Progressive since ATR rises from } 23.63\% \text{ to } 23.76\%)$$

First-Principles Proof: Correct! Joe's R420 000 annual income yields 63 853 + 0.31(114 150) = R99 239.50 (23.63%), while Bella's R426 000 yields 100 263 + 0.36(2 700) = R101 235.00 (23.76%).

PROBLEM 15.16 — FOLLOW-UP CALCULATION OF BELLA MBONAMBI'S EFFECTIVE TAX RATE FROM PREREQUISITE NET TAX PAYABLE

Follow-Up Calculation of Bella Mbonambi's Effective Tax Rate From Prerequisite Net Tax Payable • Medium

Continuing with the personal income tax analysis for Bella Mbonambi (aged 23, earning R315 850 per year as a junior economist), where the bracket R315 101 – R400 700 specifies 'R58 389 + 35% of taxable income above R315 100' and the primary rebate is R10 900: the first part of the calculation established that her gross tax before rebate is R58 651.50 and her net tax payable after deducting the R10 900 primary rebate is R47 751.50. Using this prerequisite net tax payable (R47 751.50), calculate Bella's effective (average) tax rate (ATR) and explain why it is lower than her 35% statutory marginal tax rate (MTR).

Verified Solution: Her effective (average) tax rate is 15.12% after the rebate (18.57% pre-rebate), which is below the 35% marginal rate because lower tranches face lower rates.

$$\text{Prerequisite Baseline: Annual Taxable Income} = \text{R}315\,850; \quad \text{Gross Tax} = \text{R}58\,651.50; \quad \text{Net Tax Payable} = \text{R}47\,751.50$$

$$\text{Effective (Average) Tax Rate After Rebate} = \frac{47\,751.50}{315\,850} \times 100 = 15.12\%$$

$$\text{Pre-Rebate Average Tax Rate} = \frac{58\,651.50}{315\,850} \times 100 = 18.57\%$$

$$\text{Why ATR } (15.12\%) < \text{MTR } (35.00\%): \text{ Earlier income tranches below R315 100 are taxed at lower marginal rates and the R10 900 primary rebate reduces}$$

First-Principles Proof: Correct! Dividing Bella's net tax payable (R47 751.50) by her annual taxable income (R315 850) gives 15.12% (18.57% gross), well below the 35% rate applied only to the R750 excess.

PROBLEM 15.17 — FOLLOW-UP ANALYSIS OF DEADWEIGHT LOSS AND CORRECTIVE GOVERNMENT INTERVENTION

Follow-Up Analysis of Deadweight Loss and Corrective Government Intervention • Medium

Continuing with the polluting Highveld chemical plant where Marginal Social Cost ($MSC = MPC + MEC$) lies above Marginal Private Cost (MPC), so the unregulated competitive market equilibrium produces output Q_1 at price P_1 whereas the socially efficient optimum is at lower output Q_2 and higher price P_2: (1) Why does every unit produced between Q_2 and Q_1 generate a deadweight welfare loss to society? (2) What corrective policy instrument can the government use to internalise the externality and restore output to Q_2?

Verified Solution: Between Q2 and Q1, Marginal Social Cost exceeds Marginal Social Benefit (MSC > MSB); government can impose a Pigouvian tax equal to the marginal external cost (MEC).

$$\text{Prerequisite Context: Unregulated Market Equilibrium at } Q_1 \text{ where } MPC = MSB; \quad \text{Social Optimum at } Q_2 < Q_1 \text{ where } MSC = MSB$$

$$\text{For all output units } Q \in (Q_2, Q_1): \quad MSC(Q) > MSB(Q) \implies \text{social cost of each extra unit exceeds its social benefit, creating a deadweight loss triangle}$$

$$\text{Corrective Policy: Impose a specific Pigouvian excise tax } t = MEC \text{ per unit on the polluter, shifting } MPC \text{ upward to coincide with } MSC \text{ at } (Q_2, P_2)$$

First-Principles Proof: Correct! For every unit beyond Q2 up to Q1, total cost to society (MSC) exceeds the benefit to consumers (DD = MSB), and a Pigouvian tax equal to MEC shifts private cost up to MSC.

PROBLEM 15.18 — FOLLOW-UP CALCULATION OF PERSON B'S TAX LIABILITY AND AVERAGE TAX RATE TO PROVE PROGRESSIVITY

Follow-Up Calculation of Person B's Tax Liability and Average Tax Rate to Prove Progressivity • Medium

Continuing with the progressive personal income tax schedule (with zero rebate) where income up to R20 000 is taxed at 15%, income from R20 001 to R60 000 is taxed at 'R3 000 + 20% above R20 000', and income above R60 000 is taxed at 'R11 000 + 40% above R60 000' (or 'R14 000 + 40% above R60 000' under the 25% middle-bracket variant): the first part of the analysis established that Person A (earning R25 000) pays R4 000 in tax, faces a 20.0% Marginal Tax Rate (MTR), and has a 16.0% Average Tax Rate (ATR). Now calculate the tax liability, MTR, and ATR for Person B who earns R80 000, and classify the tax schedule by comparing Person B's ATR with Person A's 16.0% ATR.

Verified Solution: Person B pays R19 000 (ATR = 23.75%, or R22 000 / 27.50% with R14 000 base) and faces a 40% MTR; since ATR rises above Person A's 16.0%, the tax is progressive.

$$\text{Prerequisite Baseline (Person A at R25 000): Tax}_A = \text{R}4\,000, \quad \text{MTR}_A = 20.0\%, \quad \text{ATR}_A = 16.0\%$$

$$\text{Person B } (Y_B = \text{R}80\,000 \text{ in the R60 001+ bracket}): \text{ Excess above R60 000} = 80\,000 - 60\,000 = \text{R}20\,000$$

$$\text{Tax}_B = 11\,000 + 0.40(20\,000) = \text{R}19\,000, \ \text{ATR}_B = \frac{19\,000}{80\,000} = 23.75\% \quad (\text{or } 14\,000 + 8\,000 = \text{R}22\,000, \ \text{ATR}_B = \frac{22\,000}{80\,000} = 27.50\%)$$

$$\text{Classification: Since ATR}_B \ (23.75\% \text{ or } 27.50\%) > \text{ATR}_A \ (16.0\%), \text{ the tax schedule is strictly PROGRESSIVE}$$

First-Principles Proof: Correct! Adding 40% of the R20 000 excess (+R8 000) to the bracket base yields R19 000 (23.75% ATR) or R22 000 (27.50% ATR); since ATR rises with income (above 16.0%), the tax is progressive.

PROBLEM 15.19 — FOLLOW-UP CALCULATION OF TAX LIABILITY, MTR, AND ATR AFTER PROMOTION TO R40 723/MONTH

Follow-Up Calculation of Tax Liability, MTR, and ATR After Promotion to R40 723/Month • Hard

Continuing with the personal income tax analysis for Muhammad, where the first part established that at his initial salary of R26 700 per month (R320 400 per year in the R226 001 – R353 100 bracket: 'R40 680 + 26% above R226 000'), his annual tax payable is R65 224, his Marginal Tax Rate (MTR) is 26.0%, and his Average Tax Rate (ATR) is 20.36%: suppose Muhammad is promoted to Senior Manager and his monthly taxable salary increases to R40 723 per month. In the statutory annual tax table, the bracket R353 101 – R488 700 specifies: 'R73 726 + 31% of taxable income above R353 100'. Calculate his new annual taxable income, new annual tax payable, new MTR, and new ATR, and state what the change in ATR from 20.36% proves.

Verified Solution: New annual income is R488 676, new tax is R115 754.56, MTR rises to 31.0%, and ATR rises from 20.36% to 23.69%, proving the tax schedule is progressive.

$$\text{Prerequisite Baseline at R26 700/mo (R320 400/yr): } \text{Tax}_0 = \text{R65 224}, \quad \text{MTR}_0 = 26.0\%, \quad \text{ATR}_0 = 20.36\%$$
$$\text{New Annual Taxable Income} = \text{R40 723} \times 12 = \text{R488 676} \quad \text{(falls in the R353 101 – R488 700 bracket)}$$
$$\text{New Annual Tax} = 73\,726 + 0.31(488\,676 - 353\,100) = 73\,726 + 0.31(135\,576) = 73\,726 + 42\,028.56 = \text{R115 754.56}$$
$$\text{New MTR} = 31.0\%; \quad \text{New ATR} = \frac{115\,754.56}{488\,676} \times 100 = 23.69\% > 20.36\% \implies \text{Proves the tax system is PROGRESSIVE}$$

First-Principles Proof: Correct! Annualising R40 723 × 12 gives R488 676; adding 31% of the R135 576 excess (R42 028.56) to R73 726 yields R115 754.56 (ATR = 23.69% > 20.36%), proving progressivity.

PROBLEM 15.20 — FOLLOW-UP CALCULATION OF EFFECTIVE TAX RATE FROM PREREQUISITE NET TAX PAYABLE ON R825 000

Follow-Up Calculation of Effective Tax Rate From Prerequisite Net Tax Payable on R825 000 • Medium

Continuing with the personal income tax analysis for an individual (under age 65) earning R825 000 per annum, where the bracket R673 001 – R857 900 specifies 'R179 147 + 39% of taxable income above R673 000' and the primary rebate is R17 235: the first part of the analysis established that gross tax before the rebate is R238 427 (179 147 + 59 280), net tax payable after the R17 235 primary rebate is R221 192, and the marginal tax rate is 39%. Using this prerequisite net tax payable (R221 192), calculate the taxpayer's effective (average) tax rate (ATR) and compare it with the pre-rebate average rate and 39% marginal rate.

Verified Solution: The effective (average) tax rate after the rebate is 26.81% (and 28.90% before the rebate), which is well below the 39.00% marginal tax rate.

$$\text{Prerequisite Baseline: Taxable Income} = \text{R825 000}; \quad \text{Gross Tax} = \text{R238 427}; \quad \text{Net Tax Payable} = \text{R221 192}$$
$$\text{Effective (Average) Tax Rate After Rebate} = \frac{221\,192}{825\,000} \times 100 = 26.81\%$$
$$\text{Pre-Rebate Average Tax Rate} = \frac{238\,427}{825\,000} \times 100 = 28.90\%$$
$$\text{Comparison: Effective ATR } (26.81\%) < \text{Pre-Rebate ATR } (28.90\%) < \text{Marginal Tax Rate } (39.00\%)$$

First-Principles Proof: Correct! Dividing net tax payable (R221 192) by taxable income (R825 000) yields an effective tax rate of 26.81% (or 238 427 / 825 000 = 28.90% pre-rebate), below the 39% marginal rate.

PROBLEM 15.21 — FINANCING, REDISTRIBUTING & REGULATING TAX FUNCTIONS AND VAT REGRESSIVITY PROOFS

Financing, Redistributing & Regulating Tax Functions and VAT Regressivity Proofs • Hard

Evaluate the fiscal functions and incidence of taxation from Core Module: (1) Classify four government tax actions into Financing, Redistributing, or Regulating: (a) taxing to subsidise basic needs for poor communities, (b) raising taxes to cool an overheating economy, (c) raising taxes to reduce annual government borrowing, and (d) cutting taxes to stimulate a slow economy; (2) Calculate Peter's tax liability and average tax rate on an income of R16 601 under the bracket 'R15 001–R20 000: R2 550 + 21% above R15 000' (given Susan earns R83 450 and pays R26 483.50 tax = 31.74% ATR); and (3) Compare the 14% VAT burden as a share of total income for Nicky (earns R1 000/month, spends 75% = R750) versus Duncan (earns R500/month, spends 100% = R500).

Verified Solution: (1) a=Redistributing, b=Regulating, c=Financing, d=Regulating; (2) Peter pays R2 886.21 (ATR = 17.39% vs Susan's 31.74%, progressive); (3) Nicky pays R105 (10.5% of income) vs Duncan's R70 (14.0% of income, regressive).

$$\text{(1) Tax Functions: (a) Redistributing; (b) Regulating (contractionary); (c) Financing; (d) Regulating (expansionary)}$$
$$\text{(2) Peter (Income R16 601): Tax} = 2\,550 + 0.21(16\,601 - 15\,000) = 2\,550 + 336.21 = \text{R2 886.21}$$
$$\text{Peter ATR} = \frac{2\,886.21}{16\,601} \times 100 = 17.39\% < \text{Susan ATR } (31.74\%) \implies \text{Progressive income tax}$$
$$\text{(3) Nicky (Income R1 000, spends R750): VAT} = 0.14 \times 750 = \text{R105} \implies \text{ATR}_{\text{Nicky}} = \frac{105}{1\,000} = 10.5\%$$
$$\text{(3) Duncan (Income R500, spends R500): VAT} = 0.14 \times 500 = \text{R70} \implies \text{ATR}_{\text{Duncan}} = \frac{70}{500} = 14.0\% > 10.5\% \implies \text{Regressive indirect tax}$$

First-Principles Proof: Correct! Subsidising basic needs is redistributive, macroeconomic stabilisation (b and d) is regulatory, replacing debt is financing, Peter's ATR is 17.39%, and Duncan's 14.0% VAT share exceeds Nicky's 10.5%.

PROBLEM 15.22 — STATUTORY PERSONAL INCOME TAX, REBATE & EFFECTIVE TAX RATE (JEROME MULLER R288 000)

Statutory Personal Income Tax, Rebate & Effective Tax Rate (Jerome Muller R288 000) • Medium

Jerome Muller is 37 years old and earns a taxable salary of R24 000 per month during the statutory tax year. According to the Core Module statutory tax table, the bracket for annual taxable income from R189 881 to R296 540 is 'R34 178 + 26% of taxable income above R189 880', and the Primary Rebate (under age 65) is R13 500. Calculate: (1) Annual taxable income and gross tax before rebates; (2) Net annual tax payable after the primary rebate; (3) Effective (average) tax rate and marginal tax rate; and (4) Annual post-tax disposable income.

Verified Solution: Annual income = R288 000; Gross tax = R59 689.20; Net tax payable = R46 189.20 (≈ R46 189); Effective tax rate = 16.04% (Marginal rate = 26%); Post-tax income = R241 810.80.

$$\text{(1) Annual Taxable Income} = 24\,000 \times 12 = \text{R}288\,000 \text{ (falls in R189 881 – R296 540 bracket)}$$
$$\text{Gross Tax Before Rebate} = 34\,178 + 0.26 \times (288\,000 - 189\,880) = 34\,178 + 0.26(98\,120) = 34\,178 + 25\,511.20 = \text{R}59\,689.20$$
$$\text{(2) Net Tax Payable After Primary Rebate} = 59\,689.20 - 13\,500 = \text{R}46\,189.20 \approx \text{R}46\,189$$
$$\text{(3) Effective (Average) Tax Rate} = \frac{46\,189.20}{288\,000} \times 100 = 16.04\% \text{ (vs } 20.73\% \text{ before rebate)}; \quad \text{Marginal Tax Rate} = 26\%$$
$$\text{(4) Post-Tax Disposable Income} = 288\,000 - 46\,189.20 = \text{R}241\,810.80 \approx \text{R}241\,811$$

First-Principles Proof: Correct! At R288 000/year, gross tax is R34 178 + 26% of R98 120 = R59 689.20; subtracting the R13 500 primary rebate yields R46 189.20 net tax (16.04% effective rate) and R241 810.80 disposable income.

PROBLEM 15.23 — TOP-BRACKET STATUTORY INCOME TAX, REBATE & EFFECTIVE RATE (SALLY MOLOI R4 500 000)

Top-Bracket Statutory Income Tax, Rebate & Effective Rate (Sally Moloi R4 500 000) • Medium

Sally Moloi is 55 years old and earns a taxable salary of R375 000 per month during the statutory tax year. According to the Core Module statutory tax table, the top bracket for annual taxable income of R1 500 001 and above is 'R533 625 + 45% of taxable income above R1 500 000', and the Primary Rebate (under age 65) is R13 500. Calculate: (1) Annual taxable income and gross tax before rebates; (2) Net annual tax payable after the primary rebate; (3) Effective (average) tax rate and marginal tax rate; and (4) Annual post-tax disposable income.

Verified Solution: Annual income = R4 500 000; Exact net tax = R1 870 125 (ETR = 41.56%, Post-tax = R2 629 875) [or R1 900 125 / 42.22% in guide]; Marginal tax rate = 45%.

$$\text{(1) Annual Taxable Income} = 375\,000 \times 12 = \text{R}4\,500\,000 \text{ (falls in R1 500 001+ top bracket)}$$
$$\text{Exact } 45\% \text{ on Excess Above R1 500 000} = 0.45 \times (4\,500\,000 - 1\,500\,000) = 0.45 \times 3\,000\,000 = \text{R}1\,350\,000 \text{ (note: R1 380 000 in guide print typo)}$$
$$\text{Exact Gross Tax} = 533\,625 + 1\,350\,000 = \text{R}1\,883\,625 \text{ (or R1 913 625 in guide print)}$$
$$\text{(2) Exact Net Tax Payable} = 1\,883\,625 - 13\,500 = \text{R}1\,870\,125 \text{ (or R1 900 125 in guide print)}$$
$$\text{(3) Effective Tax Rate} = \frac{1\,870\,125}{4\,500\,000} \times 100 = 41.56\% \text{ (or } 42.22\% \text{ in guide)}; \quad \text{Marginal Rate} = 45\%; \quad \text{Post-Tax Income} = \text{R}2\,629\,875$$

First-Principles Proof: Correct! On R4 500 000 annual income, 45% of the R3 000 000 excess is R1 350 000, giving R1 883 625 gross tax and R1 870 125 net tax after the R13 500 primary rebate (41.56% effective rate vs 45% marginal rate).

CHAPTER 16 Foreign Sector & Exchange Rates • International Trade & Foreign Exchange • 10 Hours

Chapter 16: The Foreign Sector

Globalisation, absolute vs comparative (relative) advantage, opportunity cost trade calculations, import substitution vs export promotion, GATT/WTO & trade barriers, foreign exchange market (ZAR/USD appreciation & depreciation), and the terms of trade.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Foreign Sector & Exchange Rates • 10 Hours):

- Explain globalisation, international economic interdependence, and why uneven factor endowments give rise to international trade.
- Calculate and explain absolute advantage and comparative (relative) advantage using opportunity cost ratios to determine optimal specialisation.
- Identify trade barriers (tariffs, quotas, subsidies, non-tariff barriers, exchange controls), compare import substitution with outward-oriented export promotion, and explain the role of GATT and the WTO.
- Explain and graphically illustrate how exchange rates are determined in a controlled floating exchange rate regime, analyse currency appreciation and depreciation, and define the terms of trade.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (3 IDENTITIES)

EQ 16.1 **Opportunity Cost & Comparative Advantage**

$$\text{Opp. Cost of 1 unit of } X = \frac{\text{Maximum Output of } Y}{\text{Maximum Output of } X}$$

Meaning: A country has a comparative (relative) advantage in producing good X if it sacrifices fewer units of good Y per unit of X than its trading partner.

EQ 16.2 **Terms of Trade Index**

$$\text{Terms of Trade} = \left(\frac{\text{Export Price Index}}{\text{Import Price Index}}\right) \times 100$$

Meaning: Measures the volume of imports a country can purchase per unit of exports; an increase represents an improvement in national purchasing power.

EQ 16.3 **Direct Exchange Rate Quotation**

$$E_{\text{direct}} = \frac{\text{Domestic Currency (ZAR)}}{\text{1 Unit of Foreign Currency (USD)}}$$

Meaning: Under direct quotation in SA (e.g. R18.50/1), $a decrease in the Rand price of 1$ is a Rand appreciation; an increase is a Rand depreciation.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 16.1 **Absolute vs. Comparative (Relative) Advantage** (Core Macroeconomic Concept)

Absolute advantage exists when a country can produce more of a good than another country using the same resources. Comparative (relative) advantage exists when a country can produce a good at a lower opportunity cost (sacrificing less of the alternative good).

> **Worked Proof:** In Core Module (p. 49): A Less Developed Country (LDC) produces 2 kg wheat or 1 m cloth (1 m cloth = 2 kg wheat; 1 kg wheat = 0.5 m cloth). A Developed Country (DC) produces 4 kg wheat or 8 m cloth (1 m cloth = 0.5 kg wheat; 1 kg wheat = 2 m cloth). DC has absolute advantage in both, but LDC has comparative advantage in wheat (0.5 < 2) and DC in cloth (0.5 < 2).

CONCEPT 16.2 **Trade Policy Measures, GATT/WTO & Export Promotion** (Core Macroeconomic Concept)

Governments restrict import volumes or protect domestic infant industries and jobs using six main instruments: (1) Import tariffs (specific fixed amount per unit vs ad valorem percentage of value), (2) Import quotas (quantitative limits), (3) Subsidies to domestic producers, (4) Non-tariff barriers (administrative/health standards), (5) Exchange controls, and (6) Exchange rate policy. Outward-oriented export promotion achieves higher growth than inward-looking import substitution.

> **Worked Proof:** Imposing an import tariff raises the domestic price above the world price, increasing domestic production, reducing domestic consumption, and generating tariff revenue, but creating deadweight welfare losses and hurting consumers.

CONCEPT 16.3 **Exchange Rate Appreciation, Depreciation & Cross-Currency Pricing** (Core Macroeconomic Concept)

In the ZAR/USD foreign exchange market, US dollars are supplied by buyers of SA exports and foreign investors bringing capital into SA, and demanded by SA importers and residents investing abroad. South Africa operates a controlled floating single exchange rate regime. To determine trade flows across multiple commodities, prices must first be converted into a common currency via the exchange rate.

> **Worked Proof:** In Core Module (p. 54): USA prices are A=1, B =4, C= 6, D =8, E=12 $and British prices are A = £9, B = £7, C = £6, D = £4, E = £1. At £1$ =1, USA exports A & B (C equal at 6) and imports D & E; at £1=2 (dollar depreciation), British dollar prices double so USA exports A, B, C (D equal at 8) $and imports only E; at £1$ =0.50 (dollar appreciation), USA exports only A and imports B, C, D, E.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Absolute vs. Comparative (Relative) Advantage): Concluding that no mutually beneficial trade can occur when one country has an absolute advantage in both goods.

First-Principles Law: David Ricardo's law of comparative advantage proves that as long as opportunity costs differ between two countries, each country gains by specialising in the good where its opportunity cost is lowest.

Fallacy (Trade Policy Measures, GATT/WTO & Export Promotion): Confusing import substitution with export promotion, or thinking governments can impose arbitrary tariffs without WTO constraints.

First-Principles Law: Import substitution uses tariffs/quotas to replace previously imported goods with locally manufactured goods, whereas export promotion provides incentives to produce for world markets under WTO rules.

Fallacy (Exchange Rate Appreciation, Depreciation & Cross-Currency Pricing): Thinking a rise in the number R18/ $to R20/$ represents an appreciation of the Rand.

First-Principles Law: Because R/ $is a direct quote (Rands per 1 USD), paying MORE Rands for 1$ (R18 -> R20) means the Rand has lost value (depreciated) and the US dollar has appreciated.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (12 PROBLEMS)

PROBLEM 16.1 — ABSOLUTE VS COMPARATIVE ADVANTAGE & OPPORTUNITY COST (SPAIN & GREECE)

Macroeconomics, §16.2, pp. 325–329 • Medium

Spain and Greece produce olives and tobacco. With a given unit of resources, Spain can produce either 100 kg of olives or 150 kg of tobacco, while Greece can produce either 300 kg of olives or 300 kg of tobacco. What are the opportunity costs, and which country has the comparative advantage in each product?

Verified Solution: Spain has comparative advantage in tobacco (0.67 < 1 olive) and Greece in olives (1 < 1.5 tobacco).

$$\text{Spain: } 1\text{ Tobacco} = \frac{100}{150} = 0.67\text{ Olives},\ 1\text{ Olive} = \frac{150}{100} = 1.5\text{ Tobacco}.$$
$$\text{Greece: } 1\text{ Tobacco} = \frac{300}{300} = 1\text{ Olive},\ 1\text{ Olive} = 1\text{ Tobacco}.$$

First-Principles Proof: Correct! Spain sacrifices only 0.67 kg of olives per kg of tobacco (vs 1 kg in Greece), while Greece sacrifices only 1 kg of tobacco per kg of olives (vs 1.5 kg in Spain).

PROBLEM 16.2 — OPPORTUNITY COST RATIOS & SPECIALISATION IN TRADE (AUSTRALIA & BRAZIL)

Macroeconomics, §16.2 & §16.4, pp. 326–336 • Medium

Australia can produce 150 surfboards or 900 bottles of sunscreen per month, whereas the USA can produce 200 surfboards or 1 600 bottles of sunscreen per month. Furthermore, 1 bottle of sunscreen costs US$1.50 and the exchange rate is 1 US$ = 1.70 AUS$. Which country has a relative advantage in surfboards, and what will Australia pay in AUS$ for 50 bottles of sunscreen?

Verified Solution: Australia has relative advantage in surfboards (costing 6 vs 8 bottles); 50 bottles cost AUS$127.50.

$$\text{Opp. Cost of 1 Surfboard: Australia} = \frac{900}{150} = 6\text{ bottles};\quad \text{USA} = \frac{1600}{200} = 8\text{ bottles} \implies \text{Australia } (6 < 8).$$
$$\text{Cost} = 50 \times \$1.50 \times 1.70 = \text{AUS}\$127.50.$$

First-Principles Proof: Correct! Australia gives up 6 bottles of sunscreen per surfboard compared to 8 in the USA, and 50 × US$1.50 × 1.70 = AUS$127.50.

PROBLEM 16.3 — EXCHANGE RATE APPRECIATION (ZAR/USD): TRADE BALANCE & DOMESTIC INFLATION

Macroeconomics, §16.4, pp. 336–340 • Easy

Suppose the South African Rand appreciates against the US dollar (for example, moving from R19.00/1 to R17.00/1) due to strong capital inflows and rising commodity demand. How will this appreciation impact South Africa's export prices (in foreign currency) and import prices (in Rand), ceteris paribus?

Verified Solution: South African export prices in foreign currency increase, while import prices in Rand decrease.

$$\text{Rand Appreciation } (E\downarrow) \implies P_{\text{exports in USD}}\uparrow \implies X\downarrow;\quad P_{\text{imports in ZAR}}\downarrow \implies Z\uparrow.$$

First-Principles Proof: Correct! A stronger Rand makes South African goods more expensive for foreign buyers in USD, while foreign goods become cheaper for South Africans in ZAR.

PROBLEM 16.4 — COMPARATIVE ADVANTAGE IN WAGYU BEEF & SEAFOOD (JAPAN & PORTUGAL)

Macroeconomics, §16.2, pp. 325–329 • Easy

Suppose Japan and Portugal produce only Wagyu beef and Seafood. If all resources are fully used, Japan can produce 1 000 kg of Wagyu beef or 2 000 kg of Seafood, while Portugal can produce 2 200 kg of Wagyu beef or 3 000 kg of Seafood. Compute the opportunity cost of producing 1 kg of Wagyu beef in both countries and identify who holds the comparative advantage in Wagyu beef.

Verified Solution: Japan sacrifices 2.00 kg Seafood and Portugal sacrifices 1.36 kg, giving Portugal comparative advantage.

$$\text{Japan: } 1\text{ kg Wagyu} = \frac{2\,000}{1\,000} = 2.00\text{ kg Seafood}.\quad \text{Portugal: } 1\text{ kg Wagyu} = \frac{3\,000}{2\,200} = 1.36\text{ kg Seafood}.$$

First-Principles Proof: Correct! Japan gives up 2000/1000 = 2.00 kg of Seafood per kg of Wagyu, while Portugal gives up only 3000/2200 = 1.36 kg of Seafood.

PROBLEM 16.5 — LABOUR-HOUR OPPORTUNITY COST & MUTUAL GAINS FROM TRADE (CLOTH & WINE)

Macroeconomics, §16.2, pp. 325–329 • Medium

Suppose 1 hour of labour in South Africa can produce either 6 metres of cloth or 4 cars, whereas 1 hour of labour in Zimbabwe can produce either 1 metre of cloth or 5 cars. Determine the opportunity cost of producing 1 metre of cloth and 1 car in each country, and state which country should specialise in each product.

Verified Solution: SA has comparative advantage in cloth (0.67 < 5 cars) and Zimbabwe has comparative advantage in cars (0.20 < 1.50m cloth).

$$\text{SA: } 1\text{m cloth} = \frac{4}{6} = 0.67\text{ cars},\ 1\text{ car} = \frac{6}{4} = 1.50\text{m cloth}.\quad \text{Zim: } 1\text{m cloth} = \frac{5}{1} = 5\text{ cars},\ 1\text{ car} = \frac{1}{5} = 0.20\text{m cloth}.$$

First-Principles Proof: Correct! SA sacrifices only 4/6 = 0.67 cars per metre of cloth (vs 5 cars in Zimbabwe), while Zimbabwe sacrifices only 1/5 = 0.20m of cloth per car (vs 1.50m in SA).

PROBLEM 16.6 — SIX INSTRUMENTS TO RESTRICT IMPORTS AND WELFARE EFFECTS OF AN IMPORT TARIFF

Six Instruments to Restrict Imports and Welfare Effects of an Import Tariff • Medium

List the six primary policy instruments governments use to restrict imports or influence international trade flows, and explain the domestic impact of imposing a specific import tariff on a competitive good whose world price is below the domestic autarky equilibrium price.

Verified Solution: Instruments: Tariffs, quotas, domestic subsidies, non-tariff barriers, exchange controls, and exchange rate policy; a tariff raises domestic price, expands local supply, cuts imports, and yields state revenue.

$$\text{Six Instruments: (1) Import tariffs, (2) Import quotas, (3) Subsidies to domestic producers, (4) Non-tariff barriers (administrative/health regulations), (5)}$$
$$\text{Tariff Impact: Domestic price rises from } P_w \text{ to } P_w + t \implies \text{Domestic quantity supplied rises, domestic quantity demanded falls, import volume shrinks}$$
$$\text{Welfare: Domestic producers gain surplus, government collects tariff revenue on remaining imports, and domestic consumers lose surplus (creating deadwe}$$

First-Principles Proof: Correct! All six trade policy instruments are accurately identified, and a tariff raises price from Pw to Pw+t, expanding domestic supply while contracting domestic demand and imports.

PROBLEM 16.7 — OPPORTUNITY COST OF MAIZE VS STEEL BETWEEN TWO TRADING PARTNERS

Opportunity Cost of Maize vs Steel Between Two Trading Partners • Medium

Using identical resource endowments, Country A can produce either 80 tons of Maize or 40 tons of Steel, whereas Country B can produce either 30 tons of Maize or 30 tons of Steel. Determine (1) which country holds the absolute advantage in both goods, (2) each country's opportunity cost of producing 1 ton of Steel and 1 ton of Maize, and (3) the pattern of comparative advantage.

Verified Solution: Country A has absolute advantage in both goods; Country A has comparative advantage in Maize (cost = 0.5t Steel), and Country B has comparative advantage in Steel (cost = 1.0t Maize).

$$\text{Absolute Advantage: Country A produces more Maize } (80 > 30) \text{ and more Steel } (40 > 30)$$
$$\text{Opportunity Cost of 1 ton of Steel: Country A} = \frac{80}{40} = 2.0 \text{ tons Maize}; \quad \text{Country B} = \frac{30}{30} = 1.0 \text{ ton Maize}$$
$$\text{Opportunity Cost of 1 ton of Maize: Country A} = \frac{40}{80} = 0.5 \text{ tons Steel}; \quad \text{Country B} = \frac{30}{30} = 1.0 \text{ ton Steel}$$
$$\text{Comparative Advantage: Country A in Maize } (0.5 < 1.0 \text{ Steel}); \quad \text{Country B in Steel } (1.0 < 2.0 \text{ Maize})$$

First-Principles Proof: Correct! Even though Country A is more productive in both goods, Country B sacrifices only 1 ton of Maize per ton of Steel (vs 2 tons in Country A), giving B comparative advantage in Steel.

PROBLEM 16.8 — FOLLOW-UP CALCULATION OF OPPORTUNITY COST OF OLIVES BETWEEN SPAIN AND GREECE

Follow-Up Calculation of Opportunity Cost of Olives Between Spain and Greece • Medium

Continuing with the two-country trade model where Spain can produce either 100 kg of olives or 150 kg of tobacco per unit of resources and Greece can produce either 300 kg of olives or 300 kg of tobacco (and where the first part established that 1 kg of tobacco costs 0.67 kg of olives in Spain versus 1.00 kg of olives in Greece, giving Spain the comparative advantage in tobacco): calculate each country's opportunity cost of producing 1 kg of olives and determine which country should specialise in and export olives.

Verified Solution: 1 kg of olives costs 1.50 kg of tobacco in Spain and 1.00 kg of tobacco in Greece, so Greece holds the comparative advantage in olives and exports olives.

$$\text{Prerequisite Part 1: Opp. Cost of 1 kg Tobacco} = \frac{100}{150} = 0.67 \text{ kg Olives in Spain vs } \frac{300}{300} = 1.00 \text{ kg Olives in Greece (Spain specialises in Tobacco)}$$
$$\text{Opp. Cost of 1 kg Olives in Spain} = \frac{150 \text{ kg Tobacco}}{100 \text{ kg Olives}} = 1.50 \text{ kg Tobacco}$$
$$\text{Opp. Cost of 1 kg Olives in Greece} = \frac{300 \text{ kg Tobacco}}{300 \text{ kg Olives}} = 1.00 \text{ kg Tobacco}$$
$$\text{Since } 1.00 < 1.50 \text{ kg Tobacco, Greece has the lower opportunity cost (comparative advantage) in Olives and exports Olives to Spain}$$

First-Principles Proof: Correct! Spain sacrifices 150/100 = 1.50 kg of tobacco per kg of olives while Greece sacrifices only 300/300 = 1.00 kg of tobacco, giving Greece the comparative advantage in olives.

PROBLEM 16.9 — FOLLOW-UP CALCULATION OF IMPORT COST IN DOMESTIC CURRENCY UNDER EXCHANGE RATE SHIFTS

Follow-Up Calculation of Import Cost in Domestic Currency Under Exchange Rate Shifts • Medium

Continuing with the trade model between Australia (150 surfboards or 900 bottles of sunscreen) and the USA (200 surfboards or 1 600 bottles of sunscreen)—where the first part established that Australia has the comparative advantage in surfboards (sacrificing 6 vs 8 bottles of sunscreen) while the USA has the comparative advantage in sunscreen (sacrificing 0.125 vs 0.167 surfboards): suppose Australia imports 50 bottles of sunscreen from the USA at a US price of US1.50$per bottle. Calculate (1) how much Australia pays in Australian dollars (AUS)$ when the exchange rate is US1 $= AUS$1.70, and (2) how much Australia pays if the Australian dollar appreciates to US1 $= AUS$1.50.

Verified Solution: The 50 bottles cost US$75.00, which equals AUS$127.50 at AUS1.70/US$ and falls to AUS $112.50 when the Australian dollar appreciates to AUS$1.50/US$.

$$\text{Foreign Currency Invoice for 50 Bottles} = 50 \times \text{US\$1.50} = \text{US\$75.00}$$
$$\text{Cost in AUS\$ at US\$1} = \text{AUS\$1.70}: \quad \text{US\$75.00} \times 1.70 = \text{AUS\$127.50}$$
$$\text{Cost in AUS\$ after Appreciation to US\$1} = \text{AUS\$1.50}: \quad \text{US\$75.00} \times 1.50 = \text{AUS\$112.50} \quad (\text{imports become cheaper by AUS\$15.00})$$

First-Principles Proof: Correct! 50 × US1.50 = US75.00; multiplying by 1.70 gives AUS127.50, and an appreciation of the AUS$ to 1.50 per US$ lowers the domestic import bill to AUS$112.50.

PROBLEM 16.10 — FOUR MACROECONOMIC DETERMINANTS OF DOMESTIC CURRENCY APPRECIATION AND TRADE BALANCE IMPACT

Four Macroeconomic Determinants of Domestic Currency Appreciation and Trade Balance Impact • Easy

In a floating exchange rate regime (such as the Rand against the US dollar or Russian rouble), which four fundamental macroeconomic developments cause an appreciation of the domestic currency (either by shifting foreign currency supply SS rightward or foreign currency demand DD leftward), and what is the resulting impact on the domestic trade balance?

Verified Solution: Higher export demand/commodity prices, higher relative domestic interest rates (capital inflows), inbound tourism, and lower domestic inflation; appreciation reduces net exports (X - Z).

$$\text{Rightward Shift of Foreign Currency Supply } (SS \uparrow \implies E \downarrow, \text{ Appreciation}):$$
$$(1) \text{ Stronger foreign demand for domestic exports (or rising export commodity prices)}; \quad (2) \text{ Higher domestic interest rates relative to abroad (attracting fi}$$
$$\text{Leftward Shift of Foreign Currency Demand } (DD \downarrow \implies E \downarrow, \text{ Appreciation}): (4) \text{ Lower domestic inflation relative to trading partners (reducing import d}$$
$$\text{Trade Balance Impact: Appreciation makes exports pricier in foreign currency } (X \downarrow) \text{ and imports cheaper in Rand } (Z \uparrow) \implies \text{Net Exports } (X - Z) \text{ declin}$$

First-Principles Proof: Correct! All four factors increase foreign currency inflows or reduce outflows (appreciating the Rand), which subsequently makes exports dearer abroad and imports cheaper locally.

PROBLEM 16.11 — ABSOLUTE VS COMPARATIVE ADVANTAGE & OPPORTUNITY COST RATIOS (LDC & DC WHEAT/CLOTH)

Absolute vs Comparative Advantage & Opportunity Cost Ratios (LDC & DC Wheat/Cloth) • Medium

In the Core Module trade model between a Less Developed Country (LDC) and a Developed Country (DC), one worker-day of labour can produce: in the LDC, either 2 kg of wheat OR 1 metre of cloth; in the DC, either 4 kg of wheat OR 8 metres of cloth. Determine: (1) Which country holds an absolute advantage in wheat and cloth; (2) The opportunity cost of 1 kg of wheat and 1 metre of cloth in each country; and (3) The pattern of specialisation based on comparative advantage.

Verified Solution: The DC has an absolute advantage in both goods; 1 kg wheat costs 0.5 m cloth in the LDC vs 2.0 m cloth in the DC; the LDC specialises in wheat and the DC specialises in cloth.

$$(1) \text{ Absolute Advantage: DC produces more wheat } (4 \text{ kg} > 2 \text{ kg}) \text{ AND more cloth } (8 \text{ m} > 1 \text{ m}) \text{ per worker-day} \implies \text{DC has absolute advantage in both}$$
$$(2) \text{ LDC Opportunity Costs: 1 kg wheat} = \frac{1}{2} = 0.5 \text{ m cloth}; \quad 1 \text{ m cloth} = \frac{2}{1} = 2 \text{ kg wheat}$$
$$(2) \text{ DC Opportunity Costs: 1 kg wheat} = \frac{8}{4} = 2.0 \text{ m cloth}; \quad 1 \text{ m cloth} = \frac{4}{8} = 0.5 \text{ kg wheat}$$
$$(3) \text{ Comparative Advantage: LDC sacrifices only 0.5 m cloth per kg of wheat } (< 2.0 \text{ m in DC}) \implies \text{LDC specialises in Wheat; DC specialises in Cloth } (0.5$$

First-Principles Proof: Correct! Even though the DC is more productive in both goods (4 kg > 2 kg and 8 m > 1 m), the LDC has a lower opportunity cost in wheat (0.5 m < 2.0 m cloth) and the DC has a lower opportunity cost in cloth (0.5 kg < 2.0 kg wheat).

PROBLEM 16.12 — EXCHANGE-RATE VALUATION & EXPORT/IMPORT SWITCHING ACROSS FIVE COMMODITIES (USA & UK)

Exchange-Rate Valuation & Export/Import Switching Across Five Commodities (USA & UK) • Hard

In Core Module, domestic production prices for five tradable commodities in the USA ()$and the UK (£) are: Commodity A$(1.60 vs £1.00), Commodity B ($2.00 vs £1.40), Commodity C$(2.50 vs £1.50), Commodity D ($0.40 vs £1.20), and Commodity E$(4.00 vs £2.00). Assuming zero transport costs, determine which commodities the USA will export to the UK and which the USA will import from the UK under three exchange rate regimes: (1) £1 = 1.00; (2) £1 =2.00 (dollar depreciation); and (3) £1 = $0.50 (dollar appreciation).

Verified Solution: At £1=1.00, USA exports Dand imports A, B, C, E; at £1 =2.00 (dollar depreciation), USA exports A, B, C, D (with E equal at 4.00) and imports none; at £1 =0.50, USA exports only D and imports A, B, C, E.

$$\text{Regime (1) } £1 = \$1.00 : \text{UK prices in \$ are A=\$1.00, B=\$1.40, C=\$1.50, D=\$1.20, E=\$2.00}$$
$$\implies \text{USA is cheaper only in D } (\$0.40 < \$1.20): \text{USA exports D and imports A, B, C, E}$$
$$\text{Regime (2) } £1 = \$2.00 : \text{UK prices in \$ double to A=\$2.00, B=\$2.80, C=\$3.00, D=\$2.40, E=\$4.00}$$
$$\implies \text{USA is cheaper in A } (\$1.60<\$2.00), \text{ B } (\$2.00<\$2.80), \text{ C } (\$2.50<\$3.00), \text{ D } (\$0.40<\$2.40), \text{ and equal in E } (\$4.00=\$4.00): \text{USA exports A, B, C, D}$$
$$\text{Regime (3) } £1 = \$0.50 : \text{UK prices in \$ halve to A=\$0.50, B=\$0.70, C=\$0.75, D=\$0.60, E=\$1.00} \implies \text{USA exports D } (\$0.40<\$0.60) \text{ and imports A, B, C}$$

First-Principles Proof: Correct! Converting UK pound prices into US dollars shows that dollar depreciation (£1 = 2.00) makes US goods A, B, C, and D cheaper than UK equivalents (and E equal at 4.00), boosting US exports.

CHAPTER 17 Simple Macroeconomic Model • Simple Keynesian Goods Market • 10 Hours

Chapter 17: A Simple Keynesian Model of the Economy

Core assumptions of the simple closed no-government Keynesian model, Say's Law vs Keynes's Law, the consumption function (APC vs MPC) and its three characteristics, saving function (APS vs MPS), autonomous investment, the 45° line, equilibrium income, and the simple multiplier.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Simple Macroeconomic Model • 10 Hours):

- State the restricting assumptions of the Simple Keynesian Model (closed two-sector economy with only households and firms, fixed prices and wages, constant interest rates, demand-determined output).
- Describe, formulate, and graph the consumption function (C = $\bar{C}$ + cY) and calculate the Average Propensity to Consume (APC = C/Y), Marginal Propensity to Consume (MPC = ΔC/ΔY), Average Propensity to Save (APS = S/Y), and Marginal Propensity to Save (MPS = ΔS/ΔY).

- Define and illustrate the autonomous investment function and explain its inverse relationship with the interest rate.
- Determine equilibrium income algebraically, tabularly, and graphically using the 45° line (A = Y) and saving–investment equality (S = I).
- Formulate, calculate, and graphically illustrate the simple expenditure multiplier (α = 1/(1 - c) = 1/s) and the Paradox of Thrift.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (4 IDENTITIES)

EQ 17.1 Consumption & Saving Functions

$$C = \bar{C} + cY, \quad S = -\bar{C} + (1-c)Y, \quad c + s = 1$$

Meaning: Autonomous consumption C_bar (>0) is independent of income; induced consumption cY depends on the marginal propensity to consume c (0 < c < 1).

EQ 17.2 Average vs. Marginal Propensities (APC, APS, MPC, MPS)

$$\text{APC} = \frac{C}{Y}, \quad \text{APS} = \frac{S}{Y}, \quad \text{MPC} = \frac{\Delta C}{\Delta Y} = c, \quad \text{MPS} = \frac{\Delta S}{\Delta Y} = s, \quad \text{APC}$$

Meaning: APC and APS measure the fraction of total income consumed or saved, whereas MPC (slope c) and MPS (slope s) measure the fraction of an additional Rand consumed or saved.

EQ 17.3 Simple Closed-Economy Equilibrium

$$Y_0 = \alpha \bar{A} = \left(\frac{1}{1-c}\right)(\bar{C} + \bar{I})$$

Meaning: Equilibrium occurs along the 45° line where Aggregate Spending A = C + I equals Total Production/Income Y (and Planned Saving S = I).

EQ 17.4 The Simple Multiplier Process

$$\alpha = \frac{\Delta Y}{\Delta \bar{A}} = \frac{1}{1-c} = \frac{1}{s} > 1$$

Meaning: An autonomous injection (e.g. ΔI) triggers a geometric chain of induced consumption rounds, expanding equilibrium income by a multiple of the initial spending change.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 17.1 Assumptions of the Simple Keynesian Model & Say's Law (Core Macroeconomic Concept)

The simple model assumes: (1) The economy consists only of households and firms (closed, no government), (2) Prices and wages are fixed, (3) The money market and interest rates are constant, and (4) Aggregate spending (demand) determines aggregate production (income). Unlike Classical Say's Law ('supply creates its own demand', Y → A), Keynes showed that aggregate spending drives output (A → Y) and equilibrium can persist below full employment.

> **Worked Proof:** At any output level below equilibrium (to the left of the 45° intersection), Aggregate Spending exceeds Production (A > Y), causing unplanned inventory depletion and prompting firms to expand output.

CONCEPT 17.2 Three Characteristics of the Consumption Function & APC vs. MPC (Core Macroeconomic Concept)

Keynes's consumption function C = C_bar + cY has three fundamental properties: (1) Consumption is strictly positive even when income is zero (C_bar > 0, financed by dissaving/past wealth), (2) Consumption increases as income increases (positive slope, c > 0), and (3) When income increases, consumption increases by LESS than the increase in income (0 < c < 1, because households save part of every extra Rand).

> **Worked Proof:** In Core Module (pp. 62–64): For the schedule Y = [40, 60, 80, 100, 120, 140] and C = [47, 58, 69, 80, 91, 102], every ΔY = 20 yields ΔC = 11, so MPC = 11/20 = 0.55, MPS = 0.45, and C̄ = 25 (C = 25 + 0.55Y). At Y = 100, C = 80 so APC = 0.80 and APS = 0.20.

CONCEPT 17.3 The Multiplier, Island Economy Equilibrium & Saving Identity (Core Macroeconomic Concept)

When investment increases by ΔI, total spending jumps vertically by ΔI. Because each Rand spent becomes income for someone else who spends a fraction c of it, total income expands by ΔY = [1/(1-c)] ΔI. In a closed two-sector economy, households either consume or save their income (Y = C + S), so at equilibrium (Y = C + I) planned saving must equal investment (S = I).

> **Worked Proof:** In Core Module's island economy (pp. 68–70) with C = 250 + 0.8Y and I = 400: Ye = (250 + 400)/(1 - 0.8) = 650/0.2 = 3 250; equilibrium consumption C = 250 + 0.8(3 250) = 2 850; and household saving S = Ye - C = 3 250 - 2 850 = 400 = I.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Assumptions of the Simple Keynesian Model & Say's Law): Believing that Keynesian equilibrium (A = Y) automatically guarantees full employment of all labour.

First-Principles Law: Macroeconomic equilibrium in the Keynesian model simply means planned spending equals production (A = Y); this equilibrium income Y0 can be far below full-employment output Yf.

Fallacy (Three Characteristics of the Consumption Function & APC vs. MPC): Confusing a movement along the consumption curve (caused by a change in income Y) with a vertical shift of the consumption curve, or using APC instead of MPC in the multiplier.

First-Principles Law: Income Y is on the horizontal axis, so ΔY causes a movement along C = C_bar + cY (induced consumption), whereas a change in autonomous consumption ΔC_bar shifts the entire intercept vertically.

Fallacy (The Multiplier, Island Economy Equilibrium & Saving Identity): Thinking that higher household thrift (an autonomous rise in the saving rate) always increases total national saving in the simple Keynesian model.

First-Principles Law: By the Paradox of Thrift (Box 17-9), an autonomous attempt to save more reduces C_bar and lowers the multiplier, contracting equilibrium income Y0 and leaving total saving unchanged at S = I.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (8 PROBLEMS)

PROBLEM 17.1 — CORE ASSUMPTIONS & EQUILIBRIUM LAWS OF THE SIMPLE KEYNESIAN MODEL

Macroeconomics, §17.1–17.6, pp. 348–361 • Easy

Which one of the following statements accurately describes the simple Keynesian model of income determination and its relationship to the 45-degree line?

Verified Solution: Any point on the aggregate spending line above the 45° line means spending exceeds production (A > Y).

$$\text{Keynes's Law: } A \to Y. \quad \text{Above } 45^\circ \text{ line } (Y < Y_0) : A > Y \implies \Delta\text{Inventories} < 0 \implies Y \uparrow.$$

First-Principles Proof: Correct! Above the 45° line (to the left of equilibrium), aggregate spending A exceeds output Y, causing unplanned inventory depletion and an expansion in output.

PROBLEM 17.2 — CLOSED-ECONOMY MULTIPLIER, EQUILIBRIUM INCOME & INVESTMENT INJECTION

Macroeconomics, §17.8, pp. 366–371 • Medium

In a simple closed Keynesian economy without a government, autonomous consumption is R120 million, the marginal propensity to consume (c) is 0.75, and autonomous investment is R80 million. If autonomous investment increases by R40 million to R120 million, how do the equilibrium level of income and the relative sizes of ΔY and ΔI compare?

Verified Solution: Equilibrium income rises by R160m (from R800m to R960m), so ΔY is four times larger than ΔI.

$$\alpha = \frac{1}{1-0.75} = 4. \quad Y_0 = 4(120+80) = \text{R800m}. \quad \Delta Y = 4 \times 40 = \text{R160m} \implies Y_1 = \text{R960m}, \ \Delta Y > \Delta I.$$

First-Principles Proof: Correct! With c = 0.75, the multiplier is α = 1/(1 - 0.75) = 4, so a R40m rise in investment triggers induced consumption rounds that raise equilibrium income by R160m.

PROBLEM 17.3 — DERIVING THE CONSUMPTION FUNCTION & AUTONOMOUS SPENDING FROM SAVING

Macroeconomics, §17.4–17.7, pp. 361–376 • Medium

In a simple closed economy without a government, the saving function is given by S = -120 + 0.25Y and autonomous investment is I = 80. Determine the consumption function C(Y), the simple multiplier (α), equilibrium income (Y0), and what happens if households attempt to save more by shifting autonomous saving upward (the Paradox of Thrift).

Verified Solution: C = 120 + 0.75Y, α = 4.0, Y0 = 800; higher autonomous saving reduces equilibrium income while leaving total S = I = 80.

$$S = -\bar{C} + (1-c)Y = -120 + 0.25Y \implies \bar{C} = 120, \ c = 0.75. \quad \alpha = \frac{1}{0.25} = 4.0. \quad Y_0 = 4(120+80) = 800.$$

First-Principles Proof: Correct! Since S = -$\bar{C}$ + (1-c)Y, C = 120 + 0.75Y, α = 1/0.25 = 4, Y0 = 4(200) = 800, and the Paradox of Thrift lowers Y0 without raising equilibrium saving (which remains equal to I = 80).

PROBLEM 17.4 — CLASSICAL SAY'S LAW VS KEYNESIAN EFFECTIVE DEMAND PRINCIPLE

Macroeconomics, §17.1–17.2, pp. 354–357 • Easy

Compare Classical (Say's Law) theory and Keynesian theory regarding the relationship between total production (Y) and aggregate spending (A): which statement accurately contrasts the two frameworks and interprets points off the 45-degree line?

Verified Solution: Say's Law holds that supply creates demand (Y → A), whereas Keynes holds that aggregate spending drives output (A → Y).

$$\text{Say's Law: } Y \implies A \text{ (supply creates its own demand)}. \quad \text{Keynes: } A \implies Y \text{ (spending determines production)}.$$

First-Principles Proof: Correct! Classical Say's Law states that production automatically generates sufficient demand, whereas in the Keynesian model aggregate spending determines national production.

PROBLEM 17.5 — CHARACTERISTICS OF THE CONSUMPTION FUNCTION, INCOME VS DISPOSABLE INCOME, AND INJECTIONS/LEAKAGES

Characteristics of the Consumption Function, Income vs Disposable Income, and Injections/Leakages • Easy

In the Keynesian macroeconomic model: (1) What are the three fundamental characteristics of the household consumption function $C = \bar{C} + cY_d$? (2) How does total National Income (Y) differ from Disposable Income (Y_d)? (3) How do Investment (I) and Saving (S) classify as injections or leakages, and what are their effects on the multiplier α and equilibrium income Y_e?

Verified Solution: (1) C > 0 at Yd = 0, C rises with Yd, and 0 < c < 1; (2) Disposable income is after-tax income (Yd = Y - T); (3) Investment is an injection raising Ye, while Saving is a leakage.

$$\text{Three Characteristics: } (1)\ C > 0 \text{ even when } Y_d = 0 \text{ (autonomous consumption } \bar{C} > 0);\ (2)\ C \text{ increases as } Y_d \text{ increases } (c > 0);\ (3)\ \Delta C < \Delta Y_d \text{ because } 0 < Y \text{ vs } Y_d : Y \text{ is total factor income earned across the economy, whereas } Y_d = Y - T = (1-t)Y \text{ is after-tax income available to households for consumption } c$$
$$I \text{ vs } S : I \text{ is an autonomous injection (raises } Y_e \text{ via } \Delta A, \text{ leaves } \alpha \text{ unchanged}); \quad S \text{ is a leakage (higher saving propensity } s = 1 - c \text{ lowers } c, \text{ decreasing both}$$

First-Principles Proof: Correct! These capture the three Keynesian consumption axioms, the tax wedge between Y and Yd, and the exact roles of I (autonomous injection) and S (income-induced leakage).

PROBLEM 17.6 — SIMPLIFYING ASSUMPTIONS OF THE TWO-SECTOR MODEL VS THE FOUR AGGREGATE EXPENDITURE COMPONENTS

Simplifying Assumptions of the Two-Sector Model vs the Four Aggregate Expenditure Components • Easy

In the build-up of the macroeconomic income-determination model: (1) What are the fundamental simplifying assumptions of the elementary two-sector Keynesian model? (2) When the model is expanded to the full open economy with government, what are the four components of Aggregate Expenditure (A) and which of those components are treated as purely autonomous versus induced by income?

Verified Solution: Elementary assumptions: only households/firms (no G, T, X, Z), fixed prices/wages, given interest rates, and demand-driven output; Full components: C, I, G, and Net Exports (X - Z).

$$\text{Elementary Two-Sector Assumptions: (1) Only households and firms exist (no government } G = T = 0,\ \text{no foreign sector } X = Z = 0);$$
$$\text{(2) Prices and nominal wages are fixed/given (no inflation); (3) Money market and interest rates are excluded (investment } \bar{I} \text{ is autonomous); (4) Aggregate}$$
$$\text{Full Four-Sector Expenditure: } A = C + I + G + (X - Z),\ \text{where } \bar{I}, \bar{G}, \bar{X} \text{ are autonomous while } C = \bar{C} + c(1-t)Y \text{ and } Z = \bar{Z} + mY \text{ contain induced part}$$

First-Principles Proof: Correct! The elementary model excludes the state, foreign trade, price flexibility, and financial markets to isolate C + I, before expanding to A = C + I + G + (X - Z).

PROBLEM 17.7 — EXTRACTING MPC, MPS, AUTONOMOUS CONSUMPTION, APC & APS FROM A TABULAR SCHEDULE

Extracting MPC, MPS, Autonomous Consumption, APC & APS from a Tabular Schedule • Medium

A macroeconomic household survey in macroeconomic analysis: records the following linear relationship between disposable income (Y) and total consumption expenditure (C): at Y = 40, C = 47; at Y = 60, C = 58; at Y = 80, C = 69; at Y = 100, C = 80; at Y = 120, C = 91; and at Y = 140, C = 102. Calculate: (1) The Marginal Propensity to Consume (MPC) and Marginal Propensity to Save (MPS); (2) Autonomous consumption ($\bar{C}$) and the full consumption and saving equations; and (3) The Average Propensity to Consume (APC) and Average Propensity to Save (APS) at Y = 100.

Verified Solution: (1) MPC = 0.55 and MPS = 0.45; (2) Autonomous consumption $\bar{C}$ = 25, giving C = 25 + 0.55Y and S = -25 + 0.45Y; (3) At Y = 100, APC = 0.80 and APS = 0.20.

$$(1)\ \text{Marginal Propensity to Consume: } c = \text{MPC} = \frac{\Delta C}{\Delta Y} = \frac{58-47}{60-40} = \frac{11}{20} = 0.55; \quad s = \text{MPS} = 1 - 0.55 = 0.45$$
$$(2)\ \text{Autonomous Consumption: } C = \bar{C} + 0.55Y \implies 47 = \bar{C} + 0.55(40) = \bar{C} + 22 \implies \bar{C} = 25$$
$$\text{Consumption and Saving Functions: } C = 25 + 0.55Y \quad \text{and} \quad S = -25 + 0.45Y$$
$$(3)\ \text{At } Y = 100\ (\text{where } C = 80, S = 20)\text{: APC} = \frac{C}{Y} = \frac{80}{100} = 0.80 \quad \text{and} \quad \text{APS} = \frac{S}{Y} = \frac{20}{100} = 0.20 \quad (\text{APC} + \text{APS} = 1.00)$$

First-Principles Proof: Correct! For every ΔY = 20, consumption rises by ΔC = 11 (MPC = 11/20 = 0.55, MPS = 0.45); solving 47 = $\bar{C}$ + 0.55(40) yields $\bar{C}$ = 25, and at Y = 100, APC = 80/100 = 0.80 and APS = 20/100 = 0.20.

PROBLEM 17.8 — TWO-SECTOR KEYNESIAN EQUILIBRIUM, MULTIPLIER & SAVING-INVESTMENT IDENTITY (ISLAND ECONOMY)

Two-Sector Keynesian Equilibrium, Multiplier & Saving-Investment Identity (Island Economy) • Easy

In the Core Module island economy without a government or foreign sector, consumption is given by C = 250 + 0.8Y and autonomous private investment is I = 400. Calculate: (1) Total autonomous spending (Ā) and the expenditure multiplier (α); (2) The equilibrium level of national income (Ye) and equilibrium consumption (Ce); and (3) The equilibrium level of household saving (Se), verifying the saving-investment equilibrium condition.

Verified Solution: (1) Ā = 650 and α = 5; (2) Equilibrium income Ye = 3 250 and consumption Ce = 2 850; (3) Equilibrium saving Se = -250 + 0.2(3 250) = 400, confirming Se = I = 400.

$$(1)\ \text{Autonomous Spending: } \bar{A} = \bar{C} + \bar{I} = 250 + 400 = 650; \quad \text{Multiplier: } \alpha = \frac{1}{1-c} = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$$
$$(2)\ \text{Equilibrium Income: } Y_e = \alpha \times \bar{A} = 5 \times 650 = 3\,250; \quad \text{Equilibrium Consumption: } C_e = 250 + 0.8(3\,250) = 250 + 2\,600 = 2\,850$$
$$(3)\ \text{Saving Function: } S = -\bar{C} + (1-c)Y = -250 + 0.2(3\,250) = -250 + 650 = 400 \equiv I \quad (\text{and } Y_e - C_e = 3\,250 - 2\,850 = 400)$$

First-Principles Proof: Correct! With Ā = 250 + 400 = 650 and α = 1/(1 - 0.8) = 5, equilibrium income is Ye = 3 250, consumption is Ce = 2 850, and household saving is Se = 3 250 - 2 850 = 400 = I.

CHAPTER 18 Open Economy with Government • Complete Keynesian Model (Government & Foreign) • 10 Hours

Chapter 18: The Keynesian Model with a Government and a Foreign Sector

Introducing government spending (G) and lump-sum or proportional taxes (T = $\bar{T}$ + tY), disposable income, open-economy exports (X) and induced imports (Z = Z_bar + mY), leakages vs injections, order of operations in the open-economy multiplier, and fiscal policy adjustments.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Open Economy with Government • 10 Hours):

- Explain how autonomous government expenditure (G) and lump-sum or proportional income taxes (T = $\bar{T}$ + tY) alter disposable income (Yd = Y - T), the consumption function, the multiplier, and equilibrium income.
- Apply the strict order of operations when calculating the tax-adjusted multiplier α = 1/[1 - c(1 - t)], computing (1 - t) first, multiplying by c, and subtracting from 1.
- Compare the quantitative impact on equilibrium income of an autonomous increase in government spending (ΔG) versus an equal-sized lump-sum tax cut (-ΔT).

- Incorporate autonomous exports (X) and autonomous plus induced imports (Z = Z̄ + mY) into the open-economy Keynesian model (A = C + I + G + X - Z) and calculate the open-economy multiplier α = 1/[1 - c(1 - t) + m].
- Use the multiplier to calculate the exact fiscal policy change (ΔG) required to eliminate a recessionary output gap (Yf - Ye).

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (4 IDENTITIES)

EQ 18.1 Consumption with Lump-Sum & Proportional Taxes

$$T = \bar{T} + tY, \quad Y_d = (1-t)Y - \bar{T} \implies C = \bar{C} - c\bar{T} + c(1-t)Y$$

Meaning: Lump-sum taxes T̄ reduce the autonomous intercept by cT̄, while proportional taxes t reduce disposable income Yd and flatten the consumption slope from c to c(1 - t).

EQ 18.2 Multiplier with Government Sector (Closed)

$$\alpha_G = \frac{1}{1 - c(1-t)}$$

Meaning: Order of operations: calculate (1 - t) first, multiply by c, and subtract from 1. Never evaluate as (1 - c)(1 - t).

EQ 18.3 Open-Economy Multiplier (with Induced Imports)

$$\alpha_{\text{open}} = \frac{1}{1 - c(1-t) + m}, \quad \bar{A} = \bar{C} - c\bar{T} + \bar{I} + \bar{G} + \bar{X} - \bar{Z}$$

Meaning: Both proportional taxes (t) and the marginal propensity to import (m) act as leakages from the domestic spending stream, reducing the size of the multiplier.

EQ 18.4 Fiscal Policy Gap to Full Employment

$$\Delta\bar{G} = \frac{Y_f - Y_0}{\alpha}$$

Meaning: To close a recessionary output gap (Yf - Y0), government spending must increase by the income gap divided by the multiplier.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 18.1 How G and t Affect the Aggregate Spending Curve & Order of Operations (Core Macroeconomic Concept)

Government spending G is an autonomous injection: it shifts the Aggregate Spending curve upward in parallel by ΔG without changing its slope or the multiplier. Proportional taxes T = tY are an induced leakage: an increase in the tax rate t rotates (pivots) the Aggregate Spending curve downward, reducing its slope c(1-t) and shrinking the multiplier. Furthermore, a R5 000 increase in G raises equilibrium income by more than a R5 000 tax cut because households save part of a tax cut.

> **Worked Proof:** In Core Module (pp. 77–79): Given C = 250 + 0.85Y, I = 150, G = 350 (t = 0): Ā = 750, α = 1/(1 - 0.85) = 6.67, and Ye = 5 000. If ΔG = +100, ΔY = 6.67 × 100 = 667, raising Y to 5 667.

CONCEPT 18.2 Exports, Induced Imports & the Islamabad Open-Economy Model (Core Macroeconomic Concept)

Exports X are an autonomous injection (+X_bar). Imports Z = Z_bar + mY consist of autonomous imports (Z_bar, subtracted from A_bar) and induced imports (mY, which rise as domestic income rises, reducing the slope of net spending to c(1-t) - m).

> **Worked Proof:** In Core Module's Islamabad economy (pp. 86–89): C = 150 + 0.8Yd, T = 20 + 0.2Y, I = 250, G = 450, X = 325, Z = 50 + 0.2Y. Autonomous spending Ā = 150 - 0.8(20) + 250 + 450 + 325 - 50 = 1 109. Induced slope = 0.8(1 - 0.2) - 0.2 = 0.44, so 1 - 0.44 = 0.56, α = 1/0.56 = 1.79, and Ye = 1 109 / 0.56 = 1 980. When investment rises from 250 to 310 (ΔI = 60), ΔY = 60 × 1.79 = 107.4, giving Y_new = 2 087 (or exact 2 087.5).

CONCEPT 18.3 Injections vs. Leakages Table (Core Synthesis) (Core Macroeconomic Principle)

In the open circular flow with government: Investment (I), Government Spending (G), and Exports (X) are INJECTIONS that increase equilibrium income Y0 without changing α. Saving (s), Taxation (t), and Imports (m) are LEAKAGES (withdrawals): an increase in s, t, or m decreases the multiplier α and decreases equilibrium income Y0.

> **Worked Proof:** Variable G is an Injection, α remains unchanged, and Ye increases; Variable t is a Leakage, α decreases, and Ye decreases.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (How G and t Affect the Aggregate Spending Curve & Order of Operations): Evaluating 1 - c(1 - t) as (1 - c)(1 - t), or claiming that an increase in G increases the size of the multiplier α.

First-Principles Law: In α = 1/[1 - c(1-t) + m], 1 - c is NOT in brackets: compute (1 - t) first, multiply by c, and subtract from 1. Autonomous G changes the intercept A_bar, never α.

Fallacy (Exports, Induced Imports & the Islamabad Open-Economy Model): Adding autonomous imports Z_bar (+50) into autonomous spending A_bar instead of subtracting it (-50).

First-Principles Law: Imports represent domestic spending leaking to foreign producers (X - Z = X_bar - Z_bar - mY), so autonomous imports Z_bar must always be subtracted from A_bar.

Fallacy (Injections vs. Leakages Table (Core Synthesis)): Claiming that an increase in the marginal propensity to consume (c) reduces the multiplier.

First-Principles Law: Because households re-spend a larger fraction of income domestically when c rises, α = 1/[1 - c(1-t) + m] increases and equilibrium income Y0 increases.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (13 PROBLEMS)

PROBLEM 18.1 — OPEN-ECONOMY KEYNESIAN MODEL: MULTIPLIER, EQUILIBRIUM & FISCAL GAP (PORTOFINO)

Macroeconomics, §18.2–18.3, pp. 387–393 • Hard

Suppose the following equations describe the open economy of Portofino, Italy: Autonomous consumption = 100, MPC (c) = 0.8, proportional tax rate (t) = 0.2 (so T = 0.2Y), Investment (I) = 100, Government spending (G) = 250, Exports (X) = 300, and Imports Z = 20 + 0.1Y. Calculate the open-economy multiplier (rounded to 2 decimal places) and the equilibrium level of income.

Verified Solution: The multiplier is 2.17 and equilibrium income is 1 584.10 (or 1 586.96 unrounded) with Ā = 730.

$$\bar{A} = 100 + 100 + 250 + 300 - 20 = 730.$$
$$\alpha = \frac{1}{1 - c(1-t) + m} = \frac{1}{1 - 0.8(1-0.2) + 0.1} = \frac{1}{1 - 0.64 + 0.1} = \frac{1}{0.46} \approx 2.17.$$
$$Y_0 = 2.17 \times 730 = 1\,584.10 \text{ (or } 730/0.46 = 1\,586.96).$$

First-Principles Proof: Correct! Autonomous spending is 100 + 100 + 250 + 300 - 20 = 730, and α = 1/[1 - 0.8(0.8) + 0.1] = 1/0.46 = 2.17, giving Y0 = 1 584.10 (or 1 586.96 exact).

PROBLEM 18.2 — OPEN-ECONOMY MULTIPLIER WITH PROPORTIONAL TAXES & INDUCED IMPORTS

Macroeconomics, §18.2–18.4, pp. 387–396 • Hard

Consider an open economy with government where: Government expenditure (G) = R400, Exports (X) = R250, Imports (Z) = 50 + 0.15Y, Autonomous consumption (C̄) = R150, Investment (I) = R300, Full-employment output (Yf) = R1 900, MPC (c) = 0.75, and Tax rate (t) = 0.25. Calculate the multiplier (rounded to 2 decimals), equilibrium income, and the new equilibrium income if investment falls to R250.

Verified Solution: Multiplier is 1.70, equilibrium income is R1 785 (R1 787.23 exact), falling to R1 700 when I = 250.

$$\bar{A} = 150 + 300 + 400 + 250 - 50 = 1\,050.$$
$$\alpha = \frac{1}{1 - 0.75(1-0.25) + 0.15} = \frac{1}{1 - 0.5625 + 0.15} = \frac{1}{0.5875} \approx 1.70.$$
$$Y_0 = 1.70 \times 1\,050 = 1\,785 \text{ (exact } 1\,787.23). \quad \Delta I = -50 \implies \Delta Y = -85 \implies Y_{\text{new}} = 1\,700.$$

First-Principles Proof: Correct! α = 1/0.5875 = 1.70 and Ā = 1 050, so Y0 = 1.70 × 1 050 = R1 785 (or R1 787.23 exact); a R50 drop in I reduces Y by 50 × 1.70 = R85 to R1 700.

PROBLEM 18.3 — FISCAL TRANSFER SHOCK, TAX RATE ADJUSTMENT & NEW EQUILIBRIUM OUTPUT

Macroeconomics, §18.4, pp. 393–398 • Hard

In the South African Keynesian model (in R millions): C̄ = 30, Ī = 10, Ḡ = 20, X̄ = 10, Z̄ = 5, MPC (c) = 0.6, average tax rate (t) = 0.167, and MPM (m) = 0.125. Initial equilibrium income is R104m (with α = 1.60 and Ā = 65). Suppose the government introduces a basic income grant that raises G by R4m (to R24m) and finances it by raising the tax rate t to 0.25. What are the new equilibrium income and tax revenue?

Verified Solution: New equilibrium income is R102.22 million and government tax revenue is R25.56 million.

$$\bar{A}_{\text{new}} = 30 + 10 + 24 + 10 - 5 = 69. \quad c(1 - t_{\text{new}}) = 0.6(0.75) = 0.45.$$
$$Y_{\text{new}} = \frac{69}{1 - 0.45 + 0.125} = \frac{69}{0.675} = \text{R}102.22\text{m}. \quad T_{\text{new}} = 0.25 \times 102.22 = \text{R}25.56\text{m}.$$

First-Principles Proof: Correct! With Ā_new = 69 and the new denominator 1 - 0.6(0.75) + 0.125 = 0.675, Y_new = 69 / 0.675 = R102.22m and T = 0.25 × 102.22 = R25.56m.

PROBLEM 18.4 — IMPORTS AS A FUNCTION OF DISPOSABLE INCOME (Z = Z̄ + MYD) & BUDGET BALANCE

Macroeconomics, §18.2–18.3, pp. 387–393 • Hard

You are given the following model for the Italian economy: Consumption C = 85 + 0.5Yd; Investment I = 75; Government expenditure G = 70; Taxes T = 0.25Y; Disposable income Yd = Y - T; Exports X = 30; Imports Z = 30 + 0.3Yd. Calculate equilibrium income (Y), taxes collected at equilibrium (T), the government budget balance (T - G), and household savings (S).

Verified Solution: Equilibrium Y is 270.59, Taxes are 67.65, Budget Deficit is -2.35, and Savings equal 16.47.

$$\bar{A} = 85 + 75 + 70 + 30 - 30 = 230. \quad Y = 230 + (0.5 - 0.3)(1 - 0.25)Y = 230 + 0.15Y \implies Y = \frac{230}{0.85} = 270.59.$$
$$T = 0.25(270.59) = 67.65; \quad \text{Budget} = 67.65 - 70 = -2.35; \quad S = -85 + 0.5(0.75 \times 270.59) = 16.47.$$

First-Principles Proof: Correct! With Ā = 230 and induced slope (0.5 - 0.3)(0.75) = 0.15, Y = 230 / 0.85 = 270.59, T = 67.65, Budget = 67.65 - 70 = -2.35 (deficit), and S = 16.47.

PROBLEM 18.5 — DIRECT AUTONOMOUS GOVERNMENT SPENDING INJECTION VS TAX-RATE MULTIPLIER EFFECT

Macroeconomics, §18.1, pp. 381–387 • Easy

In the Keynesian model with a government sector, compare how an increase in government spending (G) versus an increase in the proportional tax rate (t) affects (i) the circular flow (injection vs leakage), (ii) the size of the multiplier (α), and (iii) the equilibrium level of income (Ye).

Verified Solution: G is an injection that leaves α unchanged and raises Ye; t is a leakage that reduces α (flattens A) and lowers Ye.

$$G: \text{Autonomous Injection } (\alpha \text{ unchanged}, \ Y_e \uparrow). \quad t: \text{Induced Leakage } \left(\alpha = \frac{1}{1 - c(1-t) + m} \downarrow, \ Y_e \downarrow\right).$$

First-Principles Proof: Correct! Government spending G is purely autonomous (shifting A parallel upward without changing α), whereas the tax rate t reduces the slope c(1-t) and lowers both α and Ye.

PROBLEM 18.6 — CALCULATING AUTONOMOUS SPENDING, OPEN-ECONOMY MULTIPLIER, AND EQUILIBRIUM INCOME

Calculating Autonomous Spending, Open-Economy Multiplier, and Equilibrium Income • Hard

Consider an open economy with a government sector described by the following structural parameters: Autonomous consumption $\bar{C} = 400$, Marginal propensity to consume $c = 0.6$, Proportional income tax rate $t = 0.25$, Autonomous investment $\bar{I} = 400$, Government expenditure $\bar{G} = 300$, Exports $\bar{X} = 200$, and Imports $Z = 100 + 0.15Y$. Calculate (1) total autonomous expenditure $\bar{A}$, (2) the open-economy multiplier α, and (3) the equilibrium level of national income Y_e.

Verified Solution: Autonomous spending Ā is 1 200, the open-economy multiplier α is 1.43 (1/0.70), and equilibrium income Ye is 1 714.29 (or 1 716.00).

$$\text{Autonomous Expenditure: } \bar{A} = \bar{C} + \bar{I} + \bar{G} + \bar{X} - \bar{Z} = 400 + 400 + 300 + 200 - 100 = 1\,200$$
$$\text{Induced Slope: } c(1-t) - m = 0.6(1-0.25) - 0.15 = 0.6(0.75) - 0.15 = 0.45 - 0.15 = 0.30$$
$$\text{Open-Economy Multiplier: } \alpha = \frac{1}{1 - c(1-t) + m} = \frac{1}{1 - 0.45 + 0.15} = \frac{1}{0.70} \approx 1.43$$
$$\text{Equilibrium Income: } Y_e = \frac{1\,200}{0.70} = 1\,714.29 \quad (\text{or } 1.43 \times 1\,200 = 1\,716.00 \text{ using rounded } \alpha)$$

First-Principles Proof: Correct! Subtracting autonomous imports (100) gives Ā = 1 200, while 1 - 0.6(0.75) + 0.15 = 0.70 gives α = 1.43 and Ye = 1 200 / 0.70 = 1 714.29.

PROBLEM 18.7 — FOLLOW-UP CALCULATION OF EQUILIBRIUM INCOME AND ΔI = +150 SHOCK USING PREREQUISITE MULTIPLIER

Follow-Up Calculation of Equilibrium Income and ΔI = +150 Shock Using Prerequisite Multiplier • Hard

Continuing with the open economy of Portofino where Autonomous consumption $\bar{C} = 100$, MPC $c = 0.8$, proportional tax rate $t = 0.2$, Investment $\bar{I} = 100$, Government spending $\bar{G} = 250$, Exports $\bar{X} = 300$, and Imports $Z = 20 + 0.1Y$: the first part of the analysis established that autonomous expenditure is $\bar{A} = 730$ and the open-economy multiplier (rounded to two decimal places) is $\alpha = 2.17$ (unrounded $1/0.46 \approx 2.1739$). Using this prerequisite multiplier, calculate (1) the initial equilibrium level of income Y_0, (2) the change in equilibrium income ΔY and new equilibrium income Y_1 if autonomous investment increases by $\Delta I = +150$ (from 100 to 250), and (3) how this shock appears on the 45° Keynesian diagram.

Verified Solution: Initial Y0 = 1 584.10 (1 586.96 exact); ΔI = +150 raises Y by +325.50 to Y1 = 1 909.60 (1 913.04 exact) via a parallel upward shift of A from 730 to 880.

$$\text{Prerequisite Baseline: } \bar{A}_0 = 100 + 100 + 250 + 300 - 20 = 730; \quad \alpha = \frac{1}{1 - 0.8(0.8) + 0.1} = \frac{1}{0.46} \approx 2.17$$
$$\text{Initial Equilibrium Income: } Y_0 = \alpha \times \bar{A}_0 = 2.17 \times 730 = 1\,584.10 \quad \left(\text{or } \frac{730}{0.46} = 1\,586.96 \text{ unrounded}\right)$$
$$\text{Investment Shock } (\Delta I = +150): \quad \Delta Y = 2.17 \times 150 = +325.50 \implies Y_1 = 1\,584.10 + 325.50 = 1\,909.60 \quad \left(\text{or } \frac{880}{0.46} = 1\,913.04\right)$$
$$\text{Diagrammatic Effect: Parallel upward shift of the Aggregate Expenditure line } A \text{ from vertical intercept } 730 \text{ to } 880 \text{ (slope } 0.54 \text{ unchanged)}$$

First-Principles Proof: Correct! Multiplying Ā0 = 730 by α = 2.17 yields Y0 = 1 584.10, and the +150 investment shock shifts the A curve vertically upward to 880, adding 2.17 × 150 = +325.50 to reach 1 909.60.

PROBLEM 18.8 — FOLLOW-UP CALCULATION OF TAX REVENUE, BUDGET SURPLUS, AND INVESTMENT DROP USING PREREQUISITE EQUILIBRIUM

Follow-Up Calculation of Tax Revenue, Budget Surplus, and Investment Drop Using Prerequisite Equilibrium • Hard

In the open economy where $\bar{C} = \text{R}150$, $\bar{I} = \text{R}300$, $\bar{G} = \text{R}400$, $\bar{X} = \text{R}250$, $Z = 50 + 0.15Y$, $c = 0.75$, $t = 0.25$, and full-employment income $Y_f = \text{R}1\,900$: the first part of the analysis established that autonomous spending is $\bar{A} = \text{R}1\,050$, the rounded multiplier is $\alpha = 1.70$ (unrounded $1/0.5875 \approx 1.7021$), and initial equilibrium income is $Y_0 = \text{R}1\,785$ (unrounded R1 787.23). Using this prerequisite equilibrium Y_0, calculate (1) government tax revenue $T = tY_0$ and the government budget balance $(T - G)$ at initial equilibrium, (2) the new equilibrium income Y_1 if autonomous investment drops from $\text{R}300$ to $\text{R}250$ ($\Delta I = -50$), and (3) the required ΔG to reach $Y_f = \text{R}1\,900$ from $Y_0 = \text{R}1\,785$.

Verified Solution: Tax revenue is R446.25 (R446.81 exact) yielding a +R46.25 budget surplus; when I drops to R250, Y1 falls by R85 to R1 700 (and reaching Yf = R1 900 from Y0 requires ΔG = +R67.65).

$$\text{Prerequisite Baseline: } \bar{A}_0 = \text{R}1\,050, \quad \alpha = 1.70 \text{ (exact } 1.7021), \quad Y_0 = \text{R}1\,785 \text{ (exact R1 787.23)}$$
$$\text{Tax Revenue at } Y_0: \quad T = 0.25 \times 1\,785 = \text{R}446.25 \text{ (exact R446.81)}; \quad \text{Budget Balance} = T - G = 446.25 - 400 = +\text{R}46.25 \text{ Surplus}$$
$$\text{Investment Drop } (\Delta I = 250 - 300 = -50): \quad \Delta Y = 1.70 \times (-50) = -\text{R}85 \implies Y_1 = 1\,785 - 85 = \text{R}1\,700 \text{ (exact R1 702.13)}$$
$$\text{Fiscal Injection to Reach } Y_f = 1\,900: \quad \Delta Y = 1\,900 - 1\,785 = 115 \implies \Delta G = \frac{115}{1.70} = \text{R}67.65 \text{ (exact R66.25)}$$

First-Principles Proof: Correct! At Y0 = R1 785, T = 0.25(1 785) = R446.25 > G = R400 (+R46.25 surplus); a R50 drop in I reduces Y by 1.70 × 50 = R85 to R1 700, while closing the R115 output gap requires ΔG = 115/1.70 = R67.65.

PROBLEM 18.9 — CALCULATING INITIAL AUTONOMOUS SPENDING, MULTIPLIER, AND EQUILIBRIUM INCOME PRIOR TO A FISCAL SHIFT

Calculating Initial Autonomous Spending, Multiplier, and Equilibrium Income Prior to a Fiscal Shift • Medium

Consider the baseline open-economy Keynesian model (in R millions) prior to any fiscal reform: Autonomous consumption $\bar{C} = 30$, Autonomous investment $\bar{I} = 10$, Government spending $\bar{G} = 20$, Exports $\bar{X} = 10$, Autonomous imports $\bar{Z} = 5$, Marginal propensity to consume $c = 0.6$ (so marginal propensity to save $s = 0.4$), average/proportional tax rate $t = 0.167$ $(1/6)$, and Marginal propensity to import $m = 0.125$ $(1/8)$. Calculate (1) total autonomous expenditure $\bar{A}$, (2) the open-economy multiplier α, and (3) the initial equilibrium level of income Y_0 and initial tax revenue T_0.

Verified Solution: Autonomous expenditure Ā is R65m, the multiplier α is 1.60 (1/0.625), initial equilibrium income Y0 is R104m, and initial tax revenue is R17.37m.

$$\text{Autonomous Expenditure: } \bar{A} = \bar{C} + \bar{I} + \bar{G} + \bar{X} - \bar{Z} = 30 + 10 + 20 + 10 - 5 = \text{R65 million}$$
$$\text{Induced Slope: } c(1-t) - m = 0.6(1 - 0.167) - 0.125 = 0.6(0.8333) - 0.125 = 0.50 - 0.125 = 0.375$$
$$\text{Multiplier: } \alpha = \frac{1}{1 - c(1-t) + m} = \frac{1}{1 - 0.50 + 0.125} = \frac{1}{0.625} = 1.60$$
$$\text{Initial Equilibrium Income: } Y_0 = \alpha \times \bar{A} = 1.60 \times 65 = \text{R104 million}; \quad T_0 = 0.167 \times 104 = \text{R17.37m}$$

First-Principles Proof: Correct! Ā = 30 + 10 + 20 + 10 - 5 = R65m, 1 - 0.6(1 - 0.167) + 0.125 = 0.625 gives α = 1.60, and Y0 = 1.60 × 65 = R104m (which serves as the baseline before any tax-financed grant).

PROBLEM 18.10 — FOLLOW-UP CALCULATION OF TAXES, BUDGET BALANCE, AND HOUSEHOLD SAVING GIVEN EQUILIBRIUM INCOME

Follow-Up Calculation of Taxes, Budget Balance, and Household Saving Given Equilibrium Income • Hard

In the macroeconomic model where Consumption $C = 85 + 0.5Y_d$, Investment $I = 75$, Government expenditure $G = 70$, Taxes $T = 0.25Y$, Disposable income $Y_d = Y - T$, Exports $X = 30$, and Imports $Z = 30 + 0.3Y_d$: the first part of the analysis established that autonomous spending is $\bar{A} = 230$, the multiplier is $\alpha = 1/0.85 \approx 1.1765$, and equilibrium national income is $Y_0 = 270.59$ $(230/0.85)$. Using this prerequisite equilibrium income $Y_0 = 270.59$, calculate (1) equilibrium tax revenue T, (2) the government budget balance $(T - G)$, (3) equilibrium disposable income Y_d, and (4) household saving $S = Y_d - C$.

Verified Solution: Taxes T = 67.65, the government budget has a deficit of -2.35 (since G = 70 > T = 67.65), disposable income Yd = 202.94, and household saving S = 16.47.

$$\text{Prerequisite Baseline: } Y_0 = \frac{230}{1 - (0.5 - 0.3)(1 - 0.25)} = \frac{230}{0.85} = 270.59$$
$$\text{Equilibrium Taxes: } T = 0.25 \times Y_0 = 0.25 \times 270.59 = 67.65$$
$$\text{Government Budget Balance: } T - G = 67.65 - 70.00 = -2.35 \quad (\text{Budget Deficit of } 2.35, \text{ since } G > T)$$
$$\text{Disposable Income and Saving: } Y_d = 270.59 - 67.65 = 202.94; \quad C = 85 + 0.5(202.94) = 186.47; \quad S = Y_d - C = 202.94 - 186.47 = 16.47$$

First-Principles Proof: Correct! At Y0 = 270.59, T = 0.25(270.59) = 67.65, leaving a budget deficit of 67.65 - 70 = -2.35; disposable income is 202.94, consumption is 186.47, and saving is 202.94 - 186.47 = 16.47.

PROBLEM 18.11 — HOUSEHOLD DISPOSABLE INCOME, THREE-SECTOR GOVERNMENT MULTIPLIER & LUMP-SUM TAXES

Household Disposable Income, Three-Sector Government Multiplier & Lump-Sum Taxes • Medium

Solve three Keynesian fiscal exercises from Core Module: (Exercise 1) Thandi earns gross income Y = R500 and pays lump-sum tax T = R50; if her autonomous consumption is C̄ = R50 and she spends 75% of disposable income (c = 0.75), find her total consumption C; (Exercise 2) In a three-sector economy with C = 250 + 0.85Y, I = 150, and G = 350, find initial equilibrium income Ye and the new equilibrium income when G rises by ΔG = 100; (Exercise 3) In an open economy with C = 80 + 0.8Yd, I = 50, G = 30, lump-sum tax T = 10, X = 50, and Z = 20, find equilibrium income Ye.

Verified Solution: (1) Thandi's consumption is R387.50; (2) Three-sector Ye = 5 000 (α = 6.67) and rises to 5 667 when ΔG = +100; (3) Open-economy autonomous spending is Ā = 182 and Ye = 910.

$$(1)\ \text{Thandi: } Y_d = Y - T = 500 - 50 = \text{R450} \implies C = \bar{C} + cY_d = 50 + 0.75(450) = 50 + 337.50 = \text{R387.50}$$
$$(2)\ \text{Three-Sector Model: } \bar{A} = 250 + 150 + 350 = 750; \quad \alpha = \frac{1}{1 - 0.85} = \frac{1}{0.15} \approx 6.67; \quad Y_e = \frac{750}{0.15} = 5\,000$$
$$\text{When } \Delta G = +100 : \Delta Y = 6.67 \times 100 = 667 \implies Y_{\text{new}} = 5\,000 + 667 = 5\,667 \text{ (exact } 5\,666.67)$$
$$(3)\ \text{Open Economy with Lump-Sum Tax } T = 10 : C = 80 + 0.8(Y - 10) = 72 + 0.8Y$$
$$\bar{A} = 72 + 50 + 30 + 50 - 20 = 182; \quad \alpha = \frac{1}{1 - 0.8} = 5 \implies Y_e = 5 \times 182 = 910$$

First-Principles Proof: Correct! Thandi spends 50 + 0.75(450) = R387.50; the three-sector economy has Ye = 750/0.15 = 5 000 and Y_new = 5 667; and in Exercise 3, C = 80 + 0.8(Y - 10) gives Ā = 182 and Ye = 5 × 182 = 910.

PROBLEM 18.12 — OPEN-ECONOMY EQUILIBRIUM UNDER LUMP-SUM TAXES VS A 5% PROPORTIONAL INCOME TAX RATE

Open-Economy Equilibrium under Lump-Sum Taxes vs a 5% Proportional Income Tax Rate • Hard

Evaluate two open-economy models from Core Module: (Model A) Given C = 150 + 0.85Yd, I = 350, G = 250, X = 230, and Z = 250 (initially with T = 0), calculate: (i) initial equilibrium income Ye, (ii) the required increase in income ΔY if household saving increases by ΔS = 400, and (iii) the new equilibrium income if a lump-sum tax of T = 200 is introduced; (Model B) Given C = 185 + 0.7Yd, I = 220, G = 350, a proportional tax rate of t = 5% (0.05), X = 250, and Z = 205, calculate autonomous spending Ā, the tax-adjusted multiplier α, and equilibrium income Ye.

Verified Solution: (Model A) Initial Ye = 4 866.67, ΔY for ΔS = 400 is 2 666.67, and with T = 200, Y_new = 3 733.33; (Model B) Ā = 800, α = 1/0.335 ≈ 2.985, and Ye = 2 388.

$$\text{Model A (i) Initial Equilibrium } (T=0): \bar{A} = 150 + 350 + 250 + 230 - 250 = 730; \quad Y_e = \frac{730}{1 - 0.85} = \frac{730}{0.15} = 4\,866.67$$

$$\text{Model A (ii) Saving Target } \Delta S = 400: \text{MPS} = 1 - 0.85 = 0.15 \implies \Delta Y = \frac{\Delta S}{\text{MPS}} = \frac{400}{0.15} = 2\,666.67$$

$$\text{Model A (iii) With Lump-Sum Tax } T = 200: \bar{A}_{\text{new}} = 730 - 0.85(200) = 730 - 170 = 560 \implies Y_{\text{new}} = \frac{560}{0.15} = 3\,733.33$$

$$\text{Model B (t = 0.05): } \bar{A} = 185 + 220 + 350 + 250 - 205 = 800; \quad c(1-t) = 0.7(1 - 0.05) = 0.7(0.95) = 0.665$$

$$\text{Model B Multiplier and Equilibrium: } \alpha = \frac{1}{1 - 0.665} = \frac{1}{0.335} \approx 2.985; \quad Y_e = \frac{800}{0.335} = 2\,388.06 \approx 2\,388$$

First-Principles Proof: Correct! In Model A, Ye = 730/0.15 = 4 866.67, ΔY = 400/0.15 = 2 666.67, and lump-sum T = 200 lowers Ā by 0.85(200) = 170 to 560 (Y_new = 3 733.33); in Model B, 1 - 0.7(0.95) = 0.335 gives Ye = 800/0.335 = 2 388.

PROBLEM 18.13 — OPEN-ECONOMY MULTIPLIER WITH AUTONOMOUS + PROPORTIONAL TAXES & INDUCED IMPORTS (ISLAMABAD)

Open-Economy Multiplier with Autonomous + Proportional Taxes & Induced Imports (Islamabad) • Hard

The open economy of Islamabad (Core Module) is characterised by the following structural equations: Consumption C = 150 + 0.8Yd, Tax function T = 20 + 0.2Y (combining autonomous tax T̄ = 20 and proportional tax rate t = 0.2), Investment I = 250, Government spending G = 450, Exports X = 325, and Import function Z = 50 + 0.2Y (combining autonomous imports Z̄ = 50 and marginal propensity to import m = 0.2). Calculate: (1) The reduced-form Aggregate Expenditure equation A = Ā + βY; (2) The open-economy multiplier α and initial equilibrium income Ye; and (3) The new equilibrium income Y_new if autonomous investment rises from 250 to 310.

Verified Solution: (1) A = 1 109 + 0.44Y; (2) Multiplier α = 1/0.56 ≈ 1.79 and initial equilibrium income Ye = 1 980; (3) When I rises by 60 to 310, equilibrium income rises by 107.4 to Y_new = 2 087 (or 2 087.5).

$$(1)\ \text{Substitute } Y_d = Y - T = Y - (20 + 0.2Y) = -20 + 0.8Y \text{ into } C:$$

$$C = 150 + 0.8(-20 + 0.8Y) = 150 - 16 + 0.64Y = 134 + 0.64Y$$

$$A = C + I + G + X - Z = (134 + 0.64Y) + 250 + 450 + 325 - (50 + 0.2Y) = 1\,109 + 0.44Y$$

$$(2)\ \text{Multiplier and Initial Equilibrium: } 1 - 0.44 = 0.56 \implies \alpha = \frac{1}{0.56} \approx 1.79; \quad Y_e = \frac{1\,109}{0.56} = 1\,980.36 \approx 1\,980$$

$$(3)\ \text{Investment Shock } (\Delta I = 310 - 250 = +60): \Delta Y = 1.79 \times 60 = 107.4 \implies Y_{\text{new}} = 1\,980 + 107.4 = 2\,087.4 \approx 2\,087 \text{ (exact } 1\,169/0.56 = 2\,087.5)$$

First-Principles Proof: Correct! Substituting T = 20 + 0.2Y into C gives C = 134 + 0.64Y; subtracting Z = 50 + 0.2Y yields A = 1 109 + 0.44Y, α = 1/0.56 ≈ 1.79, Ye = 1 980, and ΔI = +60 raises Ye to 2 087.

CHAPTER 19 AD-AS Framework & Policy Debates • Macroeconomic Theory & Policy Transmission • 10 Hours

Chapter 19: More on Macroeconomic Theory and Policy

The Aggregate Demand–Aggregate Supply (AD–AS) model, variable price levels, real vs monetary AD shifters, why AD slopes downward, SRAS vs LRAS, imported input cost-push shocks, monetary transmission mechanism channels, four policy lags, and Monetarist vs Keynesian schools.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (AD-AS Framework & Policy Debates • 10 Hours):

- Relax the fixed-price assumption of the simple Keynesian model and analyse macroeconomic equilibrium in the Aggregate Demand–Aggregate Supply (AD–AS) framework.
- Distinguish between real factors (changes in autonomous C, I, G, X - Z) and monetary factors (changes in money supply M and interest rates i) that shift the Aggregate Demand curve.
- Explain how domestic and imported intermediate input price shocks (electricity, transport, steel, imported vehicle engines) shift Aggregate Supply upward/leftward, causing cost-push stagflation.
- Compare the policy perspectives of the Monetarist and Keynesian schools on stabilisation, velocity of circulation (MV ≡ PY), and the four policy implementation lags.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (3 IDENTITIES)

EQ 19.1 **Aggregate Demand Downward Slope Effects**

$$P \uparrow \Longrightarrow \begin{cases} (M/P) \downarrow \Longrightarrow C \downarrow & (\text{Wealth/Real Balance}) \\ i \uparrow \Longrightarrow I \downarrow & (\text{Interest Rate Effect}) \\ (X - Z) \downarrow & (\text{International Trade}) \end{cases}$$

Meaning: A rise in the domestic price level P reduces real money balances, raises interest rates, and erodes export competitiveness, reducing real output demanded.

EQ 19.2 **Monetary Transmission Mechanism**

$$\Delta i_{\text{repo}} \rightarrow \Delta i_{\text{market}} \rightarrow \Delta I \ (\text{and } \Delta C, \Delta E, \Delta P_{\text{assets}}) \rightarrow \Delta \text{AD} \rightarrow \Delta P, \Delta Y$$

Meaning: Links monetary policy changes in the repo rate to real economic output and inflation via interest rate, exchange rate, asset price, and credit channels.

EQ 19.3 **Equation of Exchange (Quantity Theory of Money)**

$$M \times V = P \times Y \implies \text{If } V, Y \text{ constant}, \%\Delta P = \%\Delta M$$

Meaning: Monetarists argue that velocity V is stable and output Y is near full capacity, so excessive money growth directly causes inflation.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 19.1 **The AD–AS Framework: Real vs. Monetary AD Shifters** (Core Macroeconomic Concept)

While the Keynesian model of Ch 17–18 assumes fixed prices, the AD–AS model places the general price level (P) on the vertical axis and real output (Y) on the horizontal axis. Core Module emphasises that Macro-Demand (AD) shifts rightward due to either real factors (autonomous increases in C, I, G, or X - Z) or monetary factors (increases in the money supply M lowering interest rates and boosting total expenditure).

Worked Proof: An expansionary fiscal boost (↑G) or monetary expansion (↑M → ↓i → ↑I) shifts AD rightward along an upward-sloping SRAS, increasing real output Y at the cost of a higher price level P.

CONCEPT 19.2 **Adverse Supply Shocks, Imported Inputs & Stagflation in AD–AS** (Core Macroeconomic Concept)

An increase in unit production costs—such as domestic transport, electricity, and steel tariffs, or more expensive imported intermediate inputs—shifts the SRAS curve upward/leftward. The economy experiences STAGFLATION: a simultaneous rise in the price level (↑P, cost-push inflation) and contraction in real output and employment (↓Y).

Worked Proof: As noted in macroeconomic analysis: (p. 97), if Ford vehicle engines are imported from England and the price of imported engines rises, the production cost of assembling vehicles in South Africa increases, shifting AS upward (raising prices and lowering output).

CONCEPT 19.3 **Four Policy Lags & Monetarist vs. Keynesian Views** (Core Macroeconomic Concept)

Stabilisation policy faces four chronological lags: (1) Recognition lag (time to collect data confirming a shock, same for both policies), (2) Decision lag (very long for fiscal policy due to annual parliamentary budget process; very short for SARB MPC), (3) Implementation lag (short for monetary policy), and (4) Impact/outside lag (long for monetary policy as investment responds slowly; shorter for direct fiscal spending).

Worked Proof: Monetarists (Milton Friedman) view the market economy as inherently stable and see money supply changes as directly affecting private spending, warning that policy lags make activist Keynesian fiscal fine-tuning destabilising.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (The AD–AS Framework: Real vs. Monetary AD Shifters): Explaining the downward slope of the macroeconomic AD curve using the microeconomic substitution effect between two individual goods.	**First-Principles Law:** When the aggregate price level P rises, all domestic goods become more expensive together; AD slopes downward due to the real balance (wealth) effect, the Keynes interest rate effect, and the international trade effect.
Fallacy (Adverse Supply Shocks, Imported Inputs & Stagflation in AD–AS): Thinking demand-management policies (monetary or fiscal policy alone) can easily cure stagflation.	**First-Principles Law:** Because monetary and fiscal policies shift AD rather than AS, they cannot simultaneously reverse both higher prices and lower output after an adverse supply shock.
Fallacy (Four Policy Lags & Monetarist vs. Keynesian Views): Claiming that monetary policy has a longer decision lag than fiscal policy.	**First-Principles Law:** Fiscal policy has a long decision/inside lag (parliamentary approval) but a relatively short impact lag, whereas monetary policy has a very short decision lag (SARB MPC meets every 2 months) but a long impact/outside lag (12–24 months).

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (5 PROBLEMS)

PROBLEM 19.1 — MONETARIST RULES VS KEYNESIAN COUNTER-CYCLICAL STABILISATION DEBATE

Macroeconomics, §19.1–19.4, pp. 408–431 • Medium

How do the views of Monetarist economists differ from Keynesian economists regarding economic fluctuations, and how do the decision lag and impact lag compare between fiscal policy and monetary policy?

Verified Solution: Keynesians see instability as endogenous; fiscal policy has a longer decision lag but shorter impact lag.

$$\text{Keynesian: Endogenous Instability} \implies \text{Active Policy.} \quad \text{Decision Lag: Fiscal} > \text{Monetary;} \quad \text{Impact Lag: Monetary} > \text{Fiscal.}$$

First-Principles Proof: Correct! Keynesians view fluctuations as inherent to market economies, and parliamentary budgets give fiscal policy a long decision lag while repo changes have a long impact lag.

PROBLEM 19.2 — WEALTH, INTEREST-RATE & INTERNATIONAL TRADE EFFECTS ON THE AD SLOPE

Macroeconomics, §19.1–19.2, pp. 409–421 • Medium

In the Aggregate Demand–Aggregate Supply (AD–AS) model, which three macroeconomic effects explain why the Aggregate Demand (AD) curve slopes downward from left to right as the general price level (P) falls?

Verified Solution: The wealth (real balance) effect, the interest rate effect, and the international trade effect.

$$P \downarrow \implies (1)\ (M/P) \uparrow \implies C \uparrow \text{ (Wealth);}\ (2)\ i \downarrow \implies I \uparrow \text{ (Interest Rate);}\ (3)\ (X - Z) \uparrow \text{ (Trade).}$$

First-Principles Proof: Correct! A lower price level raises the real value of money balances (boosting C), lowers interest rates (boosting I), and improves net export competitiveness (boosting X - Z).

PROBLEM 19.3 — STRUCTURAL TRANSMISSION OF MONETARY & FISCAL POLICY IN THE AD–AS MODEL

Macroeconomics, §19.2–19.3, pp. 421–428 • Medium

In macroeconomic policy debates (Chapter 19), how do Monetarists and Keynesians differ regarding the stability of the private sector, and how do the implementation and impact lags compare between fiscal policy and monetary policy?

Verified Solution: Monetarists view the private economy as stable; fiscal policy has a longer decision lag while monetary policy has a longer impact lag.

$$\text{Monetarists: Private sector inherently stable; avoid activist fine-tuning.} \quad \text{Fiscal: Long decision/implementation lag, short impact lag.}$$

First-Principles Proof: Correct! Monetarists believe markets are self-correcting and activist policy is destabilising; fiscal changes require parliamentary budget approval (long decision lag), while repo rate changes take 12–24 months to work (long impact lag).

PROBLEM 19.4 — EXCHANGE-RATE & ASSET-PRICE CHANNELS OF THE MONETARY TRANSMISSION MECHANISM

Macroeconomics, §19.2, pp. 416–420 • Medium

Trace the exchange-rate channel and asset-price (wealth) channel of the monetary transmission mechanism when the SARB Monetary Policy Committee (MPC) cuts the repo rate by 100 basis points:

Verified Solution: Lower interest rates trigger capital outflows that depreciate the Rand (boosting NX) and raise bond/equity prices (boosting C).

$$\Delta i < 0 \implies \text{Capital outflows} \implies \text{Rand depreciates} \implies NX \uparrow \implies AD \uparrow; \quad \Delta i < 0 \implies P_{\text{bonds/equities}} \uparrow \implies C \uparrow.$$

First-Principles Proof: Correct! Lower domestic yields reduce foreign capital inflows, depreciating the Rand (raising X - Z), while inverse bond pricing and discounted cash flows raise asset wealth and consumption.

PROBLEM 19.5 — REAL VS MONETARY AD SHIFTERS, IMPORTED ENGINE COST-PUSH AS SHOCK & THE DEFLATION TRAP

Real vs Monetary AD Shifters, Imported Engine Cost-Push AS Shock & the Deflation Trap • Medium

In the Aggregate Demand–Aggregate Supply (AD–AS) framework (Core Module): (1) Classify Aggregate Demand shifters into Real factors versus Monetary factors; (2) Trace the macroeconomic impact on South Africa's price level (P), real output (Y), and employment if the US dollar price of imported Ford vehicle engines increases sharply; and (3) Explain why sustained deflation (a falling general price level) is a dangerous macroeconomic trap rather than a benefit.

Verified Solution: (1) Real AD shifters are G, T, $\bar{C}$, $\bar{I}$, X, Z while monetary shifters are repo rate i and money supply M; (2) Higher imported engine prices shift AS leftward, raising P and lowering Y (stagflation); (3) Deflation raises real debt burdens and delays spending.

$$\begin{gathered}(1)\ \text{AD Shifters: Real factors} = \text{Fiscal policy } (G, T), \text{Autonomous consumption } (\bar{C}), \text{Autonomous investment } (\bar{I}), \text{Net exports } (X - Z); \quad \text{Monetary factors} \\ (2)\ \text{Imported Intermediate Input Shock (Ford Engines): Production cost rises} \implies \text{AS shifts Upward/Leftward from } AS_0 \text{ to } AS_1 \\ \text{Equilibrium moves along } AD_0 : \text{General Price Level } P \uparrow \text{ (cost-push inflation) and Real Output } Y \downarrow \text{ (stagflation / rising unemployment)} \\ (3)\ \text{Deflation Trap: Falling prices } (P \downarrow) \implies \text{real value of debt rises, real interest rates rise, and households delay spending} \implies AD \downarrow \text{ and recession deep}\end{gathered}$$

First-Principles Proof: Correct! Imported intermediate inputs (Ford engines) increase domestic production costs, shifting AS upward/leftward (higher P, lower real GDP Y), while deflation triggers debt-deflation and postponed spending.

CHAPTER 20 Inflation & Price Dynamics • Price Indices & Inflation Mechanics • 10 Hours

Chapter 20: Inflation

Definition and measurement of inflation (CPI, PPI, Implicit GDP Deflator), distributional/economic/social effects, bracket creep, the four approaches to inflation (Monetarist, Demand-pull/Cost-push, Structuralist, Conflict), dangers of deflation, and SARB inflation targeting.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Inflation & Price Dynamics • 10 Hours):

- Define inflation rigorously (a continuous and considerable rise in the general price level) and explain why a once-off price hike is not inflation.
- Explain how inflation is measured and calculate the inflation rate using the Consumer Price Index (CPI), Production Price Index (PPI), and the Implicit GDP Deflator.
- Explain why dividing Nominal GDP by Real GDP × 100 yields an index (the GDP Deflator) rather than an inflation rate, requiring two consecutive periods' deflators to compute percentage inflation.
- Explain the four main approaches to inflation: the Monetarist approach (MV ≡ PY), Demand-pull and Cost-push approach, the Structuralist approach (underlying, initiating, and propagating factors), and the Conflict approach.
- Explain why continuous deflation is dangerous to an economy and suggest policy measures (including inflation targeting and indexation) to fight inflation.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (3 IDENTITIES)

EQ 20.1 CPI Inflation Rate

$$\pi_t = \left(\frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}\right) \times 100\%$$

Meaning: Percentage change in the cost of a fixed representative consumer basket (measured month-on-same-month of previous year or annual average on annual average).

EQ 20.2 GDP Deflator Inflation Rate

$$\text{Deflator}_t = \frac{\text{Nom GDP}_t}{\text{Real GDP}_t} \times 100, \quad \pi_t^{\text{Defl}} = \left(\frac{\text{Deflator}_t - \text{Deflator}_{t-1}}{\text{Deflator}_{t-1}}\right) \times 1$$

Meaning: Measures economy-wide inflation across all domestically produced final goods and services using a variable current-production basket.

EQ 20.3 Real Purchasing Power & Real Interest Rate

$$r_{\text{real}} \approx i_{\text{nominal}} - \pi$$

Meaning: When inflation π exceeds the nominal interest rate i, the real interest rate is negative, redistributing real wealth from creditors (savers) to debtors (borrowers).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 20.1 CPI vs. PPI vs. Implicit GDP Deflator (Comparative Matrix) (Core Macroeconomic Concept)

Three price indicators differ in scope and basket structure: (1) CPI measures the retail cost of living for urban households using a fixed consumer basket that INCLUDES imported consumer goods and VAT; (2) PPI measures wholesale prices at the level of first significant commercial transaction (production/factory gate), EXCLUDES services and VAT, and includes capital & intermediate goods; (3) The GDP Deflator covers ALL domestically produced final goods/services (including exports & capital goods) but EXCLUDES all imports.

Worked Proof: In Core Module (pp. 95–96): When CPI rises from 102.3 to 106.9, CPI inflation is [(106.9 - 102.3)/102.3] × 100 = 4.5%. When Nominal/Real GDP gives Deflator_1 = 153.1 and Deflator_2 = 166.2, deflator inflation is [(166.2 - 153.1)/153.1] × 100 = 8.6%.

CONCEPT 20.2 Distributional Effects of Inflation, Bracket Creep & the Deflation Trap (Core Macroeconomic Concept)

Inflation redistributes income and wealth arbitrarily: (1) Debtors gain at the expense of creditors because loans are repaid in Rand with lower real purchasing power; (2) Young borrowers with mortgages gain while elderly pensioners on fixed annuities lose; (3) The government always gains via 'bracket creep' (fiscal drag). Conversely, Core Module warns that continuous deflation (falling prices) is even more dangerous because buyers and shop-owners postpone purchases expecting lower prices later, triggering recession or depression.

Worked Proof: If a worker's salary rises by 7% to match 7% inflation, their real pre-tax income is unchanged, yet under an unindexed progressive tax table their effective tax rate rises, reducing their real disposable income.

CONCEPT 20.3 Four Approaches to Inflation: Monetarist, Demand/Cost, Structuralist & Conflict (Core Macroeconomic Concept)

Core Module highlights four major theoretical approaches to inflation: (1) The Monetarist approach based on MV ≡ PY; (2) Demand-pull (↑AD) and Cost-push (←AS) inflation; (3) The Structuralist approach, which separates inflation drivers into underlying factors (structural rigidities), initiating factors (price/supply shocks), and propagating factors (indexation and wage-price spirals); and (4) The Conflict approach, where competing income claims by labour unions, business, and government exceed real output.

Worked Proof: To combat demand-pull inflation, SARB raises the repo rate and Treasury cuts G or raises T; to address structuralist and conflict inflation, supply-side reforms and incomes policies are required.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (CPI vs. PPI vs. Implicit GDP Deflator (Comparative Matrix)): Calling a one-off jump in the price of bread 'inflation', or treating the single-period ratio (Nominal GDP / Real GDP) × 100 as an inflation rate rather than a price index.

First-Principles Law: Inflation is a CONTINUOUS and CONSIDERABLE rise in prices IN GENERAL. Dividing Nominal by Real GDP × 100 yields an index; you must compare two periods' deflator indices to obtain the percentage inflation rate.

Fallacy (Distributional Effects of Inflation, Bracket Creep & the Deflation Trap): Assuming that everyone loses equally during inflation, or wishing for prices to fall continuously (deflation).

First-Principles Law: Inflation redistributes real wealth from lenders/fixed-income holders to borrowers and the state, while continuous deflation paralyses consumption and investment (as seen in late-20th-century Japan).

Fallacy (Four Approaches to Inflation: Monetarist, Demand/Cost, Structuralist & Conflict): Using contractionary monetary/fiscal policy aggressively against pure cost-push or structural inflation without side effects.

First-Principles Law: Restricting AD when inflation is cost-push curbs price rises only by worsening the contraction in real output Y and increasing unemployment further.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (20 PROBLEMS)

PROBLEM 20.1 — IMPLICIT GDP DEFLATOR & ANNUAL INFLATION RATE CALCULATION

Macroeconomics, §20.2, pp. 438–442 • Medium

The macroeconomic table below shows Nominal GDP and Real GDP (in R millions) for two consecutive years: in Period 1, Nominal GDP = 5 130 703 and Real GDP = 4 028 475; in Period 2, Nominal GDP = 5 643 773 and Real GDP = 4 391 038. Calculate the implicit GDP deflator for both years and the resulting annual inflation rate (rounded to 2 decimal places).

Verified Solution: The GDP deflator rises from 127.36 to 128.53, yielding an annual inflation rate of 0.92%.

$$\text{Deflator}_1 = \frac{5\,130\,703}{4\,028\,475} \times 100 = 127.36; \quad \text{Deflator}_2 = \frac{5\,643\,773}{4\,391\,038} \times 100 = 128.53.$$
$$\pi = \frac{128.53 - 127.36}{127.36} \times 100 = 0.92\%.$$

First-Principles Proof: Correct! Deflator Year 1 = (5 130 703 / 4 028 475) × 100 = 127.36; Deflator Year 2 = (5 643 773 / 4 391 038) × 100 = 128.53; inflation = [(128.53 - 127.36)/127.36] × 100 = 0.92%.

PROBLEM 20.2 — THREE-GOOD ECONOMY: NOMINAL GDP, REAL GDP & DEFLATOR INFLATION

Macroeconomics, §13.2 & §20.2, pp. 243, 441 • Medium

An economy produces three goods (Period 1 is the base year): Pizza (Period 1: Q=24, P=R22.50; Period 2: Q=36, P=R24.75), Soda (Period 1: Q=50, P=R4.00; Period 2: Q=75, P=R4.40), and Ice Cream (Period 1: Q=15, P=R8.00; Period 2: Q=22.5, P=R8.80). Calculate Period 1 Nominal/Real GDP, Period 2 Nominal and Real GDP, the Period 2 GDP Deflator, and the Period 2 inflation rate.

Verified Solution: Period 1 GDP is R860; Period 2 Nominal is R1 419 and Real is R1 290; Period 2 Deflator is 110 and inflation is 10%.

$$\text{GDP}_1^{\text{Nom}} = \text{GDP}_1^{\text{Real}} = 24(22.5) + 50(4) + 15(8) = \text{R}860.$$
$$\text{GDP}_2^{\text{Nom}} = 36(24.75) + 75(4.4) + 22.5(8.8) = \text{R}1\,419; \quad \text{GDP}_2^{\text{Real}} = 36(22.5) + 75(4) + 22.5(8) = \text{R}1\,290.$$
$$\text{Deflator}_2 = \frac{1419}{1290} \times 100 = 110.0 \implies \pi_2 = 10.0\%.$$

First-Principles Proof: Correct! Period 1 GDP = 540 + 200 + 120 = R860; Period 2 Nominal = 891 + 330 + 198 = R1 419; Period 2 Real = 810 + 300 + 180 = R1 290; Deflator = 110, so inflation = 10%.

PROBLEM 20.3 — STRUCTURAL COMPARISON OF CPI, PPI & THE IMPLICIT GDP DEFLATOR

Macroeconomics, §20.2, pp. 438–442 • Easy

Which statement accurately contrasts the Consumer Price Index (CPI), the Producer Price Index (PPI), and the Implicit GDP Deflator in South Africa?

Verified Solution: CPI includes services and imported consumer goods; PPI excludes services and includes capital goods.

$$\text{CPI: Urban consumer basket (incl. imports and services).} \quad \text{PPI: Production gate (excl. services and VAT, incl. capital/intermediate).} \quad \text{Deflator: All dome}$$

First-Principles Proof: Correct! CPI measures household cost of living (including services, VAT, and imported consumer goods), whereas PPI measures producer prices (excluding services and VAT, including capital/intermediate goods).

PROBLEM 20.4 — HOUSEHOLD BASKET COST-OF-LIVING ESCALATION UNDER HEADLINE CPI INFLATION

Macroeconomics, §20.2–20.3, pp. 438–445 • Medium

A household spent R4 200 on Food, R1 900 on Transport, R5 800 on Rent, and R1 300 on Utilities last year (total R13 200). If CPI inflation during the year was 6.8%, how much must total household expenditure increase to maintain the exact same real consumption basket, and why does CPI inflation often exceed GDP Deflator inflation when global crude oil prices spike?

Verified Solution: Expenditure rises by R897.60 (to R14 097.60); CPI includes imported fuel while the Deflator excludes imports.

$$\text{Initial Total} = 4\,200 + 1\,900 + 5\,800 + 1\,300 = \text{R}13\,200. \quad \Delta E = 13\,200 \times 0.068 = \text{R}897.60 \implies E_{\text{new}} = \text{R}14\,097.60.$$

First-Principles Proof: Correct! 6.8% of R13 200 is R897.60 (new total R14 097.60), and CPI uses a fixed consumer basket that includes imported oil whereas the GDP Deflator covers only domestic production.

PROBLEM 20.5 — CONVERTING NOMINAL GDP GROWTH & REAL GDP GROWTH INTO DEFLATOR INFLATION

Macroeconomics, §20.2, pp. 441–443 • Medium

According to central bank national accounting statistics, Nominal GDP was R2 917 539 million in Period 1 and R3 155 195 million in Period 2, while Real GDP (at constant base prices) was R1 905 736 million in Period 1 and R1 954 303 million in Period 2. Calculate the implicit GDP Deflator for Period 1 and Period 2 and the resulting annual inflation rate.

Verified Solution: The Period 1 GDP Deflator is 153.09, the Period 2 GDP Deflator is 161.45, and the inflation rate is 5.46%.

$$\text{Defl}_1 = \frac{2\,917\,539}{1\,905\,736} \times 100 = 153.09; \quad \text{Defl}_2 = \frac{3\,155\,195}{1\,954\,303} \times 100 = 161.45.$$
$$\pi_2 = \frac{161.45 - 153.09}{153.09} \times 100 = 5.46\%.$$

First-Principles Proof: Correct! Deflator 1 = (2 917 539 / 1 905 736) × 100 = 153.09 and Deflator 2 = (3 155 195 / 1 954 303) × 100 = 161.45; percentage change = (161.45 - 153.09) / 153.09 × 100 = 5.46%.

PROBLEM 20.6 — CALCULATING ANNUAL HEADLINE CPI INFLATION RATES AND MULTI-YEAR AVERAGE INFLATION

Calculating Annual Headline CPI Inflation Rates and Multi-Year Average Inflation • Medium

A country's headline Consumer Price Index (CPI) over five consecutive periods is recorded as follows: Period 1 = 118.4, Period 2 = 124.6, Period 3 = 131.2, Period 4 = 138.5, and Period 5 = 145.8. Calculate the annual percentage CPI inflation rates for Period 2, Period 3, Period 4, and Period 5, determine which period experienced the highest inflation rate, and compute the four-period average inflation rate.

Verified Solution: Annual inflation rates are 5.24%, 5.30%, 5.56% (highest in Period 4), and 5.27%, yielding a four-period average inflation rate of 5.34%.

$$\text{Period 2 Inflation: } \pi_2 = \frac{124.6 - 118.4}{118.4} \times 100 = 5.24\%; \quad \text{Period 3 Inflation: } \pi_3 = \frac{131.2 - 124.6}{124.6} \times 100 = 5.30\%$$
$$\text{Period 4 Inflation: } \pi_4 = \frac{138.5 - 131.2}{131.2} \times 100 = 5.56\% \text{ (Highest)}; \quad \text{Period 5 Inflation: } \pi_5 = \frac{145.8 - 138.5}{138.5} \times 100 = 5.27\%$$
$$\text{Four-Period Average Inflation Rate: } \bar{\pi} = \frac{5.24 + 5.30 + 5.56 + 5.27}{4} = 5.34\%$$

First-Principles Proof: Correct! Dividing each period-on-period index change by the preceding period's CPI gives 5.24%, 5.30%, 5.56% (peak in Period 4), and 5.27%, averaging 5.34%.

PROBLEM 20.7 — IDENTIFYING THE BASE PERIOD FROM CURRENT VS CONSTANT GNI AND COMPUTING DEFLATOR INFLATION

Identifying the Base Period from Current vs Constant GNI and Computing Deflator Inflation • Medium

National accounting statistics show that in Period 1, Gross National Income (GNI) at current prices is R2 946 328 million and GNI at constant prices is also R2 946 328 million. In Period 2, GNI at current prices is R3 165 941 million and GNI at constant prices is R3 094 761 million. Explain why current and constant GNI are identical in Period 1, and calculate the implicit price deflator for Period 1 and Period 2 as well as the Period 2 inflation rate.

Verified Solution: Period 1 is the base period (so current and constant prices coincide at Deflator = 100.00); Period 2 Deflator is 102.30, giving 2.30% inflation.

$$\text{Why Identical in Period 1: Period 1 is the designated base period, so current prices and constant base-period prices are identical}$$
$$\text{Period 1 Implicit Deflator} = \frac{2\,946\,328}{2\,946\,328} \times 100 = 100.00$$
$$\text{Period 2 Implicit Deflator} = \frac{3\,165\,941}{3\,094\,761} \times 100 = 102.30 \implies \pi_2 = \frac{102.30 - 100.00}{100.00} \times 100 = 2.30\%$$

First-Principles Proof: Correct! Current and constant aggregates are always equal in the base period (Deflator = 100.00), and in Period 2 (3 165 941 / 3 094 761) × 100 = 102.30 (+2.30% inflation).

PROBLEM 20.8 — FOLLOW-UP CALCULATION OF GDP DEFLATOR INFLATION USING PREREQUISITE NOMINAL AND REAL GDP

Follow-Up Calculation of GDP Deflator Inflation Using Prerequisite Nominal and Real GDP • Medium

Continuing with the three-good economy producing Pizza (Period 1: Q=24, P=R22.50; Period 2: Q=36, P=R24.75), Soda (Period 1: Q=50, P=R4.00; Period 2: Q=75, P=R4.40), and Ice Cream (Period 1: Q=15, P=R8.00; Period 2: Q=22.5, P=R8.80) where Period 1 is the base period: the first part of the analysis established that Period 1 Nominal GDP = Period 1 Real GDP = R860, Period 2 Nominal GDP = R1 419, and Period 2 Real GDP (at constant Period 1 base prices) = R1 290. Using these prerequisite totals, calculate the Period 1 and Period 2 GDP Deflators and the Period 2 percentage inflation rate.

Verified Solution: The GDP Deflator rises from 100.00 in Period 1 to 110.00 in Period 2 (1 419 / 1 290 × 100), yielding a Period 2 inflation rate of 10.00%.

$$\text{Prerequisite Totals: Nominal GDP}_1 = \text{Real GDP}_1 = \text{R860}; \quad \text{Nominal GDP}_2 = \text{R1 419}; \quad \text{Real GDP}_2 = \text{R1 290}$$
$$\text{Period 1 GDP Deflator} = \frac{860}{860} \times 100 = 100.00$$
$$\text{Period 2 GDP Deflator} = \frac{1\,419}{1\,290} \times 100 = 110.00$$
$$\text{Period 2 Inflation Rate} = \frac{110.00 - 100.00}{100.00} \times 100 = 10.00\% \quad \left(\text{while real output grew by } \frac{1\,290 - 860}{860} = +50.00\%\right)$$

First-Principles Proof: Correct! Dividing Period 2 Nominal GDP (R1 419) by Period 2 Real GDP (R1 290) and multiplying by 100 gives a Deflator of 110.00, which is 10.00% above the base index of 100.00.

PROBLEM 20.9 — CALCULATING PERCENTAGE PRICE INCREASES ACROSS FIVE CONSUMER GOODS AND SERVICES

Calculating Percentage Price Increases Across Five Consumer Goods and Services • Easy

A consumer price survey tracks the average retail prices of five everyday items between Period 1 and Period 2: (1) Petrol (1 litre): R22.10 to R25.80; (2) White bread (1 loaf): R15.40 to R17.25; (3) Local minibus taxi fare (single trip): R14.00 to R18.00; (4) Prepaid electricity tariff (per kWh): R2.05 to R2.48; and (5) Student laptop computer: R14 500 to R16 800. Calculate the percentage price increase for each of the five items and identify which item experienced the highest percentage inflation rate and which experienced the lowest.

Verified Solution: Petrol +16.74%, Bread +12.01% (lowest), Taxi fare +28.57% (highest), Electricity +20.98%, and Laptop +15.86%.

$$\text{Petrol: } \frac{25.80 - 22.10}{22.10} \times 100 = \frac{3.70}{22.10} \times 100 = +16.74\%; \quad \text{Bread: } \frac{17.25 - 15.40}{15.40} \times 100 = \frac{1.85}{15.40} \times 100 = +12.01\% \text{ (Lowest)}$$

$$\text{Taxi Fare: } \frac{18.00 - 14.00}{14.00} \times 100 = \frac{4.00}{14.00} \times 100 = +28.57\% \text{ (Highest)}; \quad \text{Electricity: } \frac{2.48 - 2.05}{2.05} \times 100 = \frac{0.43}{2.05} \times 100 = +20.98\%$$

$$\text{Student Laptop: } \frac{16\,800 - 14\,500}{14\,500} \times 100 = \frac{2\,300}{14\,500} \times 100 = +15.86\%$$

First-Principles Proof: Correct! Dividing each price change by its Period 1 price yields +16.74% (Petrol), +12.01% (Bread, lowest), +28.57% (Taxi fare, highest), +20.98% (Electricity), and +15.86% (Laptop).

PROBLEM 20.10 — CALCULATING THE GDP DEFLATOR AND INFLATION RATE FROM NOMINAL AND REAL GDP

Calculating the GDP Deflator and Inflation Rate From Nominal and Real GDP • Medium

National accounting data for an economy show that Nominal GDP (at current prices) increased from R58 421 million in Period 1 to R60 798 million in Period 2, while Real GDP (at constant base-period prices) increased from R46 324 million in Period 1 to R47 760 million in Period 2. Calculate (1) the implicit GDP Deflator for Period 1 and Period 2, and (2) the annual percentage inflation rate in Period 2 based on the GDP Deflator.

Verified Solution: The Period 1 GDP Deflator is 126.11, the Period 2 GDP Deflator is 127.30, and the Period 2 annual inflation rate is 0.94%.

$$\text{Period 1 GDP Deflator} = \frac{\text{Nominal GDP}_1}{\text{Real GDP}_1} \times 100 = \frac{58\,421}{46\,324} \times 100 = 126.11$$

$$\text{Period 2 GDP Deflator} = \frac{\text{Nominal GDP}_2}{\text{Real GDP}_2} \times 100 = \frac{60\,798}{47\,760} \times 100 = 127.30$$

$$\text{Period 2 Inflation Rate} = \frac{127.30 - 126.11}{126.11} \times 100 = \frac{1.19}{126.11} \times 100 = 0.94\%$$

First-Principles Proof: Correct! Deflator 1 = (58 421 / 46 324) × 100 = 126.11 and Deflator 2 = (60 798 / 47 760) × 100 = 127.30; the percentage increase is (127.30 - 126.11) / 126.11 × 100 = 0.94%.

PROBLEM 20.11 — CALCULATING THREE-PERIOD IMPLICIT GDP DEFLATORS ALONGSIDE HEADLINE CPI

Calculating Three-Period Implicit GDP Deflators Alongside Headline CPI • Medium

A national economy records the following macroeconomic series over three consecutive periods: in Period 1, Nominal GDP = R5.80 trillion, Real GDP = R5.42 trillion, and CPI = 131.5; in Period 2, Nominal GDP = R6.32 trillion, Real GDP = R5.68 trillion, and CPI = 139.8; in Period 3, Nominal GDP = R6.91 trillion, Real GDP = R5.97 trillion, and CPI = 147.6. As the first stage of comparing domestic production price changes against consumer price changes, calculate the implicit GDP Deflator for Period 1, Period 2, and Period 3.

Verified Solution: The implicit GDP Deflator is 107.01 in Period 1, 111.27 in Period 2, and 115.75 in Period 3.

$$\text{Formula: GDP Deflator}_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} \times 100$$

$$\text{Period 1 GDP Deflator} = \frac{5.80}{5.42} \times 100 = 107.01$$

$$\text{Period 2 GDP Deflator} = \frac{6.32}{5.68} \times 100 = 111.27$$

$$\text{Period 3 GDP Deflator} = \frac{6.91}{5.97} \times 100 = 115.75$$

First-Principles Proof: Correct! Dividing Nominal GDP by Real GDP and multiplying by 100 yields (5.80/5.42)×100 = 107.01, (6.32/5.68)×100 = 111.27, and (6.91/5.97)×100 = 115.75.

PROBLEM 20.12 — FOLLOW-UP COMPARISON OF ANNUAL GDP DEFLATOR INFLATION VS CPI INFLATION RATES

Follow-Up Comparison of Annual GDP Deflator Inflation vs CPI Inflation Rates • Hard

Continuing with the three-period macroeconomic series where Period 1 has Nominal GDP = R5.80T, Real GDP = R5.42T, CPI = 131.5; Period 2 has Nominal GDP = R6.32T, Real GDP = R5.68T, CPI = 139.8; and Period 3 has Nominal GDP = R6.91T, Real GDP = R5.97T, CPI = 147.6: the first stage of the analysis established that the implicit GDP Deflators are 107.01 in Period 1, 111.27 in Period 2, and 115.75 in Period 3. Using these prerequisite GDP Deflators alongside the CPI series, calculate (1) the annual GDP Deflator inflation rates for Period 2 and Period 3, (2) the annual CPI inflation rates for Period 2 and Period 3, and (3) why CPI inflation exceeded GDP Deflator inflation during a global crude oil spike.

Verified Solution: Deflator inflation is 3.98% and 4.03% while CPI inflation is 6.31% and 5.58%; CPI is higher because imported fuel is in the CPI basket but excluded from GDP.

$$\text{Prerequisite GDP Deflators: } \text{Defl}_1 = 107.01, \quad \text{Defl}_2 = 111.27, \quad \text{Defl}_3 = 115.75$$

$$\text{GDP Deflator Inflation: } \pi_2^{\text{Defl}} = \frac{111.27 - 107.01}{107.01} \times 100 = 3.98\%; \quad \pi_3^{\text{Defl}} = \frac{115.75 - 111.27}{111.27} \times 100 = 4.03\%$$

$$\text{CPI Inflation: } \pi_2^{\text{CPI}} = \frac{139.8 - 131.5}{131.5} \times 100 = 6.31\%; \quad \pi_3^{\text{CPI}} = \frac{147.6 - 139.8}{139.8} \times 100 = 5.58\%$$

$$\text{Why } \pi^{\text{CPI}} > \pi^{\text{Defl}}: \text{ Imported consumer goods (like crude oil/petrol) are included in the fixed CPI basket but excluded from domestic GDP}$$

First-Principles Proof: Correct! Using the prerequisite deflators (107.01, 111.27, 115.75) gives 3.98% and 4.03% inflation vs 6.31% and 5.58% for CPI, as imported oil directly hits CPI but not domestic GDP.

PROBLEM 20.13 — CALCULATING SEQUENTIAL MONTH-ON-MONTH INFLATION RATES FROM MONTHLY CPI DATA

Calculating Sequential Month-on-Month Inflation Rates From Monthly CPI Data • Easy

A statistical agency releases the following monthly Consumer Price Index (CPI) series for the first five months of the year: January = 146.3, February = 147.0, March = 148.2, April = 149.1, and May = 150.8. Calculate the month-on-month percentage inflation rate for February, March, April, and May (each measured relative to the immediately preceding month), and identify which month recorded the sharpest monthly price acceleration.

Verified Solution: Monthly inflation is 0.48% in February, 0.82% in March, 0.61% in April, and 1.14% in May (with May recording the highest monthly jump).

$$\text{February: } \pi_{\text{Feb}} = \frac{147.0 - 146.3}{146.3} \times 100 = \frac{0.7}{146.3} \times 100 = 0.48\%$$

$$\text{March: } \pi_{\text{Mar}} = \frac{148.2 - 147.0}{147.0} \times 100 = \frac{1.2}{147.0} \times 100 = 0.82\%$$

$$\text{April: } \pi_{\text{Apr}} = \frac{149.1 - 148.2}{148.2} \times 100 = \frac{0.9}{148.2} \times 100 = 0.61\%$$

$$\text{May: } \pi_{\text{May}} = \frac{150.8 - 149.1}{149.1} \times 100 = \frac{1.7}{149.1} \times 100 = 1.14\% \quad (\text{Sharpest monthly jump in May})$$

First-Principles Proof: Correct! Dividing each monthly index gain (0.7, 1.2, 0.9, 1.7) by the prior month's CPI yields 0.48%, 0.82%, 0.61%, and 1.14%, peaking in May.

PROBLEM 20.14 — CALCULATING IMPLICIT GDP DEFLATORS ACROSS FOUR CONSECUTIVE PERIODS

Calculating Implicit GDP Deflators Across Four Consecutive Periods • Medium

National accounts data for an economy over four consecutive periods (in R billions) are as follows: Period 1: Nominal GDP = 6 240, Real GDP = 5 920; Period 2: Nominal GDP = 6 680, Real GDP = 6 150; Period 3: Nominal GDP = 7 210, Real GDP = 6 515; and Period 4: Nominal GDP = 7 745, Real GDP = 6 830. Calculate the implicit GDP Deflator for Period 1, Period 2, Period 3, and Period 4 (rounded to two decimal places).

Verified Solution: The implicit GDP Deflators are 105.41 in Period 1, 108.62 in Period 2, 110.67 in Period 3, and 113.40 in Period 4.

$$\text{Period 1 GDP Deflator} = \frac{6\,240}{5\,920} \times 100 = 105.41$$

$$\text{Period 2 GDP Deflator} = \frac{6\,680}{6\,150} \times 100 = 108.62$$

$$\text{Period 3 GDP Deflator} = \frac{7\,210}{6\,515} \times 100 = 110.67$$

$$\text{Period 4 GDP Deflator} = \frac{7\,745}{6\,830} \times 100 = 113.40$$

First-Principles Proof: Correct! Applying (Nominal GDP / Real GDP) × 100 gives 105.41 (6 240/5 920), 108.62 (6 680/6 150), 110.67 (7 210/6 515), and 113.40 (7 745/6 830).

PROBLEM 20.15 — FOLLOW-UP CALCULATION OF ANNUAL INFLATION RATES FROM PREREQUISITE FOUR-PERIOD GDP DEFLATORS

Follow-Up Calculation of Annual Inflation Rates From Prerequisite Four-Period GDP Deflators • Medium

Continuing with the four-period National Accounts table where Nominal GDP is 6 240, 6 680, 7 210, and 7 745 (R billions) and Real GDP is 5 920, 6 150, 6 515, and 6 830 (R billions): the first part of the calculation established that the implicit GDP Deflators are 105.41 in Period 1, 108.62 in Period 2, 110.67 in Period 3, and 113.40 in Period 4. Using these prerequisite GDP Deflators, calculate the annual percentage inflation rate for Period 2, Period 3, and Period 4.

Verified Solution: Annual GDP Deflator inflation is 3.05% in Period 2, slows to 1.89% in Period 3, and is 2.47% in Period 4.

$$\text{Prerequisite GDP Deflators: } \text{Defl}_1 = 105.41, \quad \text{Defl}_2 = 108.62, \quad \text{Defl}_3 = 110.67, \quad \text{Defl}_4 = 113.40$$

$$\text{Period 2 Inflation Rate: } \pi_2 = \frac{108.62 - 105.41}{105.41} \times 100 = \frac{3.21}{105.41} \times 100 = 3.05\%$$

$$\text{Period 3 Inflation Rate: } \pi_3 = \frac{110.67 - 108.62}{108.62} \times 100 = \frac{2.05}{108.62} \times 100 = 1.89\%$$

$$\text{Period 4 Inflation Rate: } \pi_4 = \frac{113.40 - 110.67}{110.67} \times 100 = \frac{2.73}{110.67} \times 100 = 2.47\%$$

First-Principles Proof: Correct! Using the prerequisite deflators, [(108.62 - 105.41)/105.41]×100 = 3.05%, [(110.67 - 108.62)/108.62]×100 = 1.89%, and [(113.40 - 110.67)/110.67]×100 = 2.47%.

PROBLEM 20.16 — CALCULATING CURRENT-PERIOD CPI INFLATION RATES FROM CONSECUTIVE INDEX SERIES

Calculating Current-Period CPI Inflation Rates From Consecutive Index Series • Easy

Solve two headline Consumer Price Index (CPI) calculation exercises: (Exercise A) A country's annual CPI over four consecutive periods is: Period 1 = 92.2, Period 2 = 105.7, Period 3 = 102.3, and Period 4 = 106.9. Calculate the inflation rate in the final period (Period 4). (Exercise B) A second economy records a CPI of 100 in Period 1 (base period), 125 in Period 2, and 135 in Period 3. Calculate the annual inflation rate between Period 2 and Period 3.

Verified Solution: In Exercise A, Period 4 inflation is 4.50% [(106.9 - 102.3) / 102.3 × 100]; in Exercise B, Period 3 inflation is 8.00% [(135 - 125) / 125 × 100].

$$\text{Exercise A (Period 4 Inflation): } \pi_4 = \frac{\text{CPI}_4 - \text{CPI}_3}{\text{CPI}_3} \times 100 = \frac{106.9 - 102.3}{102.3} \times 100 = \frac{4.6}{102.3} \times 100 = 4.50\%$$

$$\text{Exercise B (Period 3 Inflation): } \pi_3 = \frac{\text{CPI}_3 - \text{CPI}_2}{\text{CPI}_2} \times 100 = \frac{135 - 125}{125} \times 100 = \frac{10}{125} \times 100 = 8.00\%$$

$$\text{Common Pitfall in Exercise B: Subtracting } 135 - 125 = 10\% \text{ forgets to divide by the Period 2 base of } 125$$

First-Principles Proof: Correct! In Exercise A, (4.6 / 102.3) × 100 = 4.50%; in Exercise B, dividing the 10-point index gain by the Period 2 CPI of 125 gives (10 / 125) × 100 = 8.00%.

PROBLEM 20.17 — COMPUTING IMPLICIT GDP DEFLATORS AND INFLATION FROM OFFICIAL CURRENT AND CONSTANT GDP

Computing Implicit GDP Deflators and Inflation From Official Current and Constant GDP • Medium

Official statistical tables report GDP at current prices (Nominal GDP, in R millions) and GDP at constant base-period prices (Real GDP, in R millions) over three consecutive periods: Period 1: Current GDP = 888 057, Constant GDP = 624 379; Period 2: Current GDP = 982 944, Constant GDP = 642 038; and Period 3: Current GDP = 1 098 714, Constant GDP = 661 147. Calculate the implicit GDP Deflator for Period 2 and Period 3, and determine the annual inflation rate in Period 3.

Verified Solution: Period 2 Deflator is 153.10, Period 3 Deflator is 166.18 (166.2), and the Period 3 inflation rate is 8.55% (or 8.6% rounded to one decimal place).

$$\text{Period 1 Deflator} = \frac{888\,057}{624\,379} \times 100 = 142.23$$

$$\text{Period 2 Deflator} = \frac{982\,944}{642\,038} \times 100 = 153.10$$

$$\text{Period 3 Deflator} = \frac{1\,098\,714}{661\,147} \times 100 = 166.18 \quad (\text{or } 166.2 \text{ rounded to 1 decimal place})$$

$$\text{Period 3 Inflation Rate} = \frac{166.18 - 153.10}{153.10} \times 100 = 8.55\% \quad \left(\text{or } \frac{166.2 - 153.1}{153.1} \times 100 = 8.56\% \approx 8.6\%\right)$$

First-Principles Proof: Correct! Deflator 2 = (982 944 / 642 038) × 100 = 153.10 and Deflator 3 = (1 098 714 / 661 147) × 100 = 166.18 (166.2), giving [(166.18 - 153.10)/153.10] × 100 = 8.55% (8.6%).

PROBLEM 20.18 — CALCULATING YEAR-ON-YEAR MONTHLY HEADLINE INFLATION RATES FOR JANUARY THROUGH APRIL

Calculating Year-on-Year Monthly Headline Inflation Rates for January Through April • Medium

The monthly Consumer Price Index (CPI) for the first four months of Period 1 and Period 2 is recorded as follows: January (Period 1 = 103.8, Period 2 = 109.0); February (Period 1 = 104.1, Period 2 = 110.2); March (Period 1 = 104.8, Period 2 = 111.3); and April (Period 1 = 105.3, Period 2 = 113.1). Calculate the year-on-year headline inflation rate for each of the four months in Period 2 (comparing each month in Period 2 against the corresponding month in Period 1).

Verified Solution: Year-on-year monthly inflation rates in Period 2 are: January = 5.01% (5.0%), February = 5.86% (5.9%), March = 6.20% (6.2%), and April = 7.41% (7.4%).

$$\text{January Year-on-Year Inflation} = \frac{109.0 - 103.8}{103.8} \times 100 = \frac{5.2}{103.8} \times 100 = 5.01\% \ (5.0\%)$$
$$\text{February Year-on-Year Inflation} = \frac{110.2 - 104.1}{104.1} \times 100 = \frac{6.1}{104.1} \times 100 = 5.86\% \ (5.9\%)$$
$$\text{March Year-on-Year Inflation} = \frac{111.3 - 104.8}{104.8} \times 100 = \frac{6.5}{104.8} \times 100 = 6.20\% \ (6.2\%)$$
$$\text{April Year-on-Year Inflation} = \frac{113.1 - 105.3}{105.3} \times 100 = \frac{7.8}{105.3} \times 100 = 7.41\% \ (7.4\%)$$

First-Principles Proof: Correct! Comparing each month in Period 2 with the same month in Period 1 gives 5.2/103.8 = 5.01%, 6.1/104.1 = 5.86%, 6.5/104.8 = 6.20%, and 7.8/105.3 = 7.41%.

PROBLEM 20.19 — CALCULATING IMPLICIT GDP DEFLATORS AND INFLATION RATES ACROSS TWO MULTI-YEAR DATASETS

Calculating Implicit GDP Deflators and Inflation Rates Across Two Multi-Year Datasets • Medium

Perform two GDP Deflator inflation calculations: (Dataset 1) Over five periods, an economy's Nominal GDP and Real GDP (in R millions) end with Period 4 (Nominal GDP = 364 225, Real GDP = 388 431) and Period 5 (Nominal GDP = 397 651, Real GDP = 402 522). Calculate the Period 4 and Period 5 GDP Deflators and the Period 5 inflation rate. (Dataset 2) A second economy records: Period 1 (Nominal GDP = 280, Real GDP = 280), Period 2 (Nominal GDP = 315, Real GDP = 260), and Period 3 (Nominal GDP = 305, Real GDP = 300). Calculate its Period 2 GDP Deflator and Period 2 inflation rate.

Verified Solution: Dataset 1: Deflator rises from 93.77 to 98.79, yielding 5.35% inflation in Period 5; Dataset 2: Period 2 Deflator is 121.15, yielding 21.15% inflation.

$$\text{Dataset 1 Deflators: } \text{Defl}_4 = \frac{364\,225}{388\,431} \times 100 = 93.77; \quad \text{Defl}_5 = \frac{397\,651}{402\,522} \times 100 = 98.79$$
$$\text{Dataset 1 Period 5 Inflation: } \pi_5 = \frac{98.79 - 93.77}{93.77} \times 100 = 5.35\%$$
$$\text{Dataset 2 Deflators: } \text{Defl}_1 = \frac{280}{280} \times 100 = 100.00; \quad \text{Defl}_2 = \frac{315}{260} \times 100 = 121.15 \implies \pi_2 = \frac{121.15 - 100.00}{100.00} \times 100 = 21.15\%$$

First-Principles Proof: Correct! In Dataset 1, Deflators are 93.77 and 98.79, giving [(98.79 - 93.77)/93.77]×100 = 5.35%; in Dataset 2, Deflator 2 is (315/260)×100 = 121.15 (+21.15% inflation).

PROBLEM 20.20 — CPI VS IMPLICIT GDP DEFLATOR INFLATION SERIES & THE STRUCTURALIST / CONFLICT APPROACHES

CPI vs Implicit GDP Deflator Inflation Series & the Structuralist / Conflict Approaches • Hard

Analyse the inflation series and theoretical frameworks in macroeconomic analysis: (1) Given a four-period CPI series (Period 1 = 92.2, Period 2 = 105.7, Period 3 = 102.3, Period 4 = 106.9), calculate the Period 4 headline CPI inflation rate; (2) Given Nominal GDP (Period 1 = R232 234m, Period 2 = R297 899m, Period 3 = R327 071m) and Real GDP (Period 1 = R110 551m, Period 2 = R126 169m, Period 3 = R123 326m), calculate the Implicit GDP Deflator for all three periods and the Period 2 and Period 3 inflation rates, diagnosing Period 3's macroeconomic condition; and (3) Distinguish the Structuralist approach (underlying, initiating, and propagating factors) and the Conflict approach to inflation.

Verified Solution: (1) Period 4 CPI inflation is 4.5%; (2) Deflators are 210.07, 236.11, and 265.21 (inflation = 12.40% and 12.32%, with stagflation in Period 3); (3) Structuralist links underlying/initiating/propagating factors while Conflict cites competing claims.

$$(1)\ \text{Period 4 CPI Inflation: } \pi_4 = \frac{106.9 - 102.3}{102.3} \times 100 = \frac{4.6}{102.3} \times 100 = 4.497\% \approx 4.5\%$$
$$(2)\ \text{Implicit GDP Deflators } \left(\frac{\text{Nominal}}{\text{Real}} \times 100\right): \text{Def}_1 = \frac{232\,234}{110\,551} \times 100 = 210.07; \quad \text{Def}_2 = \frac{297\,899}{126\,169} \times 100 = 236.11; \quad \text{Def}_3 = \frac{327\,071}{123\,326} \times 100 = 265.21$$
$$\text{Deflator Inflation Rates: } \pi_2 = \frac{236.11 - 210.07}{210.07} \times 100 = 12.40\%; \quad \pi_3 = \frac{265.21 - 236.11}{236.11} \times 100 = 12.32\%$$
$$\text{Period 3 Diagnosis: High inflation } (12.32\%) \text{ combined with contracting Real GDP } (126\ 169 \ \rightarrow \ 123\ 326) = \text{Stagflation}$$
$$(3)\ \text{Structuralist Approach: Underlying factors (structural rigidities)} + \text{Initiating factors (shocks)} + \text{Propagating factors (wage-price spiral); Conflict Appr}$$

First-Principles Proof: Correct! CPI rises by (106.9 - 102.3)/102.3 = 4.5%; the GDP Deflator rises from 210.07 to 236.11 (+12.40%) to 265.21 (+12.32%) while Real GDP contracts in Period 3 (stagflation); and the Structuralist/Conflict definitions match macroeconomic principles.

CHAPTER 21 Employment & Income Distribution • Labour Market & Unemployment Dynamics • 10 Hours

Chapter 21: Unemployment, Poverty and Inequality

The unemployment pool inflows & outflows, strict (narrow) vs expanded (broad) definitions, labour force participation & absorption rates, four types of unemployment, the Eclectic Approach to job creation, the Phillips curve trade-off, incomes policy for stagflation, and SA's Triple Challenge.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Employment & Income Distribution • 10 Hours):

- Describe and discuss the strict (narrow) and expanded (broad) definitions of unemployment and calculate the unemployment rate from labour force data.
- Distinguish between frictional, seasonal, cyclical (demand-deficient), and structural unemployment and explain why structural unemployment involves a greater danger to the economy than frictional unemployment.
- Explain how unemployment and job creation fit into the Simple Macroeconomic Model (Y = C + I + G + X - Z) and apply an Eclectic Approach combining demand-side and supply-side measures.
- Explain the Phillips curve trade-off between unemployment and inflation, how stagflation shifts the Phillips curve rightward/upward, and how incomes policy can be applied to curb stagflation.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (3 IDENTITIES)

EQ 21.1 Strict vs Expanded Labour Force & Unemployment Rate

$$\text{LF}_{\text{strict}} = E + U_{\text{active}}, \quad u_{\text{strict}} = \frac{U_{\text{active}}}{E + U_{\text{active}}} \times 100\%$$

Meaning: The strict definition requires active job search in the past 4 weeks; the expanded definition adds discouraged work-seekers to both the numerator and denominator.

EQ 21.2 Expanded (Broad) Unemployment Rate

$$u_{\text{broad}} = \left(\frac{U_{\text{active}} + U_{\text{discouraged}}}{E + U_{\text{active}} + U_{\text{discouraged}}} \right) \times 100\%$$

Meaning: Includes discouraged work-seekers who want work and are available to start, even if they stopped actively searching.

EQ 21.3 Labour Force Participation & Absorption Rates

$$\text{LFPR} = \frac{\text{Labour Force}}{\text{Working-Age Pop (15–64)}} \times 100\%, \quad \text{AR} = \frac{\text{Employed}}{\text{Working-Age Pop}}$$

Meaning: Note the denominator trap: Unemployment rate divides by the Labour Force, whereas LFPR and Absorption Rate divide by the total Working-Age Population (aged 15–64).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 21.1 The Unemployment Pool: Four Inflows & Two Outflows (Core Macroeconomic Concept)

The level of unemployment is a stock that rises when inflows exceed outflows. People ENTER the unemployment pool for 4 reasons: (1) New entrants looking for a first job (school leavers/graduates), (2) Re-entrants returning to the labour force, (3) Job losers laid off or retrenched, and (4) Voluntary quitters searching for a better job. People LEAVE the pool by: (1) Finding employment, or (2) Withdrawing from the labour force (retiring or becoming discouraged under the strict definition).

> **Worked Proof:** In Core Module (p. 104): If Estonia has an active labour force of 250 000 people and 7 500 of them are unemployed, the unemployment rate is (7 500 / 250 000) × 100 = 3.0%.

CONCEPT 21.2 Four Types of Unemployment, Eclectic Policy & Phillips Curve (Core Macroeconomic Concept)

Unemployment is classified into four types: (1) Frictional (short-term search between jobs or after graduation), (2) Seasonal (predictable off-season layoffs in agriculture/tourism), (3) Cyclical / Demand-deficient (caused by recessions in aggregate demand), and (4) Structural (long-term mismatch between worker skills/location and job requirements, or automation). Structural unemployment is far more dangerous than frictional unemployment because it reflects deep structural mismatches that do not vanish when demand recovers.

> **Worked Proof:** In Core Module (p. 106): Duncan (recent graduate searching) and Thandi (starting a new job next month) are Frictional; Robert (lost his job in a recession) is Cyclical; Jan (unemployed economist in Durban while work is in Pretoria) and a shortage of managers alongside an oversupply of unskilled labourers are Structural. Core Module advocates an Eclectic Approach pooling demand-side (C+I+G+X-Z) and supply-side (skills training, SMME support, appropriate technology) measures, plus incomes policy against stagflation.

CONCEPT 21.3 Strict vs. Expanded Unemployment & Calculation Trap (Core Macroeconomic Concept)

Stats SA's strict (official/narrow) definition excludes discouraged work-seekers because they did not take active steps to look for work in the four weeks prior to the survey. The expanded (broad) definition includes discouraged work-seekers.

> **Worked Proof:** In Elaria: Employed = 6,750,000, Active Unemployed = 2,220,000, Discouraged = 1,030,000 (plus students, homemakers, etc.). Broad Unemployed = 2,220,000 + 1,030,000 = 3,250,000; Broad Labour Force = 6,750,000 + 3,250,000 = 10,000,000; Broad Unemployment Rate = (3.25m / 10.0m) × 100 = 32.50%.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (The Unemployment Pool: Four Inflows & Two Outflows): Assuming a rising unemployment rate proves that zero new jobs are being created or that GDP is shrinking.

First-Principles Law: The unemployment rate rises whenever the labour force grows faster than employment creation (jobless growth)—it measures the proportion of the economically active population unable to find work.

Fallacy (Four Types of Unemployment, Eclectic Policy & Phillips Curve): Classifying a worker permanently displaced by automation or geographic immobility as 'frictional' or 'cyclical'.

First-Principles Law: Technological displacement, geographic immobility, and skills mismatches persist even during economic booms; they represent structural unemployment, requiring holistic supply-side human capital development.

Fallacy (Strict vs. Expanded Unemployment & Calculation Trap): Including students, homemakers, pensioners, children (<15), or the medically disabled in the labour force denominator.

First-Principles Law: Students, homemakers, retirees, and children under 15 are Not Economically Active (outside the labour force) under both strict and expanded definitions.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (15 PROBLEMS)

PROBLEM 21.1 — STRICT UNEMPLOYMENT RATE & LABOUR FORCE PARTICIPATION RATE (ELARIA)

Macroeconomics, §21.1, pp. 469–471 • Medium

In the economy of Elaria (quarterly survey), workforce statistics are: Employed = 6 750 000; Unemployed (actively seeking work) = 2 220 000; Discouraged work seekers = 1 030 000; Students = 2 950 000; Homemakers = 1 280 000; Too old/young to work = 740 000; Illness = 640 000; Disability = 560 000. Calculate the strict (narrow) and broad (expanded) unemployment rates.

Verified Solution: The broad unemployment rate is 32.50% (3.25m / 10.00m) and the strict rate is 24.75%.

$$u_{\text{strict}} = \frac{2\,220\,000}{6\,750\,000 + 2\,220\,000} \times 100 = \frac{2.22\text{m}}{8.97\text{m}} \times 100 = 24.75\%.$$
$$u_{\text{broad}} = \frac{2\,220\,000 + 1\,030\,000}{6\,750\,000 + 2\,220\,000 + 1\,030\,000} \times 100 = \frac{3.25\text{m}}{10.00\text{m}} \times 100 = 32.50\%.$$

First-Principles Proof: Correct! Broad unemployment adds discouraged work-seekers (1.03m) to both active unemployed (2.22m -> 3.25m) and the strict labour force (8.97m -> 10.00m), giving 32.50%.

PROBLEM 21.2 — STRICT VS EXPANDED (BROAD) UNEMPLOYMENT & DISCOURAGED WORK-SEEKERS (TIGERIA)

Macroeconomics, §21.1, pp. 469–471 • Medium

Labour statistics for Tigeria are as follows: Total adult population = 34 000 000; Total employed = 18 600 000; Unemployed (active seekers) = 8 900 000; Unemployed (discouraged work-seekers) = 2 500 000. Calculate Tigeria's strict labour force, broad labour force, and broad unemployment rate.

Verified Solution: Strict labour force is 27.5m, broad labour force is 30.0m, and broad unemployment is 38.00%.

$$\text{LF}_{\text{strict}} = 18.6\text{m} + 8.9\text{m} = 27.5\text{m}; \quad \text{LF}_{\text{broad}} = 27.5\text{m} + 2.5\text{m} = 30.0\text{m}.$$
$$u_{\text{broad}} = \frac{8.9\text{m} + 2.5\text{m}}{30.0\text{m}} \times 100 = \frac{11.4\text{m}}{30.0\text{m}} \times 100 = 38.00\%.$$

First-Principles Proof: Correct! Strict LF = 18.6m + 8.9m = 27.5m; Broad LF adds 2.5m discouraged = 30.0m; Broad unemployment rate = (11.4m / 30.0m) × 100 = 38.00%.

PROBLEM 21.3 — DIAGNOSING FRICTIONAL, SEASONAL, STRUCTURAL & CYCLICAL UNEMPLOYMENT

Macroeconomics, §21.1, pp. 472–477 • Easy

Stats SA surveyed three unemployed individuals: (1) Yetunde worked on a soft-fruit farm in Elgin during the picking season and is now off-season; (2) Cipriano was a club bouncer retrenched during an economic recession and loadshedding downturn until business recovers; (3) Anouska worked at a travel agency that closed because online algorithms replaced human booking staff. Classify their unemployment types.

Verified Solution: Yetunde is seasonally unemployed, Cipriano is cyclically unemployed, and Anouska is structurally unemployed.

$$\text{Yetunde} \to \text{Seasonal}; \quad \text{Cipriano} \to \text{Cyclical}; \quad \text{Anouska} \to \text{Structural}.$$

First-Principles Proof: Correct! Agricultural harvest work is seasonal, recession/downturn layoffs are cyclical, and permanent displacement by online algorithms is structural.

PROBLEM 21.4 — STAGFLATION & RIGHTWARD SHIFT OF THE SHORT-RUN PHILLIPS CURVE

Macroeconomics, §20.6 & §21.1, pp. 459–471 • Medium

In macroeconomic labour analysis, what does a rightward shift of the short-run Phillips curve indicate, and what can we definitely conclude when a country's unemployment rate increases?

Verified Solution: Every inflation level has higher unemployment; the unemployed share of the labour force rose.

$$\text{Rightward Phillips Shift} \implies \text{Stagflation (higher } \pi \text{ at every } U). \quad u \uparrow \iff \frac{U}{LF} \uparrow .$$

First-Principles Proof: Correct! A rightward Phillips curve shift means a worse trade-off (more unemployment at every inflation rate), and a rising unemployment rate means the unemployed share of the labour force grew.

PROBLEM 21.5 — MULTI-PERIOD LABOUR FORCE GROWTH VS JOB CREATION & UNEMPLOYMENT RATE SHIFT

Macroeconomics, §21.1, pp. 468–471 • Easy

Labour market survey statistics for an economy show the following: in Period 1, 14 500 000 people were employed and 1 500 000 were strictly unemployed; in Period 2, 15 000 000 people were employed and 2 500 000 were strictly unemployed. Calculate the total labour force in Period 1 and Period 2, and the official unemployment rate in Period 2.

Verified Solution: The labour force was 16.0 million in Period 1 and 17.5 million in Period 2, and Period 2 unemployment was 14.29%.

$$LF_1 = 14.5\text{m} + 1.5\text{m} = 16.0\text{m}; \quad LF_2 = 15.0\text{m} + 2.5\text{m} = 17.5\text{m}. \quad u_2 = \frac{2.5}{17.5} \times 100 = 14.29\%.$$

First-Principles Proof: Correct! Summing employed and unemployed yields LF = 16.0m (14.5m + 1.5m) in Period 1 and 17.5m (15.0m + 2.5m) in Period 2, so u = 2.5 / 17.5 × 100 = 14.29%.

PROBLEM 21.6 — FILTERING NON-WORKING-AGE COHORTS TO CALCULATE THE STRICT LABOUR FORCE AND UNEMPLOYMENT RATE

Filtering Non-Working-Age Cohorts to Calculate the Strict Labour Force and Unemployment Rate • Medium

A demographic survey of an economy with a total population of 80 million people reveals the following breakdown: 15 million children under age 15; 20 million retired pensioners and full-time students outside the labour force; 15 million other economically inactive adults (homemakers and discouraged work-seekers not actively looking); 20 million employed workers; and 10 million unemployed people actively seeking work. Calculate (1) the working-age population, (2) the strict (official) Labour Force, (3) the strict Unemployment Rate, and (4) the Labour Force Participation Rate (LFPR).

Verified Solution: Working-age population is 65 million, strict Labour Force is 30 million, strict unemployment rate is 33.33%, and LFPR is 46.15%.

$$\text{Working-Age Population} = 80\text{m} - 15\text{m (children under 15)} = 65 \text{ million}$$
$$\text{Strict Labour Force (Economically Active Population)} = E + U = 20\text{m} + 10\text{m} = 30 \text{ million}$$
$$\text{Strict Unemployment Rate } (u) = \frac{U}{LF} \times 100 = \frac{10\text{m}}{30\text{m}} \times 100 = 33.33\%$$
$$\text{Labour Force Participation Rate (LFPR)} = \frac{LF}{\text{Working-Age Pop}} \times 100 = \frac{30\text{m}}{65\text{m}} \times 100 = 46.15\%$$

First-Principles Proof: Correct! Excluding the 15m children leaves 65m working-age adults; only the 20m employed + 10m active job-seekers form the 30m strict labour force, giving u = 10/30 = 33.33% and LFPR = 30/65 = 46.15%.

PROBLEM 21.7 — FOLLOW-UP CALCULATION OF TIGERIA'S BROAD UNEMPLOYMENT RATE FROM PREREQUISITE LABOUR FORCE TOTALS

Follow-Up Calculation of Tigeria's Broad Unemployment Rate From Prerequisite Labour Force Totals • Medium

Continuing with the labour market statistics for Tigeria where Total adult population = 34 000 000, Total employed = 18 600 000, Actively seeking unemployed = 8 900 000, and Discouraged work-seekers = 2 500 000: the first part of the analysis established that Tigeria's strict labour force is 27 500 000 (18.6m + 8.9m) and its broad (expanded) labour force is 30 000 000 (27.5m + 2.5m). Using these prerequisite labour force totals, calculate Tigeria's broad unemployment rate and compare it with the strict unemployment rate.

Verified Solution: Tigeria's broad unemployment rate is 38.00% (11.4m / 30.0m × 100), compared with a strict unemployment rate of 32.36% (8.9m / 27.5m × 100).

$$\text{Prerequisite Labour Force Totals: } \text{LF}_{\text{strict}} = 27.5 \text{ million}; \quad \text{LF}_{\text{broad}} = 30.0 \text{ million}$$
$$\text{Total Broad Unemployed} = \text{Active Unemployed } (8.9\text{m}) + \text{Discouraged Work-Seekers } (2.5\text{m}) = 11.4 \text{ million}$$
$$\text{Broad Unemployment Rate } (u_{\text{broad}}) = \frac{11.4\text{m}}{30.0\text{m}} \times 100 = 38.00\%$$
$$\text{Strict Unemployment Rate } (u_{\text{strict}}) = \frac{8.9\text{m}}{27.5\text{m}} \times 100 = 32.36\% \quad \text{(Broad rate is 5.64 percentage points higher)}$$

First-Principles Proof: Correct! Dividing total broad unemployed (8.9m + 2.5m = 11.4m) by the prerequisite broad labour force (30.0m) gives 38.00%, compared to 8.9m / 27.5m = 32.36% strict.

PROBLEM 21.8 — CALCULATING STRICT AND EXPANDED UNEMPLOYMENT RATES AND LFPR FOR A 55 MILLION WORKING-AGE COHORT

Calculating Strict and Expanded Unemployment Rates and LFPR for a 55 Million Working-Age Cohort • Medium

A national economy has a working-age population (aged 15–64) of 55 million people. According to the quarterly labour force survey, 26 million people are employed, 7 million are strictly unemployed (actively searching for work), and 5 million are discouraged work-seekers (who want work but have stopped actively searching). Calculate (1) the strict Labour Force and strict Unemployment Rate, and (2) the expanded (broad) Labour Force, expanded Labour Force Participation Rate (LFPR), and expanded Unemployment Rate.

Verified Solution: Strict LF is 33m (strict u = 21.21%, LFPR = 60.00%); Expanded LF is 38m, Expanded LFPR is 69.09% (38/55), and Expanded unemployment is 31.58% (12/38).

$$\text{Strict Labour Force} = E + U_{\text{strict}} = 26\text{m} + 7\text{m} = 33 \text{ million}; \quad u_{\text{strict}} = \frac{7\text{m}}{33\text{m}} \times 100 = 21.21\% \quad (\text{Strict LFPR} = \frac{33}{55} = 60.00\%)$$

$$\text{Expanded Labour Force} = 26\text{m} + 7\text{m} + 5\text{m} = 38 \text{ million}; \quad \text{Expanded LFPR} = \frac{38\text{m}}{55\text{m}} \times 100 = 69.09\%$$

$$\text{Expanded Unemployment Rate} = \frac{7\text{m} + 5\text{m}}{38\text{m}} \times 100 = \frac{12\text{m}}{38\text{m}} \times 100 = 31.58\%$$

First-Principles Proof: Correct! Strict LF = 26m + 7m = 33m (u = 7/33 = 21.21%); adding 5m discouraged work-seekers gives Expanded LF = 38m (LFPR = 38/55 = 69.09%) and Expanded u = 12/38 = 31.58%.

PROBLEM 21.9 — CALCULATING LABOUR FORCE, UNEMPLOYED POOL, AND UNEMPLOYMENT RATE (BRAZIL SURVEY A)

Calculating Labour Force, Unemployed Pool, and Unemployment Rate (Brazil Survey A) • Medium

A household labour survey in Brazil records the following demographic figures: Working-age population = 3 201 600; Persons not in the labour force (economically inactive) = 1 551 700; and Employed persons = 1 491 900. Calculate (1) the size of the active Labour Force, (2) the number of Unemployed persons, and (3) the official Unemployment Rate.

Verified Solution: The Labour Force is 1 649 900, the number of Unemployed persons is 158 000, and the official Unemployment Rate is 9.58% (with LFPR = 51.53%).

$$\text{Labour Force (LF)} = \text{Working-Age Population} - \text{Not in Labour Force} = 3\,201\,600 - 1\,551\,700 = 1\,649\,900$$

$$\text{Unemployed Persons (U)} = \text{Labour Force} - \text{Employed} = 1\,649\,900 - 1\,491\,900 = 158\,000$$

$$\text{Unemployment Rate (u)} = \frac{U}{\text{LF}} \times 100 = \frac{158\,000}{1\,649\,900} \times 100 = 9.58\% \quad (\text{LFPR} = \frac{1\,649\,900}{3\,201\,600} \times 100 = 51.53\%)$$

First-Principles Proof: Correct! Subtracting inactive persons (1 551 700) from working-age population (3 201 600) gives LF = 1 649 900; subtracting employed (1 491 900) leaves U = 158 000, so u = 158 000 / 1 649 900 = 9.58%.

PROBLEM 21.10 — WORKING BACKWARDS FROM THE UNEMPLOYMENT RATE AND LABOUR FORCE TO FIND TOTAL EMPLOYMENT (BRAZIL SURVEY B)

Working Backwards From the Unemployment Rate and Labour Force to Find Total Employment (Brazil Survey B) • Medium

In a subsequent labour market survey in Brazil, the working-age population is 4 486 600, the active labour force is 2 435 526, and the official unemployment rate is 8.8%. Calculate (1) the number of unemployed people, (2) the number of employed people, and (3) the Labour Force Participation Rate (LFPR).

Verified Solution: There are 214 326 unemployed people, 2 221 200 employed people, and the Labour Force Participation Rate is 54.28%.

$$\text{Since } u = \frac{U}{\text{LF}} = 0.088, \quad \text{Unemployed (U)} = 0.088 \times 2\,435\,526 = 214\,326 \text{ people}$$

$$\text{Employed (E)} = \text{LF} - U = 2\,435\,526 - 214\,326 = 2\,221\,200 \text{ people} \quad (\text{or } (1 - 0.088) \times 2\,435\,526 = 2\,221\,200)$$

$$\text{Labour Force Participation Rate (LFPR)} = \frac{2\,435\,526}{4\,486\,600} \times 100 = 54.28\%$$

First-Principles Proof: Correct! 8.8% of the 2 435 526 labour force is 214 326 unemployed; subtracting 214 326 from 2 435 526 leaves 2 221 200 employed, and LFPR = 2 435 526 / 4 486 600 = 54.28%.

PROBLEM 21.11 — DERIVING WORKING-AGE POPULATION, ACTIVE LABOUR FORCE, AND UNEMPLOYED POOL (NEVERLAND PART 1)

Deriving Working-Age Population, Active Labour Force, and Unemployed Pool (Neverland Part 1) • Medium

In the country of Neverland, the total population is 30 million. Of this total population, 10 million people are either under 16 years of age or over 64 years of age (outside the working-age cohort). Among the working-age population, 9 million people do not want to work (full-time students, homemakers, and retirees), and 8 million people are currently employed. Calculate (1) Neverland's working-age population, (2) the active Labour Force, and (3) the number of Unemployed people.

Verified Solution: Neverland's working-age population is 20 million, its active Labour Force is 11 million, and 3 million people are unemployed.

$$\text{Working-Age Population (aged 16–64)} = 30\text{m} - 10\text{m} = 20 \text{ million}$$
$$\text{Active Labour Force (LF)} = \text{Working-Age Population (20m)} - \text{Economically Inactive (9m)} = 11 \text{ million}$$
$$\text{Unemployed Persons (U)} = \text{Labour Force (11m)} - \text{Employed (8m)} = 3 \text{ million}$$

First-Principles Proof: Correct! Subtracting 10m outside working age from 30m leaves 20m working-age adults; subtracting 9m inactive adults leaves a Labour Force of 11m, and subtracting 8m employed leaves 3m unemployed.

PROBLEM 21.12 — FOLLOW-UP CALCULATION OF NEVERLAND'S UNEMPLOYMENT RATE AND LFPR USING PREREQUISITE TOTALS

Follow-Up Calculation of Neverland's Unemployment Rate and LFPR Using Prerequisite Totals • Medium

Continuing with the demographic data for Neverland (Total population = 30 million; Outside working age = 10 million; Economically inactive not wanting work = 9 million; Employed = 8 million): the first part of the analysis established that Neverland's working-age population is 20 million, its active Labour Force is 11 million, and the number of Unemployed people is 3 million. Using these prerequisite totals, calculate Neverland's official Unemployment Rate and Labour Force Participation Rate (LFPR).

Verified Solution: Neverland's official unemployment rate is 27.27% (3m / 11m × 100) and its Labour Force Participation Rate is 55.00% (11m / 20m × 100).

$$\text{Prerequisite Totals: Working-Age Population} = 20\text{m}; \quad \text{Labour Force (LF)} = 11\text{m}; \quad \text{Unemployed (U)} = 3\text{m}$$
$$\text{Official Unemployment Rate } (u) = \frac{U}{\text{LF}} \times 100 = \frac{3\text{m}}{11\text{m}} \times 100 = 27.27\%$$
$$\text{Labour Force Participation Rate (LFPR)} = \frac{\text{LF}}{\text{Working-Age Population}} \times 100 = \frac{11\text{m}}{20\text{m}} \times 100 = 55.00\%$$
$$\text{Employment-to-Population Ratio (Absorption Rate)} = \frac{8\text{m}}{20\text{m}} \times 100 = 40.00\%$$

First-Principles Proof: Correct! Dividing prerequisite Unemployed (3m) by the prerequisite Labour Force (11m) gives u = 27.27%, and dividing the Labour Force (11m) by Working-Age Population (20m) gives LFPR = 55.00%.

PROBLEM 21.13 — CLASSIFYING CYCLICAL, SEASONAL, FRICTIONAL, AND STRUCTURAL UNEMPLOYMENT CASES

Classifying Cyclical, Seasonal, Frictional, and Structural Unemployment Cases • Easy

Classify the type of unemployment experienced in each of the following four labour market cases: (1) An assembly-line autoworker who is laid off because a nationwide economic downturn caused a sharp decline in new vehicle sales; (2) Migrant agricultural workers who are unemployed after the summer crop harvest has been completed; (3) A qualified real estate agent who resigns from her agency in Johannesburg and spends six weeks searching for a position after relocating to Cape Town; and (4) A mining geologist who is permanently retrenched after a steep union-negotiated wage hike induces the mine to automate exploration.

Verified Solution: (1) Cyclical unemployment; (2) Seasonal unemployment; (3) Frictional unemployment; and (4) Structural unemployment.

$$(1) \text{ Autoworker laid off due to aggregate sales slump} \implies \text{Cyclical (demand-deficient) unemployment}$$
$$(2) \text{ Migrant farm workers unemployed after harvest season} \implies \text{Seasonal unemployment}$$
$$(3) \text{ Real estate agent relocating between cities} \implies \text{Frictional (search) unemployment}$$
$$(4) \text{ Geologist permanently displaced by wage-induced automation} \implies \text{Structural unemployment}$$

First-Principles Proof: Correct! Recessionary sales declines cause cyclical unemployment, post-harvest layoffs are seasonal, job search during geographic relocation is frictional, and permanent wage/automation displacement is structural.

PROBLEM 21.14 — LABOUR FORCE EXITS, IMPORT LEAKAGES, AND DEMAND-SIDE VS SUPPLY-SIDE EMPLOYMENT POLICIES

Labour Force Exits, Import Leakages, and Demand-Side vs Supply-Side Employment Policies • Medium

Evaluate three core labour-market propositions: (1) Which action represents a person leaving the active labour force (an outflow that reduces the size of the labour force)? (2) Why does an increase in South African imports of foreign consumer goods NOT generate domestic employment in South Africa? (3) Which combination of demand-side and supply-side policies is required to reduce structural and cyclical unemployment?

Verified Solution: (1) Retiring from a job at age 62 (or becoming a full-time student); (2) Imports are a circular-flow leakage paid to foreign producers; (3) Demand stimulus for cyclical plus skills/education for structural.

$$\text{(1) Leaving the Labour Force: A worker retiring at age 62 (or leaving employment to enter full-time university study) exits the active labour force}$$
$$\text{Note: Moving from Employed to Unemployed (or vice versa) shifts workers WITHIN the labour force without changing } LF = E + U$$
$$\text{(2) Imports } (Z): \text{ Spending on foreign-made goods is a leakage from the domestic circular flow that supports foreign (not domestic) jobs}$$
$$\text{(3) Policy Mix: Demand-side fiscal/monetary stimulus counters cyclical unemployment, while education/skills training and labour-market flexibility count}$$

First-Principles Proof: Correct! Retirement exits the active labour force, rising imports leak spending abroad rather than creating local jobs, and structural/cyclical unemployment require targeted supply-side skills and demand-side policies.

PROBLEM 21.15 — ESTONIA UNEMPLOYMENT RATE, CLASSIFYING FIVE UNEMPLOYMENT SCENARIOS & THE ECLECTIC APPROACH

Estonia Unemployment Rate, Classifying Five Unemployment Scenarios & the Eclectic Approach • Medium

Evaluate the labour market exercises in macroeconomic analysis: (1) In Estonia, 7 500 people are unemployed out of an economically active population (labour force) of 250 000; calculate the strict unemployment rate; (2) Classify five cases: (a) Duncan quits his job at Pick n Pay to look for a better-paying job; (b) Jan is retrenched from a gold mine due to a permanent fall in gold demand and lacks skills for other industries; (c) Thandi completes her BCom degree and is searching for her first graduate position; (d) Robert is laid off from his construction job during a general economic recession; and (e) an economy simultaneously experiences a shortage of managers and an oversupply of unskilled labourers; and (3) Define the Eclectic Approach and Incomes Policy in the Phillips-curve debate.

Verified Solution: (1) Estonia's unemployment rate is 3.0%; (2) a=Frictional, b=Structural, c=Frictional, d=Cyclical, e=Structural; (3) The Eclectic Approach combines theoretical insights while an Incomes Policy restrains wage/price growth to match productivity.

$$\text{(1) Estonia Unemployment Rate: } u = \frac{\text{Unemployed}}{\text{Economically Active Population}} \times 100 = \frac{7\,500}{250\,000} \times 100 = 3.0\%$$
$$\text{(2) Scenario Classifications: (a) Duncan} = \text{Frictional; (b) Jan} = \text{Structural; (c) Thandi} = \text{Frictional; (d) Robert} = \text{Cyclical; (e) Skill mismatch} = \text{Structura}$$
$$\text{(3) Eclectic Approach: Synthesises demand-management and supply-side insights across economic conditions; Incomes Policy: Tripartite agreement (gover}$$

First-Principles Proof: Correct! 7 500 / 250 000 × 100 = 3.0%; voluntary job search (Duncan) and initial graduate search (Thandi) are frictional, permanent industry decline (Jan) and skill mismatch (e) are structural, and recession layoffs (Robert) are cyclical.

CHAPTER 22 Growth & Economic Fluctuations • Economic Growth & Business Cycles • 10 Hours

Chapter 22: Economic Growth and Business Cycles

Definition and measurement of real and inclusive economic growth, problems with GDP as a welfare measure, the four elements of the business cycle, Classical vs Keynesian vs Structuralist cycle theories, SARB leading/coincident/lagging indicators, supply vs demand growth drivers, and fundamental causes of low growth in SA.

OFFICIAL STUDY GUIDE LEARNING OUTCOMES (Growth & Economic Fluctuations • 10 Hours):

- Define economic growth, explain how it is measured using constant-price Real GDP, and explain the necessity of inclusive, shared growth in an unequal developing society.
- Identify the five major problems associated with GDP as a measure of total production and economic welfare.
- Define the business cycle, illustrate and discuss its four elements (trough, upswing/expansion, peak, downswing/contraction), and explain how business cycles are measured.
- Identify the supply-side and demand-side sources of economic growth, diagnose the fundamental causes of low economic growth in South Africa, and formulate remedial policy measures.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS (3 IDENTITIES)

EQ 22.1 **Annual Economic Growth Rate**

$$g_t = \left(\frac{\text{Real GDP}_t - \text{Real GDP}_{t-1}}{\text{Real GDP}_{t-1}} \right) \times 100\%$$

Meaning: Economic growth MUST be calculated using Real GDP (constant prices) to eliminate price inflation; per capita real GDP adjusts further for population growth.

EQ 22.2 **Rule of 70 (Doubling Time)**

$$\text{Years to Double Real GDP} \approx \frac{70}{g\,(\%)}$$

Meaning: Approximates how many years it takes for real output or living standards to double at a steady annual growth rate g%.

EQ 22.3 **Aggregate Production Function & Capital Intensity**

$$Y = A \cdot f(K, L), \quad \text{Capital Deepening: } \uparrow \left(\frac{K}{L}\right), \quad \text{Capital Widening: } \Delta$$

Meaning: Capital widening equips new workers at a constant capital-to-labour ratio (K/L); capital deepening raises capital per worker, boosting labour productivity.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CONCEPT 22.1 **Five Problems with GDP as a Measure of Welfare** (Core Macroeconomic Concept)

Real GDP is flawed as a measure of societal well-being because: (1) Non-market production (e.g. unpaid household work, subsistence farming) is difficult or impossible to value, (2) Unrecorded/informal sector activity and illegal trade are omitted, (3) GDP data are frequently revised after initial estimates, (4) Aggregate or even per capita GDP ignores the distribution of income (inequality), and (5) GDP makes no distinction between economic 'goods' (hospitals, schools) and 'bads' (pollution, crime expenditure, weapons)—negative externalities are not subtracted.

> **Worked Proof:** If Country A has twice the per capita GDP of Country B, residents of A are NOT necessarily twice as well off because per capita GDP is a simple mean that hides extreme inequality and environmental costs.

CONCEPT 22.2 **The Business Cycle: Four Elements, Theories & SARB Indicators** (Core Macroeconomic Concept)

A complete business cycle has four elements: Trough (lowest turning point) → Expansion / Upswing (rising real GDP) → Peak / Boom (upper turning point where almost all resources are in use) → Contraction / Downswing / Recession (declining real GDP). Classical economists view the economy as inherently stable (cycles caused by exogenous shocks); Keynesian economists view cycles as endogenous (inherent to modern market economies). SARB tracks Leading indicators (turn before the economy, e.g. share prices, building plans passed, real M1), Coincident indicators (move with the economy, e.g. manufacturing volume), and Lagging indicators (turn after, e.g. prime rate, unemployment duration).

> **Worked Proof:** On a business cycle diagram where Point A is a Trough, Point B is a Peak, and Point C is the next Trough: A → B is the Expansion/Upswing phase, and B → C is the Contraction/Downswing phase.

CONCEPT 22.3 **Inclusive Growth, Supply/Demand Drivers & Causes of Low Growth in SA** (Core Macroeconomic Concept)

Core Module emphasises that in a highly unequal society like South Africa, the goal is inclusive growth—accelerating growth while sharing its benefits via broad job creation. Growth requires both Supply factors (natural resources, human capital skills, capital widening & deepening, entrepreneurship/technology) and Demand factors (domestic demand C+I+G, export promotion X, import substitution Z). Fundamental causes of low growth in South Africa include skills shortages, electricity and logistics bottlenecks, low domestic saving and investment, and structural unemployment.

> **Worked Proof:** South Africa's mineral wealth shows that natural resources are neither necessary nor sufficient for sustained growth: commodity price volatility and exchange rate swings can create a 'resource curse' unless revenues are invested in human and physical capital.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy	First-Principles Law
Fallacy (Five Problems with GDP as a Measure of Welfare): Thinking 'GDP pertains only to production inside the country' is a welfare flaw rather than its literal geographic definition.	**First-Principles Law:** GDP is by definition a geographic production total; its true welfare limitations are unrecorded informal activity, non-market valuation, data revisions, inequality blindness, and counting 'bads' alongside 'goods'.
Fallacy (The Business Cycle: Four Elements, Theories & SARB Indicators): Confusing a slowdown in the positive rate of growth with a recession (contraction).	**First-Principles Law:** During a contraction (downswing), the level of real economic activity declines between a Peak and a Trough.
Fallacy (Inclusive Growth, Supply/Demand Drivers & Causes of Low Growth in SA): Believing that higher government spending that drives up local interest rates (crowding out private investment) always guarantees higher long-run growth.	**First-Principles Law:** When supply-side capacity is constrained (e.g. skills shortages or electricity bottlenecks) or higher G crowds out private investment I via higher interest rates, demand expansion fuels inflation rather than long-run real growth.

STANDALONE ANALYTICAL PROBLEMS & STEP-BY-STEP DERIVATIONS (6 PROBLEMS)

PROBLEM 22.1 — BUSINESS CYCLE PHASES: TROUGH, UPSWING (EXPANSION), PEAK & DOWNSWING

Macroeconomics, §22.1–22.3, pp. 485–500 • Easy

Which phase of the business cycle occurs when almost all available resources in the economy are in use, and what does a strategy of 'import substitution' involve?

Verified Solution: The peak (boom) phase; using import tariffs to encourage local production of formerly imported goods.

$$\text{Maximum Capacity Utilisation} \rightarrow \text{Peak (Boom) Phase.} \quad \text{Import Substitution} \rightarrow \text{Tariffs/Quotas to produce former imports locally.}$$

First-Principles Proof: Correct! Resource utilisation reaches its maximum at the peak (boom) of the cycle, and import substitution uses trade protection to manufacture previously imported goods domestically.

PROBLEM 22.2 — LIMITATIONS OF REAL GDP PER CAPITA AS A MEASURE OF SOCIAL WELFARE

Macroeconomics, §22.1, pp. 486–488 • Easy

If per capita real GDP in Country A is twice as high as per capita real GDP in Country B, why does this NOT imply that every resident in Country A is twice as well off, and which item is NOT a welfare limitation of GDP?

Verified Solution: Per capita GDP is an average that hides inequality; 'pertaining only to domestic production' is not a flaw.

$$\text{Per Capita GDP} = \frac{\text{GDP}}{\text{Population}} \text{ (ignores Gini inequality, informal sector, non-market output, pollution).}$$

First-Principles Proof: Correct! Per capita GDP contains zero information about income distribution, and 'pertaining only to production inside the country' is simply the geographic definition of GDP (Ch 22 MCQ 5).

PROBLEM 22.3 — CAPITAL WIDENING VS CAPITAL DEEPENING IN LONG-RUN ECONOMIC GROWTH

Macroeconomics, §22.1 & §22.3, pp. 486–498 • Medium

Distinguish between 'capital widening' and 'capital deepening' in the context of long-run economic growth, and apply the Rule of 70 to determine how long it takes for a nation's real GDP per capita to double if it grows steadily at 3.5% per year.

Verified Solution: Capital widening keeps the K/L ratio constant while capital deepening raises K/L; doubling takes 20 years.

$$\text{Capital Widening: } \frac{\Delta K}{K} = \frac{\Delta L}{L} \implies \frac{K}{L} \text{ constant.} \quad \text{Capital Deepening: } \frac{K}{L} \uparrow. \quad T_{\text{double}} \approx \frac{70}{3.5} = 20 \text{ years.}$$

First-Principles Proof: Correct! Capital widening expands capital in proportion to labour growth (constant K/L), capital deepening increases capital per worker (higher K/L), and 70 / 3.5 = 20 years.

PROBLEM 22.4 — LEADING, COINCIDENT & LAGGING BUSINESS CYCLE INDICATORS & RULE OF 70

Macroeconomics, §22.2, pp. 490–494 • Easy

When economists at the South African Reserve Bank (SARB) analyse business cycle turning points, how do they classify (1) new passenger vehicle sales and building plans passed, (2) industrial production and retail sales, and (3) employment in the non-agricultural sector and hours worked in construction?

Verified Solution: (1) Leading indicators (peak before the cycle), (2) Coincident indicators, and (3) Lagging indicators (turn after the cycle).

$$\text{Building plans / JSE / New car sales} \rightarrow \text{Leading;} \quad \text{Industrial output} \rightarrow \text{Coincident;} \quad \text{Formal employment} \rightarrow \text{Lagging.}$$

First-Principles Proof: Correct! Building plans and new vehicle sales lead economic turning points, industrial production moves coincidentally with the cycle, and formal employment lags behind turning points.

PROBLEM 22.5 — HUMAN CAPITAL PRODUCTIVITY, INTEREST-RATE CROWDING OUT, AND THE NATURAL RESOURCE CURSE

Human Capital Productivity, Interest-Rate Crowding Out, and the Natural Resource Curse • Medium

In the analysis of long-run economic growth (Chapter 22): (1) Why does higher government spending that drives up domestic interest rates NOT necessarily lead to higher long-run economic growth? (2) How does investment in human capital (education, health, and vocational training) affect the Aggregate Supply (AS) curve? (3) Why is an abundance of natural mineral resources considered neither necessary nor sufficient for sustained economic growth?

Verified Solution: (1) Higher interest rates crowd out private investment I; (2) Human capital raises labour productivity and shifts AS rightward; (3) Mineral wealth alone cannot guarantee growth due to commodity volatility and Dutch disease.

$$\begin{gathered}\text{(1) Interest-Rate Crowding Out: } G \uparrow \implies i \uparrow \implies \text{private physical capital investment } I \downarrow \text{ (impeding capital deepening)} \\ \text{(2) Human Capital Investment: Raises labour productivity (output per worker), shifting the long-run Aggregate Supply (AS) curve rightward} \\ \text{(3) Natural Resources / Minerals: Commodity price volatility, exchange-rate appreciation (Dutch disease), and rent-seeking can slow non-mining growth u}\end{gathered}$$

First-Principles Proof: Correct! Debt-financed government spending that raises interest rates crowds out private capital formation, human capital shifts AS rightward via higher productivity, and resource-rich nations still require human/institutional capital.

PROBLEM 22.6 — INCLUSIVE ECONOMIC GROWTH, STRUCTURAL GROWTH CONSTRAINTS IN SOUTH AFRICA & BUSINESS CYCLE ELEMENTS

Inclusive Economic Growth, Structural Growth Constraints in South Africa & Business Cycle Elements • Medium

In the analysis of economic growth and business cycles in macroeconomic analysis: (Chapter 22): (1) Define 'inclusive economic growth' and explain why headline real GDP growth alone is insufficient for socioeconomic transformation; (2) Identify the primary structural causes of low economic growth in South Africa highlighted in the curriculum; and (3) State the four sequential elements of a complete business cycle between two successive troughs.

Verified Solution: (1) Inclusive growth requires broad-based employment creation and poverty/inequality reduction; (2) Low SA growth stems from skills shortages, infrastructure/energy bottlenecks, and low saving/investment; (3) Cycle elements: Trough, Upswing, Peak, Downswing.

(1) Inclusive Economic Growth: Sustained real output expansion that is broadly shared across the population, generating productive employment and redu
(2) Structural Causes of Low Growth in SA: Severe skills shortages / mismatch, inadequate infrastructure and electricity supply constraints, low domestic
(3) Four Elements of a Complete Business Cycle: Trough (lower turning point) $\rightarrow$ Upswing / Expansion (boom) $\rightarrow$ Peak (upper turning point) $\rightarrow$ Downs

First-Principles Proof: Correct! Inclusive growth ensures that rising real GDP translates into broad job creation and reduced inequality, addressing South Africa's structural skills, energy/infrastructure, and investment constraints across the Trough-Upswing-Peak-Downswing cycle.

Comprehensive Macroeconomic Synthesis Suites & Analytical Derivations

SYNTHESIS SUITE I — NATIONAL ACCOUNTS, VALUE-ADDED & OPEN-ECONOMY MULTIPLIERS — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 17, CHAPTER 18, CHAPTER 20, CHAPTER 21

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points • 30 Linked Problems

Conceptual Foundations — Core Definitions: Define the following macroeconomic concepts: Fiscal policy, Autonomous consumption, Comparative advantage, and Unit of account.

Analytical Solution: Fiscal policy: Government's policy in respect of the level and composition of government spending, taxation, and borrowing to influence economic activity. Autonomous consumption ($\bar{C}$): That part of consumption expenditure that is independent of the level of income (consumption when Y = 0). Comparative advantage: The ability of a country to produce a good at a lower opportunity cost (sacrificing fewer units of the alternative good) than another country. Unit of account: An agreed standard measure for stating and comparing the prices of goods and services.

Theoretical Synthesis — Cost-Push Inflation & Price Indices: Identify cost-push inflation from a leftward Aggregate Supply (AS) shift diagram, state its primary causes, and contrast the Consumer Price Index (CPI) with the Producer Price Index (PPI).

Analytical Solution: Cost-push inflation (AS shifts left/up, raising the price level P and lowering real output Y) is triggered by: wage increases exceeding productivity growth, higher imported capital/intermediate input costs, expanded profit margins, falling productivity, or natural disasters. CPI vs PPI: CPI measures the cost of living for households vs PPI measures the cost of production; CPI covers consumer goods and services vs PPI covers goods only; CPI includes VAT vs PPI excludes VAT; CPI excludes capital and intermediate goods vs PPI includes them.

Theoretical Synthesis — Labour Market Flows & Business Cycle Theories: Classify frictional, structural, cyclical, and seasonal unemployment scenarios, identify inflows and outflows of the unemployment pool, and contrast Classical versus Keynesian business cycle theories across expansion and contraction phases.

Analytical Solution: Inflows to unemployment: new entrants, re-entrants, retrenched job losers, and voluntary quitters. Outflows: finding employment or withdrawing from the labour force. Classical theory views the economy as inherently stable (cycles driven by exogenous shocks), whereas Keynesian theory views fluctuations as endogenous to unstable private investment. Upswing (Trough to Peak) features rising real GDP and employment; Downswing (Peak to Trough) features contracting real GDP.

SYNTHESIS SUITE II — REAL GROWTH, FISCAL TRANSFERS & MONETARY EQUILIBRIUM — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 17, CHAPTER 18, CHAPTER 20, CHAPTER 21, CHAPTER 22

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points • 31 Linked Problems

Conceptual Foundations — Trade, Exchange Rates & Consumption: Distinguish between Absolute and Comparative advantage, explain the determinants of currency appreciation, describe the three characteristics of the Keynesian consumption function, and explain why autonomous government spending does not alter the size of the multiplier.

Analytical Solution: Absolute advantage compares total output from given inputs; comparative advantage compares opportunity costs. Currency appreciation is driven by stronger export demand, higher domestic interest rates attracting capital inflows, lower relative inflation, or rising commodity prices. Consumption function: $C > 0$ at $Y = 0$ ($\bar{C} > 0$), C rises with Y ($c > 0$), and $\Delta C < \Delta Y$ ($0 < c < 1$). Government spending G is autonomous (shifting A upward) whereas $\alpha = 1/[1 - c(1-t) + m]$ depends strictly on c, t, and m.

Theoretical Synthesis — Endogenous Money, Public Sector & Inequality: Explain the effect of a decline in money demand under a repo-rate monetary regime with no independent money supply curve, list the four components and three functions of the public sector, and identify the three standard measures of income inequality.

Analytical Solution: A leftward shift of money demand L along the horizontal interest rate line i0 reduces the equilibrium quantity of money M endogenously. The public sector comprises central government, provincial government, local municipalities, and public corporations, performing allocative, distributive, and stabilisation functions. Income inequality is measured via the Lorenz curve, the Gini coefficient, and quantile ratios.

SYNTHESIS SUITE III — DEFLATORS, STATUTORY TAX TABLES & AGGREGATE EXPENDITURE — CHAPTER 13, CHAPTER 15, CHAPTER 18, CHAPTER 19, CHAPTER 20, CHAPTER 21

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points • 29 Linked Problems

Conceptual Foundations — Stabilisation Policy & National Accounts Identity: Contrast Monetarist and Keynesian stabilisation views, explain why the income and production methods of GDP yield identical totals after a wage change, and identify the supply-side drivers of long-run economic growth.

Analytical Solution: Keynesians advocate active counter-cyclical fiscal and monetary intervention during downswings, whereas Monetarists favour stable rules. A minimum-wage change alters the distribution between wages and operating surplus without creating a discrepancy between the income and production approaches to GDP. Economic growth is driven by human capital expansion, physical capital accumulation/technology, and natural resource/institutional efficiency.

Theoretical Synthesis — Market Failure, Open-Market Operations & Disposable Income: Illustrate negative externalities (social vs private marginal cost), explain central bank open-market bond purchases, and distinguish total national income from household disposable income.

Analytical Solution: With negative externalities, Marginal Social Cost (MSC) lies above Marginal Private Cost (MPC), causing unregulated competitive markets to overproduce at a socially inefficiently low price; corrective Pigouvian taxes internalise the external cost. Open-market bond purchases inject reserves into commercial banks, expanding credit and deposits. Disposable income Yd = Y - T is income remaining after direct taxes.

SYNTHESIS SUITE IV — VALUE-ADDED CHAINS, TRADE PROTECTION & LEAKAGES — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 17, CHAPTER 18, CHAPTER 20, CHAPTER 21, CHAPTER 22

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points • 30 Linked Problems

Conceptual Foundations — Inflation Targeting, Phillips Curve & Trade Policy: Explain why per capita real GDP is preferred for welfare comparisons, state the central bank inflation target range (3%–6%), illustrate the short-run Phillips curve trade-off, and list the policy instruments used to restrict imports.

Analytical Solution: Real GDP per capita adjusts for both price inflation and population size. The inflation target band is 3% to 6%. The short-run Phillips curve depicts an inverse relationship between inflation and unemployment. Import restriction instruments include tariffs, quotas, domestic production subsidies, non-tariff barriers, exchange controls, and exchange-rate policy.

SYNTHESIS SUITE V — INDUCED IMPORTS, DEMOGRAPHIC FILTERING & LORENZ GEOMETRY — CHAPTER 13, CHAPTER 14, CHAPTER 15, CHAPTER 16, CHAPTER 17, CHAPTER 18, CHAPTER 20, CHAPTER 21, CHAPTER 22

Scope: Macroeconomic Curriculum Synthesis | Verification: Quantitative & Theoretical Verification | 180 Mins • 100 Points • 30 Linked Problems

Conceptual Foundations — Lorenz Geometry, Say's Law & Public Sector Structure: Interpret the Lorenz curve 45-degree diagonal and Gini coefficient, contrast Say's Law (supply creates its own demand) with Keynesian effective demand (demand determines output), and explain why financial cash balances are not a physical factor of production.

Analytical Solution: The 45-degree diagonal represents perfect income equality, the bowed area represents inequality, and a Gini coefficient near 0.67 indicates severe inequality. Along the 45-degree line in the Keynesian diagram, aggregate expenditure equals total income (A = Y); above it, A > Y (inventories fall, output expands). Money is a financial medium of exchange, whereas economic capital consists strictly of physical machinery, equipment, and buildings.